



Domino Ax-Series Product Manual

Ax150i
Ax350i
Ax550i

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DOMINO Ax-SERIES PRODUCT MANUAL

This product manual, Domino Part No. EPT019297, is for use in the operation and maintenance of Domino Ax-Series printers.

This product manual is the official authority for the operation and maintenance of the Domino Ax-Series range of printers. It is the “Original Instructions” for the purposes of the Machinery Directive.

This product manual is based on software version 01.00.30.

Users of this printer are warned that it is essential to read, understand and act according to the information given in Part 1: Health and Safety. This part of the product manual also specifies a set of symbols which are used elsewhere in the product manual to convey special warnings or requirements. It is, therefore, essential that users are also familiar with these symbols and act accordingly.

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Domino Printing Sciences plc. has a policy of continuous product improvement, the company therefore reserves the right to modify the specification contained in this product manual without notice.

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CONTENTS OF EU DECLARATION OF CONFORMITY

No. Doc-0010447_R04

Manufacturers name: Domino UK Limited

Manufacturers address: Bar Hill, Cambridge CB23 8TU.

This declaration of conformity is issued under the sole responsibility of the manufacturer.

Model Number : Ax150i, Ax350i, Ax550i (from serial number AX0000000335)

Product Name : Ax-Series Continuous Inkjet Printer

The object of the declaration described above is in conformity with the relevant Union harmonisation legislation:

2011/65/EU : RoHS Directive

2014/53/EU : RED Directive

EN 300 330-2 V1.6.1	Electromagnetic compatibility and Radio spectrum Matters (ERM); Short Range Devices (SRD); Radio equipment in the frequency range 9 kHz to 25 MHz and inductive loop systems in the frequency range 9 kHz to 30 MHz; Part 2: Harmonized EN covering the essential requirements of article 3.2 of the R&TTE Directive
EN 301 489-1 V1.9.2	Electromagnetic compatibility and Radio spectrum Matters (ERM); Electromagnetic Compatibility (EMC) standard for radio equipment and services; Part 1: Common technical requirements
EN 301 489-3 V1.6.1	Electromagnetic compatibility and Radio spectrum Matters (ERM); Electromagnetic Compatibility (EMC) standard for radio equipment and services; Part 3: Specific conditions for Short-Range Devices (SRD) operating on frequencies between 9 kHz and 246 GHz
EN 61000-6-2:2005	Electromagnetic compatibility (EMC) – Part 6-2: Generic standards – Immunity for industrial environments
EN 60950-1:2006/A2:2013	Information technology equipment – safety – Part 1: General requirements.

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Signed for and on behalf of
Domino UK Limited.
Bar Hill,
Cambridge,

Date:

23rd November 2016

Signature:

Name and Job title: Carl Reynaud, Group Chief Engineer

FCC Notice

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Any changes or modifications not expressly approved by the manufacturer could void the user's authority to operate the equipment.

European EMC Statement

This product may cause interference if used in residential areas. Such use must be avoided unless the user takes special measures to reduce electromagnetic emissions to prevent interference to the reception of radio and television broadcasts.

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AMENDMENT RECORD

Amendment	Date
All parts at Issue 1	November 2016
All parts at Issue 2	May 2017
All parts at Issue 3	July 2018
All parts at Issue 4	February 2019

PART 1 : HEALTH AND SAFETY

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WARNINGS, CAUTIONS AND NOTES

The warnings, cautions and notes used throughout this manual are highlighted by the use of international hazard symbols. The following definitions for all three of these notices are described below in the format they are presented in this product manual.

WARNING: **A warning is used to alert the reader to possible hazards which may cause loss of life, physical injury or ill health.**

CAUTION: *A caution is used to alert the reader to possible hazards which may cause damage to equipment or the environment.*

Note: *A note is used to alert the reader to important information.*

Symbols

The following symbols are used in this product manual to highlight specific warnings and cautions used in the procedure below the symbols.



Eye protection must be worn.



Protective clothing must be worn. Use adequate protective gloves. Consult the relevant Safety Data Sheet (SDS).



Disconnect power before carrying out maintenance or repair.



Connect an earth terminal from the product to an appropriate ground source.



Only trained personnel should carry out this procedure.



Warning or Caution, read and comply with the text underneath this symbol to avoid physical injury or damage to equipment.



Risk of fire by igniting flammable material.



Risk of coming into contact with electricity.



Beware of Electrostatic Discharge (ESD). Electrostatic precautions must be used:

- Switch off machine first.
- Wear a wristband connected to the ESD connector provided.
- Avoid wearing clothing which can build up electrostatic voltages.
- Use ESD protective bags to transport PCBs.
- Only place PCBs on a mat made from a material which will dissipate electrostatic voltages and which is connected to ground.

INKS AND FLUIDS INFORMATION

Domino supplies Safety Data Sheets (SDS) giving specific safety information for each of its ink, make-up and wash fluids. Safety Data Sheets can be found on the MSDS section of the Domino website. There are also warnings on each container. The following are general basic requirements:

**WARNING:**

Do not install or operate the printer in closed or sealed rooms of cabinets. The printer must be installed and operated in a well ventilated environment. Insufficient ventilation may result in an accumulation of vapour which can present a risk of explosion.

Basic Requirements

When used correctly, printing inks do not cause problems. However, everybody using them should be familiar with the appropriate safety standards and be aware of the precautions that should be taken. The following are basic requirements:

- Proper standards of industrial practice relating to cleanliness and tidiness must be maintained.
- Inks and their containers must be stored and handled with care.
- Do not smoke or allow naked flames (or other sources of ignition) in the vicinity of any inks or solvents as this is highly dangerous.
- All who come into contact with inks must be properly instructed in their use.

Directions for safe working practices vary according to the environment. The following are broad principles so that necessary precautions may be taken:

- Contact with the mouth must be avoided. Therefore eating, drinking or smoking, or any personal habits or actions which may transfer ink to the mouth, must be avoided.
- Contact with the eyes must be avoided. Suitable eye protection must always be worn whenever there is any risk of splashing or misting. If ink does get into the eyes, first aid treatment is to flood the affected eye for 15 minutes with saline solution, (or clean water if saline solution is not available), taking care not to allow the water to run into an unaffected eye. Medical aid must be obtained immediately.
- Most inks contain solvents which may injure the skin. Good working practice must always be employed and risk assessments carried out. Safety Data Sheets are available that give advice on personal protective equipment. Most gloves only offer limited and short term exposure protection and must be changed after any splashing and on a frequent basis.
- Many inks contain materials which vaporise easily and can be inhaled. Good ventilation is necessary.
- Any used cleaning materials, e.g. rags, paper wipes, are a potential fire hazard. They must be collected for safe disposal after use. After exposure to ink, all possible traces must be washed off as soon as possible at the nearest washing facility.
- It is possible to control the printer remotely. If operated in this manner it is vital that the remote UI is disconnected before carrying out any cleaning or maintenance on the printer.

Storage

Printing inks must be stored in well ventilated buildings, in areas set aside for the purpose and chosen for safety in case of fire. All fluids must be stored in accordance with local regulations.

Fire Risk

Observe all warnings given on the machine and the following safety instructions.

For an electrical fire, do not use water.

If there has been an accumulation of dried ink, do not use metal scrapers to remove it, as they can produce sparks.

Fire risk is a most important consideration where printing inks are stored and used. The degree of fire hazard will vary considerably from one type of ink or wash to another.

Water-based inks will not burn, although inks based on water-alcohol mixtures may burn if there is sufficient alcohol present.

Prolonged exposure of water-based systems to high temperatures may evaporate the water to give a flammable residue.

Solvent-based inks offer a greater degree of hazard depending on the particular solvent or solvent combination. When there is a particular hazard the appropriate information is given on the SDS.

The printers place small electrostatic charges on the ink drops used for printing. In most circumstances, such as when they arrive at the print surface, these electrostatic charges are either conducted away or cannot accumulate. However, during maintenance, print drops may be collected in a container, such as a beaker. It is essential that this container is made of conducting material and is securely connected to ground/earth. The electrostatic charges will then be safely conducted to ground/earth.

If there is a fire, there is a likelihood that dangerous fumes will arise from printing inks. For this reason ink must be stored where it can be reached quickly by the fire fighting service, and where it will not spread beyond the store.

Spillages and Disposal



WARNING:

Do not allow the ink to dry or allow any build-up of dried ink spills. Some dried inks are highly flammable. Clean up all ink spillages immediately.

Spillages must be cleaned up as soon as possible with the appropriate solvent materials and with regard to the safety of personnel. Care must be taken to prevent spillages or residue from cleaning up entering drains or sewage systems.

Inks and associated fluids are materials which conduct electricity. Therefore, power to the printer must be switched off while spillages inside the printer cabinet are being cleaned up.

Printing inks and associated fluids must not be treated as ordinary waste. They must be disposed of using approved methods according to local regulations.

ELECTRO-STATIC DISCHARGE

To avoid static discharge from production line equipment, the equipment and everything on it must be grounded correctly. While it is not Domino's responsibility to maintain the customer's equipment, the customer must be aware of the importance of grounding. At the time of installation of the printer the Domino (or Domino distributor) technician should ensure that the line the printer is installed on is appropriately grounded. This means that all metallic items within 12 inches (300mm) of the print head location must be securely grounded. This should be checked with respect to the print head chassis ground using an Ohm Meter, DVM or other suitable device. It is essential that grounding is checked periodically and that if the printer is relocated to a different production line, then the new production line is properly grounded. Additional static dissipating devices, such as static brushes on the line, etc. are recommended.

OPERATION

For safe operation of the printer, follow all instructions given in this product manual.

**WARNINGS:**

- (1) **Pressurised ink system. Risk of ink spray. Read the procedure on [page 5-10](#) before removing the ITM.**
- (2) **Do not access the electronics enclosure when the printer is powered on. When powered on, the electronics enclosure contains live electrical components. Physical contact with live electrical components may result in an electric shock.**

PART 2 : SYSTEM DESCRIPTION

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SYSTEM DESCRIPTION

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GENERAL

This document contains operating instructions for the Domino Ax-Series continuous ink jet printer range.

Only engineers trained and certified by Domino should carry out repairs. Genuine Domino parts should always be used to ensure quality and performance.

Description

The Domino Ax-Series is a range of industrial ink jet printers, intended for printing fixed and variable data on products on manufacturing production lines.

Each printer uses a continuous ink jet of either 1 or 2 jets (depending on configuration) for printing on a user supplied substrate.

Each printer consists of:

- A print head and flexible conduit connecting it to a cabinet.
- A cabinet containing the ink supply and electronic control systems.
- A user interface for operator interaction.

Ax150i Printer



Ax150i Cabinet

The Ax150i is the smallest of the Ax-Series range. It features an IP55 cabinet with integrated TouchScreen, Domino Design ink system, i-Techx advanced electronics system and user interface and an i-Pulse print head.

To maintain optimum performance the Ax150i only requires the colour coded consumables to be replenished. Consumables such as the air and make-up filter, ink and make-up cartridges and ITM are colour coded yellow for easy identification and user convenience.

The i-Techx system monitors ink and make-up consumption to calculate when consumables are due for replacement. This is then indicated on both the TouchScreen user interface and on LEDs on the front of the cabinet.

Note: ITM replacement intervals will vary according to use and ink type. Contact Domino for details.



WARNING:

Pressurised ink system. Risk of ink spray. Read the procedure on [page 5-10](#) before removing the ITM.

Ax350i Printer



Ax350i Printer Cabinet with Remote TouchScreen 7"

The Ax350i is a highly configurable printer featuring a stackable IP55 cabinet design with an IP66 electronics enclosure, a range of TouchScreen options, Domino Design ink system, i-Techx advanced electronics and user interface and an i-Pulse or i-Pulse RS print head.

The Ax350i can be controlled using a Remote TouchScreen 7" or a Remote TouchScreen 10". The TouchScreen can be mounted on top of the stainless steel cabinet. The TouchScreen can also be mounted using a VESA 75 standard mounting, anywhere within range of the connecting cable.

Additional printer cabinets may be stacked and controlled via one TouchScreen.

The Ax350i printer can use a wide variety of inks. This gives the Ax350i flexibility to print on a wide range of print surfaces. Where appropriate, a heavy duty ink system is fitted which is constructed of materials resilient to inks with corrosive properties, this enables reliable operation and minimal down time.

To maintain optimum performance the Ax350i only requires the colour coded consumables to be replenished. Consumables such as the air and make-up filter, ink and make-up cartridges and ITM are colour coded yellow for easy identification and user convenience.

The i-Techx system monitors ink and make-up consumption to calculate when consumables are due for replacement. This is then indicated on both the TouchScreen user interface and on the cabinet status lights.

Note: ITM replacement frequency will vary according to the type specified. Contact Domino for details.



WARNING:

Pressurised ink system. Risk of ink spray. Read the procedure on [page 5-10](#) before removing the ITM.

Ax550i Printer



Ax550i Printer Cabinet with Remote TouchScreen 10"

The Ax550i is visually similar to the Ax350i. However, the cabinet of the Ax550i is constructed from a higher grade of stainless steel (316 Grade) for use in more corrosive operating environments.

Printer Configuration Labels

External Printer Configuration Label

The label illustrated below is an example of the configuration label which is located on the rear of all Ax-Series printers.



Printer Configuration Label

The external configuration label contains the following information:

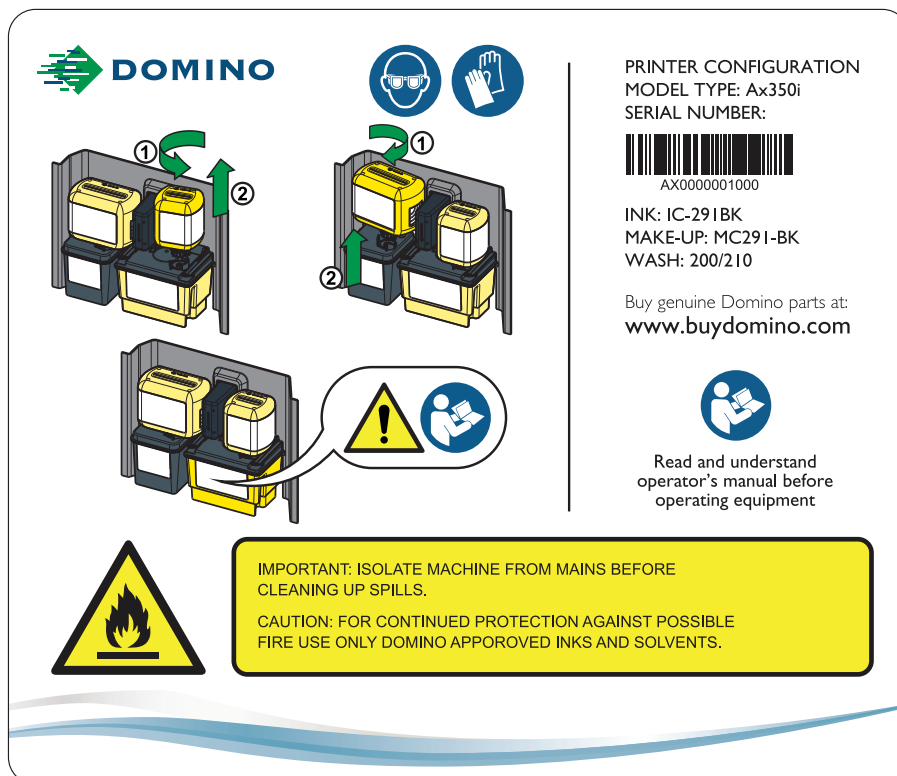
- The product model name
- The product serial number
- The name, address and contact details of the manufacturer
- IP rating of the product
- IP rating of the electronics compartment
- Power supply requirements
- FCC notice
- Local conformance information
- Warning information.

Internal Printer Configuration Label



- WARNINGS:**
- (1) Protective equipment such as gloves and glasses must be worn when opening the printer's ink compartment to access the internal configuration label. The ink compartment contains printer ink and make-up. Physical contact with printer ink or make-up can cause skin or eye damage.
 - (2) Pressurised ink system. Risk of ink spray. Read the procedure on [page 5-10](#) before removing the ITM.

The label illustrated below is an example of the internal configuration label which is located inside the front door of all Ax-Series printers.



Internal Printer Configuration Label

The internal configuration label contains the following information:

- The product model name
- The product serial number
- The ink type which the printer is configured to use
- The make-up type which the printer is configured to use
- The wash type which should be used when cleaning the print head
- The address of the BuyDomino website
- Ink and make-up cartridge removal instructions
- Warning information.

PRINTER SPECIFICATION

Ax150i

Cabinet

Standard Finish:	GRP front panel, 304 grade brushed stainless steel cabinet.
IP Rating:	Designed to IP55
Dimensions:	Width: 400mm (15.75") Depth: 394mm (15.50") Height: 373mm (14.70")
Weight (dry):	16kg (35.3lb)
Control Panel:	7 inch TouchScreen and polyester membrane touch buttons.

Standard External Connections

Product Detect / Shaft Encoder Connector:	8-Way Socket x2
Power connector:	3-Way Plug, cable supplied

Optional External Connections

Status Pack:	Beacon Connector: 5-Way Socket Alarm Connector: 7-Way Socket, 1A, 30V maximum
RS232 Pack:	Serial (RS232) Connector: 8-Way Plug
GPIO Pack:	Basic GPIO Port: 14-Way Socket
Comms Pack:	Ethernet Port USB Port: Type A
Safeguard:	DCI Port: 7-Way Socket

Ink System

ITM Type 2 Capacity:	1222ml maximum
ITM Type 3 (for heavy duty ink system) Capacity:	1118ml maximum
Make-up Module Capacity:	888ml maximum
Ink Cartridge Capacity:	825ml
Ink Cartridge (for heavy duty ink system) Capacity:	555ml
Make-up Cartridge Capacity:	1200ml
Ink Viscosity Control:	Automatic Viscometer
Ink Bleed Control:	Automatic Start-Up/Shut-Down

Environment

Temperature Range (working):	+5° to +45°C (42°F to 112°F)
Temperature Range (Storage):	-20° to +60°C (-4°F to +140°F) (machine dry - storage, wet dependent upon fluids)
Humidity:	0 - 90% RH (non-condensing)
Electrical Supply:	100-240V 50-60Hz (nominal), 4A, single phase Auto ranging
Power consumption	40 Watt (idle) 80 Watt (printing)
Acoustic Noise Level:	Not more than 70 dBA

i-Pulse Print Head

Standard Finish:	Chassis: Stainless steel Wire box: Acetal Holster: Polymer (PTFE) coated aluminium
Dimensions:	Width: 50.3mm (1.98") Depth: 50.3mm (1.98") Height including conduit retaining nut: 243mm (9.57") Operating height with conduit at 90 degrees: 320mm (12.6") Weight including 3m conduit: 1.8kg (4lbs)
Conduit length:	3m (10ft)
Conduit minimum inside bend radius:	65mm (2.55")
Nozzle Size	60 micron
Recommended distance from print surface:	8mm (0.31")
Working height relative to cabinet base:	±1m

Ax350i/Ax550i**Cabinet**

Standard Finish:	Ax350i: 304 grade stainless steel Ax550i: 316 grade stainless steel
IP Rating:	IP55 (electronics enclosure to IP66)
Dimensions:	Width: 430mm (16.9") Depth: 381mm (15.0") Height: 411mm (16.2") Height (with TouchScreen): 643mm (25.3")
Cabinet Controls:	Buttons for Standby, Start/Stop, Single print and Alerts
Weight (dry):	22kg (48.5lb)

Standard External Connection

Interface communication:	1 x non standard DVI connection providing 24V DC, 500 mA
Product Detect / Shaft Encoder Connector:	8-Way Socket x 2
Power connector:	3-Way Plug, cable supplied

Optional External Connections

Extended Comms Pack:	Powered HMI Port: 7-Way Socket USB Port: Type A
Extended IO Pack:	User Port: 25-Way D-Sub Socket User Port: 37-Way D-Sub Socket
Status Pack:	Beacon Connector: 5-Way Socket Alarm Connector: 7-Way Socket, 1A, 30V maximum
RS232 Pack:	Serial (RS232) Connector: 8-Way Plug
GPIO Pack:	Basic GPIO Port: 14-Way Socket
Comms Pack:	Ethernet Port USB Port: Type A
Safeguard:	DCI Port: 7-Way Socket

Ink System

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Humidity:	0 - 90% RH (non-condensing)
Electrical Supply:	100-240V 50-60Hz (nominal), 4A, single phase Auto ranging
Power consumption	40 Watt (idle) 80 Watt (printing)
Acoustic Noise Level:	Not more than 70 dBA

i-Pulse Print Head

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Dimensions:	Width: 50.3mm (1.98") Depth: 50.3mm (1.98") Height including conduit retaining nut: 243mm (9.57") Operating height with conduit at 90 degrees: 320mm (12.6") Weight including 3m conduit: 1.8kg (4lbs)
Conduit minimum inside bend radius:	65mm (2.55")
Conduit Length and Nozzle Size:	3m (10ft) conduit with 40, 50, 60 or 75 micron nozzle 4m (13ft) conduit with 40 or 50 micron nozzle 6m (20ft) conduit with 60 or 75 micron nozzle
Recommended distance from print surface:	75 micron nozzle: 14mm (0.55") 60 micron nozzle: 8mm (0.31") 50 micron nozzle: 6mm (0.23") 40 micron nozzle: 4mm (0.15")
Working height relative to cabinet base:	3m conduit: ± 1 m 4m conduit: ± 1 m 6m conduit: ± 3 m

i-Pulse Duo Print Head

Standard Finish:	Chassis: Stainless Steel Wire box: Moulded nylon Holster: Plated aluminium
Dimensions:	Width: 57mm (2.24") Depth (Max): 53mm (2.01") Height: 255mm (10") Weight including 3m Conduit: 2.8kg (6.17lbs)
Conduit Length:	3m (10ft) or 6m (20ft)
Conduit minimum inside bend radius:	65mm (2.55")
Nozzle size:	60 or 75 micron
Recommended distance from print surface:	75 micron nozzle: 14mm (0.55") 60 micron nozzle: 8mm (0.31")
Working height relative to cabinet base:	3m conduit: ± 1 m 6m conduit: ± 3 m

i-Pulse RS Print Head

Standard Finish:	Stainless Steel
Dimensions:	Width: 40mm (1.57") Depth: 40mm (1.57") Height (60 micron nozzle): 243mm (9.56") Height (75 micron nozzle): 252mm (9.92") Weight including 3m conduit: 1.8kg (4lbs)
Conduit Length:	3m (10ft) or 6m (20ft)
Conduit minimum inside bend radius:	Fixed installation: 70mm (2.75") Continuous flex installation: 150mm (5.90")
Nozzle size:	60 or 75 micron
Nominal / recommended distance from print surface:	75 micron nozzle: 14mm (0.55") 60 micron nozzle: 8mm (0.31")
Working height relative to cabinet base:	3m conduit: ± 1 m 6m conduit: ± 3 m

User Interface

Remote TouchScreen 10" (Ax350i/Ax550i Only)

Standard Finish:	Cast aluminium (rear), nylon plastic (front)
Mounting:	Via angled bracket to cabinet top. VESA 75 standard mounting.
Display:	10.4 inch SVGA full colour touch screen
Dimensions:	Width: 307mm (12.1") Height: 232mm (9.1") Depth: 75mm (3")
Weight:	2.5Kg (5.5lb)
IP Rating:	IP65
Temperature:	5 - 45°C
Humidity:	10 - 90% non-condensing
Power Supply:	24Vdc, 15W, 625mA over non standard DVI
Connectivity:	Non standard DVI cable to rear of printer
Data Transfer:	2 x USB Type A

Remote TouchScreen 7" (Ax350i/Ax550i Only)

Standard Finish:	Nylon plastic
Mounting:	Via angled bracket to cabinet top. VESA 75 standard mounting.
Display:	7 inch WVGA full colour touch screen
Dimensions:	Width: 245mm (9.6") Height: 160mm (6.3") Depth: 50mm (2")
Weight:	0.75Kg (1.65lbs)
IP Rating:	IP55
Temperature:	5 - 45°C
Humidity:	10 - 90% non-condensing
Power Supply:	24Vdc, 12W, 500mA over non standard DVI
Connectivity:	Non standard DVI cable to rear of printer
Data Transfer:	2 x USB Type A

Printer Control

Cabinet Buttons

There are four hardware buttons on the printer cabinet as illustrated below. It is necessary to press the buttons for at least 2-3 seconds as a precaution against accidental use. The cabinet buttons also carry an indicator light.

All other functions are available using the QuickStep interface on the TouchScreen.

Cabinet Button	Explanation
Standby Button 	The standby button is pressed to start up the printer and enter the Idle state. In the Idle state the printer's ink jet is off and the printer will not print. However, the user interface can still be used. The standby button can also be pressed for two to three seconds to shut down the printer. See "START-UP" on page 4-19.
Start/Stop Button 	The start/stop button is pressed to start up the printer and enter the Ready state. In the Ready state the ink jet is running, and the printer is ready to print. If a label is on-line, it will print on receiving a product detect signal. If the button is pressed when in a Ready state, the printer will enter the Idle state. In the Idle state the printer's ink jet is off and the printer will not print. However, the user interface can still be used. See "START-UP" on page 4-19.
Single Print Button 	The single print button is pressed to produce a single test print of the current label. Typically used to identify what label is being printed when a UI is not present.
Alert Button 	The alert button is pressed to display the current list of alerts on the TouchScreen. The alert button also carries two status lights described on page 2-18.

Cabinet Status Lights

There are a number of status lights on the printer cabinet as illustrated below.

Cabinet Status Light	Explanation
Printer Off, Power Connected 	The red light on the standby button will illuminate to indicate that the printer is connected to a power supply, but the printer is not turned on.
Start Up 	The red light on the standby button will illuminate and the green light on the standby button will flash to indicate that the printer is starting up.
Printer On / Shut Down 	The green light on the standby button will illuminate to indicate that the printer is on. The light will flash on and off to indicate that the printer is shutting down.
Red Alert 	The red alert light will illuminate to indicate a red alert. A red alert occurs when the printer has a fault which causes printing to stop. If a TouchScreen is connected to the printer, a description of the alert will be displayed on the TouchScreen's status tab.
Amber Alert 	The amber alert light will illuminate to indicate an amber alert. An amber alert occurs when the printer has a fault which requires attention. If a TouchScreen is connected to the printer, a description of the alert will be displayed on the TouchScreen's status tab.
Make-up Level Alert 	If the make-up level is ok, the light will be off. The make-up light will illuminate to indicate when the make-up level is low, and the cartridge requires replacement.

SYSTEM DESCRIPTION

Cabinet Status Light	Explanation
Ink Level Alert 	If the ink level is ok, the light will be off. The ink light will illuminate to indicate when the ink level is low, and the cartridge requires replacement.

QMM (Quality Management Module) Status Lights

The printer contains a QMM (Quality Management Module) to read and write data to RFID (Radio Frequency IDentification) tags on the make-up cartridge, ink cartridge and ITM. This enables the printer to alert the user when the ITM, ink or make-up cartridge need to be replaced to maintain reliable and efficient printer operation. The printer will also display an alert if the wrong ink or make-up or ITM type is inserted.

In addition, there are 3 status lights on the QMM inside the printer cabinet. The colour of each light indicates whether the RFID tags on the consumables have been correctly read and identified.

The QMM status lights are shown below.



Quality Management Module Status Lights

QMM Status Light Colour	Explanation
Red	The RFID tag has been read and a fault has been detected. An alert will be displayed on the QuickStep status tab to explain the fault.
Amber	The RFID tag has not been read, either because an RFID tag is not present, or the RFID tag is faulty. An alert will also be displayed on the QuickStep status tab to explain the fault.
Flashing Amber	The QMM is in the process of reading the RFID tag.
Green	The RFID tag has been correctly read and identified.

QuickStep

The QuickStep interface as displayed on the TouchScreen is shown below:



QuickStep Interface

Domino has developed QuickStep to be easy and intuitive to use. Training times, set up times and coding errors are all reduced with QuickStep.

The operation of the printer using QuickStep is described in [Part 4 : OPERATION](#).

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PART 3 : INSTALLATION

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INTRODUCTION



- WARNINGS:**
- (1) **Risk of injury. There are high voltage components inside the printer's electronics compartment. The electronics compartment must not be opened by unskilled personnel.**
 - (2) **The printer is Class 1 equipment and requires a protective earth.**
 - (3) **For the safety of personnel, all items in the printer requiring connection to earth/ground have cables for this purpose. Any earth/ground cables disconnected at any time must be refitted before the printer is returned to operation.**
 - (4) **The printer requires an H05 RR-F approved mains lead.**

Installation of the printer consists of:

- Checking that all items have been delivered.
- Installing the printer and its associated equipment.
- Preparing the printer for use.

There is a wide range of possible work areas and it is not possible to give specific instructions for each area. The information provided in this document consists of the basic requirements and general information. Ensure that the basic requirements can be met before installing the printer.

PRINTER INSTALLATION

The following is general information for installing the printer in a wide range of working areas.

Cabinet Positioning



WARNING:

Care must be taken when moving the printer to avoid injury. Only personnel who have completed manual handling training and have the appropriate equipment should carry out this task.

Note: For printer dry weight, see the specification on [page 2-9](#).

This equipment is not suitable for use in locations where children are likely to be present.

The cabinet must be positioned in a location with enough clearance to open the doors and give access to the cabinet's front and rear. The cabinet must also be positioned on a level surface and electrically isolated from other equipment, except for a normal data interface.

The printer must be positioned in an area where the temperatures will remain within +5°C and +45°C (42°F to 112°F) and the relative humidity must remain within 10% to 90% (non-condensing). The printer draws in and expels cooling air through vents underneath of the cabinet. The air vents must not be obstructed.

To ensure stability and minimise vibration, mount the printer on a Domino Printer Stand with the Domino Stacking Kit. If a Domino Printer Stand is not used, then the printer base must be secured to the mounting surface using M6 mounting bolts. Use spacers to ensure the mounting bushes are flush with the mounting surface.

Printer cabinets can be stacked, see [page 3-6](#) for more details.

Conduit and Print Head Positioning

Keep the conduit and print head away from power supply cables and other wiring capable of producing electrical noise. Use the print head mounting bracket supplied with the printer to reduce electrical noise. Keep the conduit and print head as free as possible from vibration. Mount the print head perpendicular to the printing surface.

If it is possible, position the print head to the recommended distance away from the print surface as described in the table below.

Nozzle Diameter	i-Pulse/i-Pulse RS	i-Pulse Duo (Standard Labels)	i-Pulse Duo (Stitched Labels)
75 micron	14mm	14mm	20mm
60 micron	8mm	6mm	12mm
50 micron	6mm	N/A	N/A
40 micron	4mm	N/A	N/A

Ventilation



WARNING:

Do not install or operate the printer in closed or sealed rooms or cabinets. Insufficient ventilation may result in an accumulation of vapour which can present a risk of explosion. The printer must be installed and operated in a well ventilated environment.

Solvent based printing inks and their vapours are flammable. As well as ensuring that all equipment is earthed, and that anti-static precautions are followed, an extraction system must be provided to remove fumes and vapours from the production line and anywhere else that the printer may be operated.

The ventilation system must achieve a minimum of 10 air changes per hour (20 air changes per hour recommended).

Electrical Supply



WARNINGS:

- (1) **Electricity. Risk of electric shock.**
Earth the printer cabinet for the safety of personnel working on the printer.
- (2) **Electricity. Risk of fire. Risk of injury from electric shock.**
Tighten both screws on the power supply plug to prevent moisture entering the power supply socket. Use a screwdriver to ensure the screws are correctly tightened.



The printer should be connected using a suitable plug and socket outlet which is accessible and close to the equipment so that power can be quickly disconnected. If a fused power connector is used, it should be fitted with a 5A fuse. If a fused power connector is not used, then the supply circuit should have a circuit breaker or fuse rated at 5A.

The supply must be free from electrical noise. Domino can give advice on suitable devices to ensure trouble-free operation.

Printer Cabinet Stacking Requirements



- WARNINGS:**
- (1) **Topple Hazard. Risk of injury. Do not stack two printer cabinets on the top shelf of a type 1 printer stand. Two printer cabinets can only be stacked on the top shelf of a type 2 printer stand.**
 - (2) **Topple Hazard. Risk of injury. Do not stack two printer cabinets on the top shelf of the printer stand if the bottom shelf is empty. Mount one printer on the top shelf and one printer on the bottom shelf.**
 - (3) **Topple Hazard. Risk of injury. Use a stacking kit to secure the top two printer cabinets if two printer cabinets are stacked on the top shelf of a type 2 printer stand (with a third printer or a cupboard assembly on the bottom shelf).**

Printer cabinet stacking configurations are dependent on the type of printer stand and if a stacking kit is used or not.

The possible printer stacking configurations are listed in the table below:

Table 1: Printer Cabinet Stacking Configurations

Stacking Configuration	Type 1 Printer Stand	Type 2 Printer Stand	Stacking Kit Required?
1 x Ax150i 1 x Cupboard Assembly	✓	✓	✗
2 x Ax150i	✓	✓	✗
2 x Ax350i/Ax550i	✓	✓	✗
2 x Ax350i/Ax550i 1 x Cupboard Assembly	✗	✓	✓
3 x Ax350i/Ax550i	✗	✓	✓

Printer Stand Types



Type 1 Printer Stand



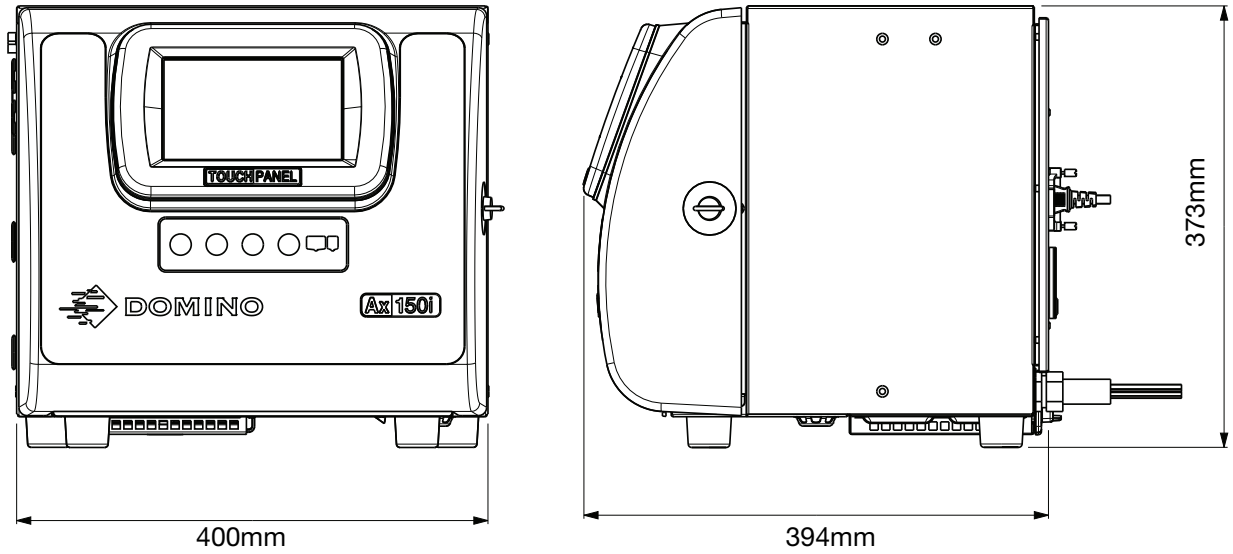
Type 2 Printer Stand



Stacking Kit

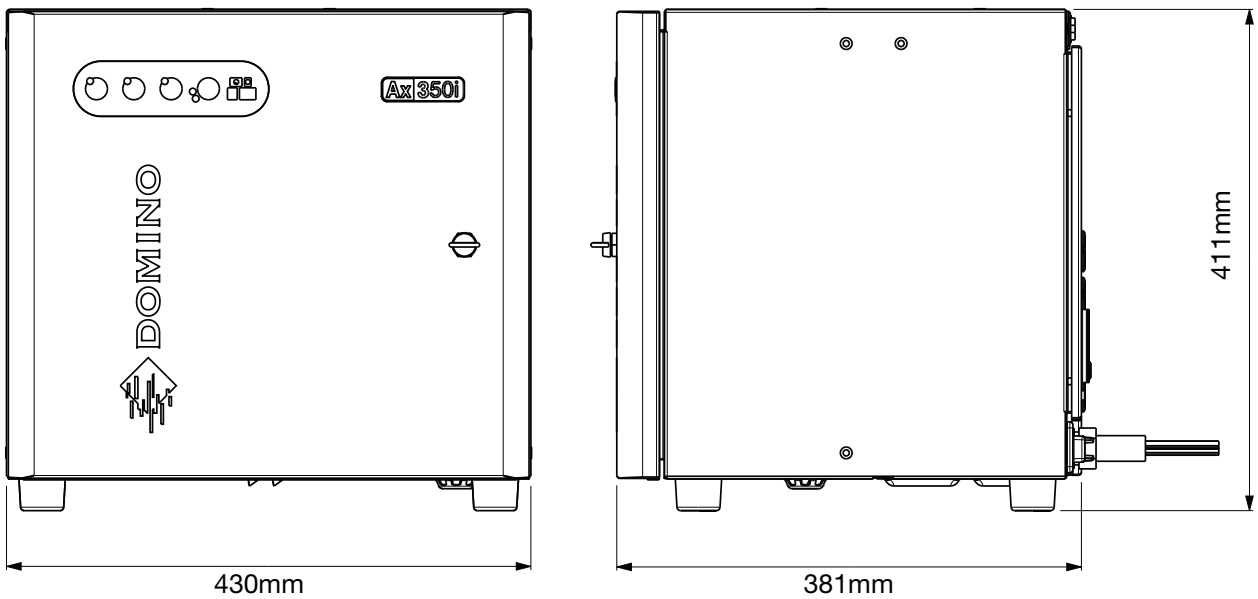
Installation Dimensions

Ax150i



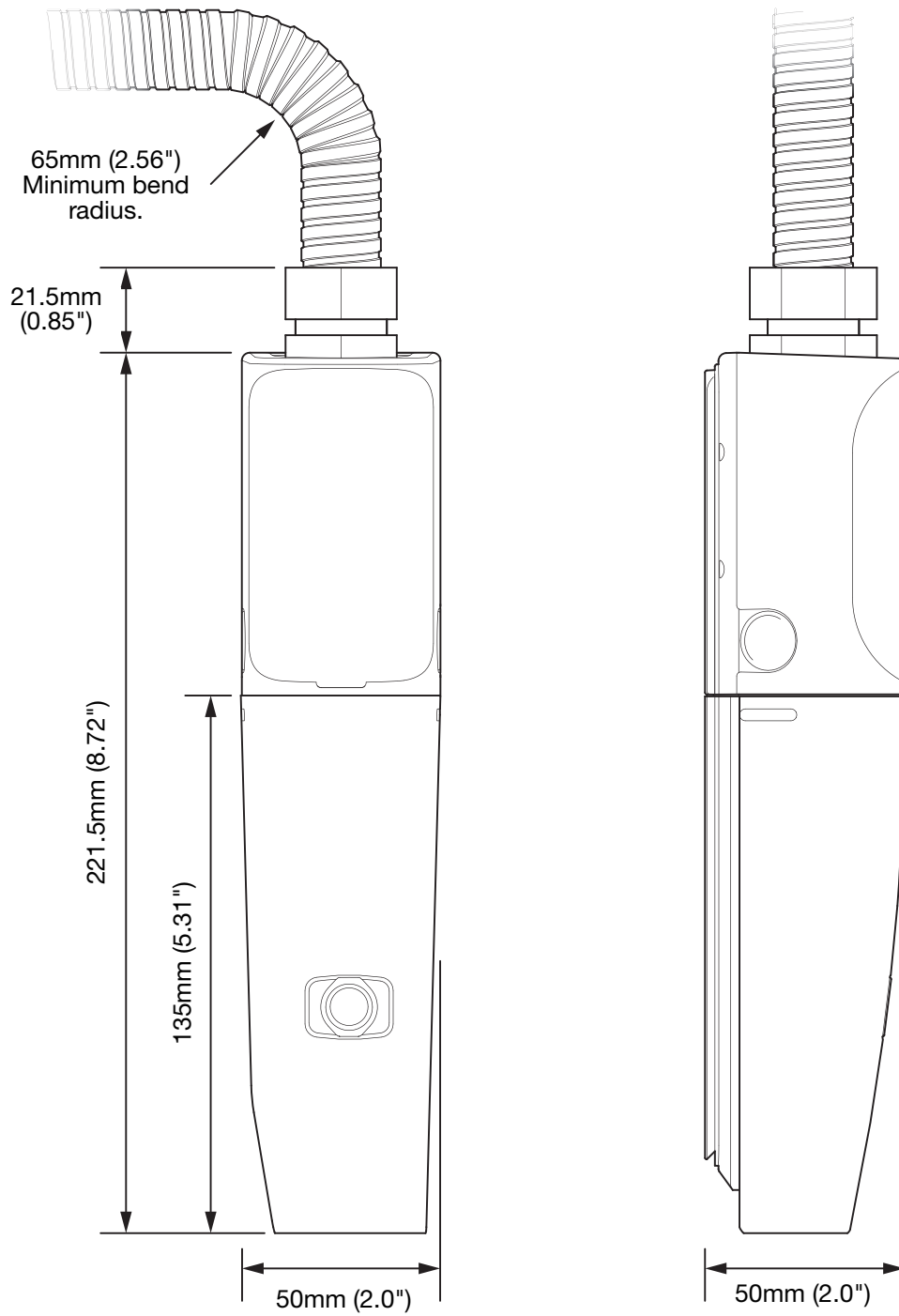
Ax150i Cabinet

Ax350i/Ax550i



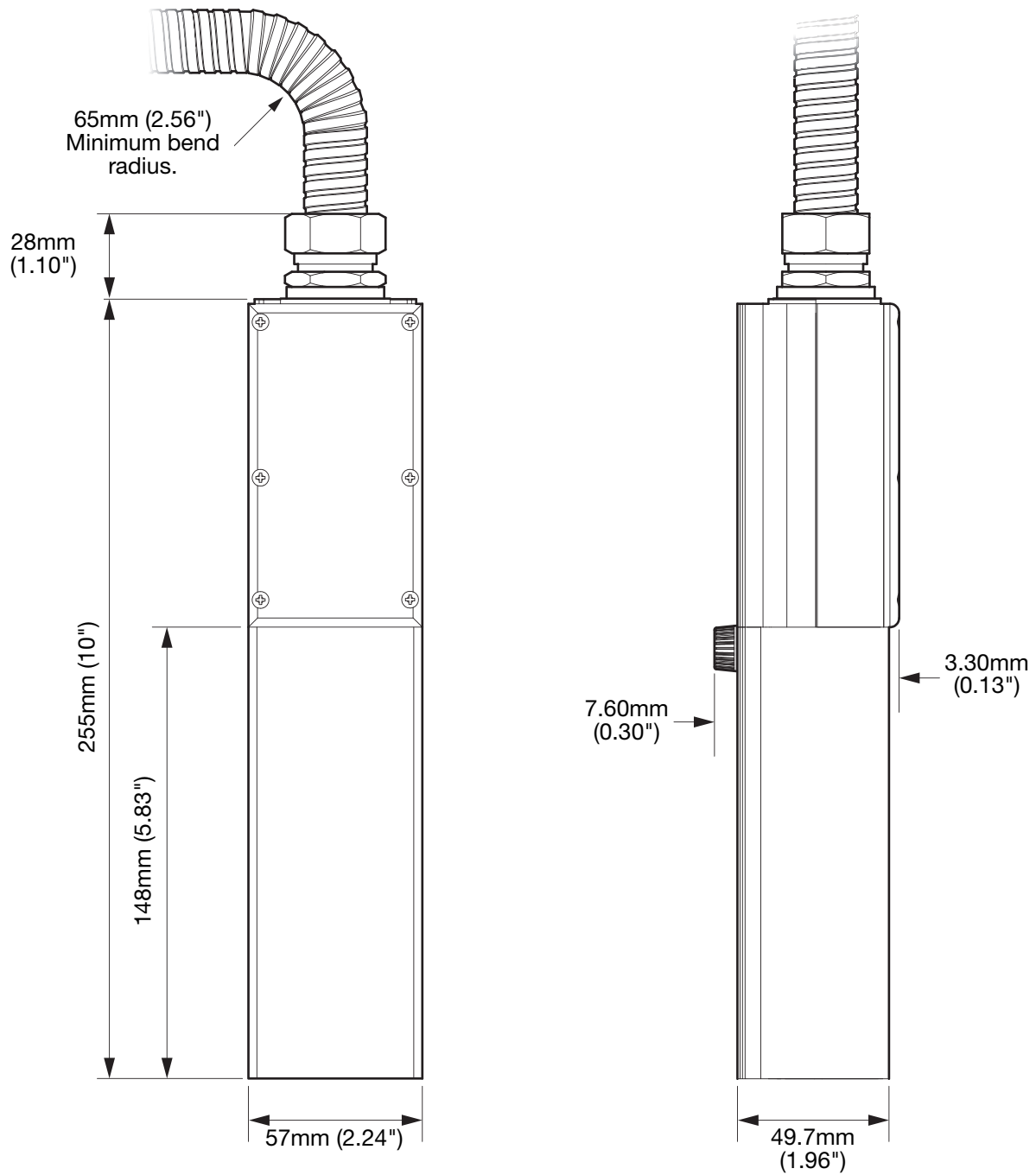
Ax350i/Ax550i Cabinet

i-Pulse Print Head



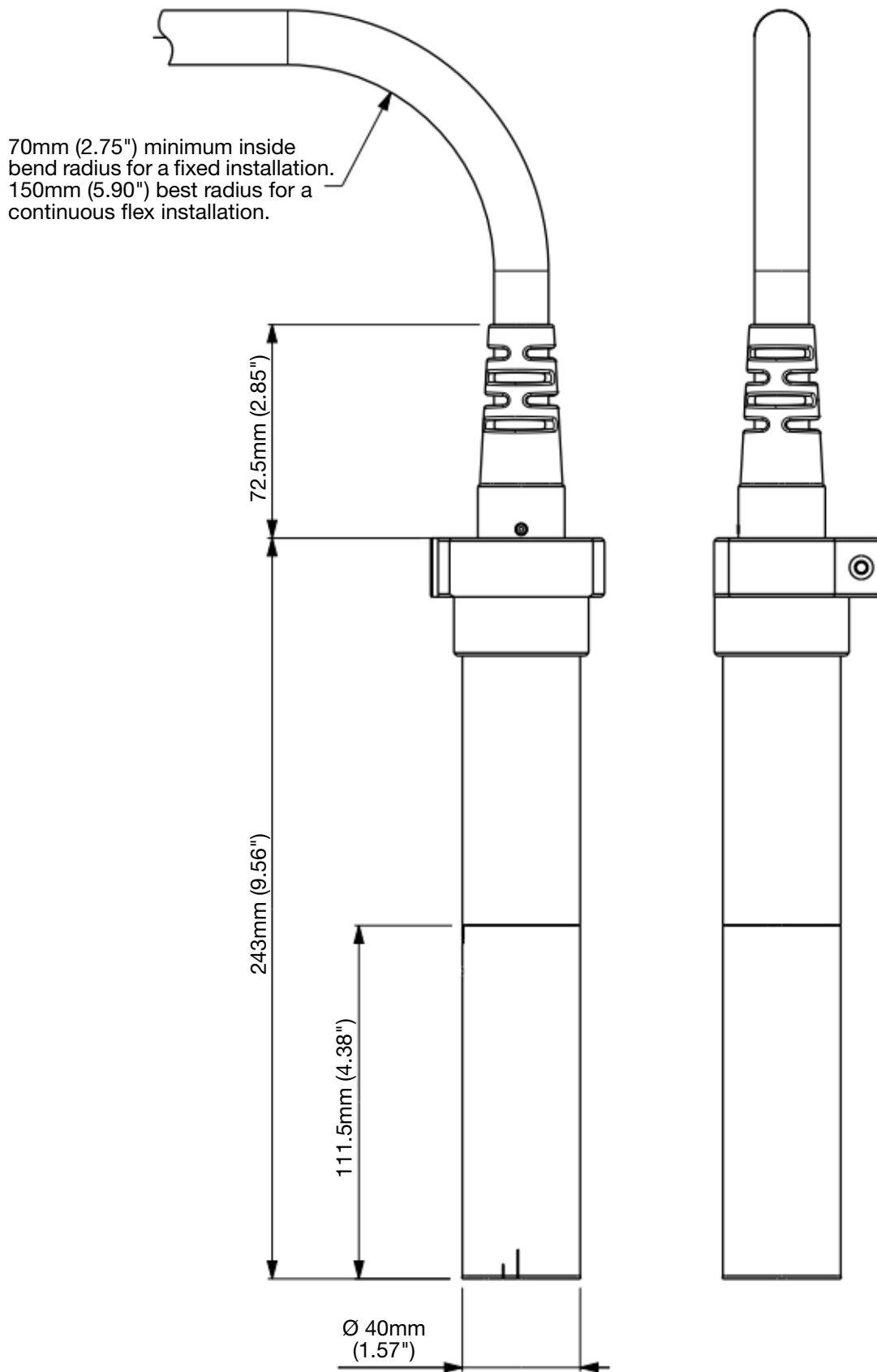
i-Pulse Print Head Dimensions

i-Pulse Duo Print Head



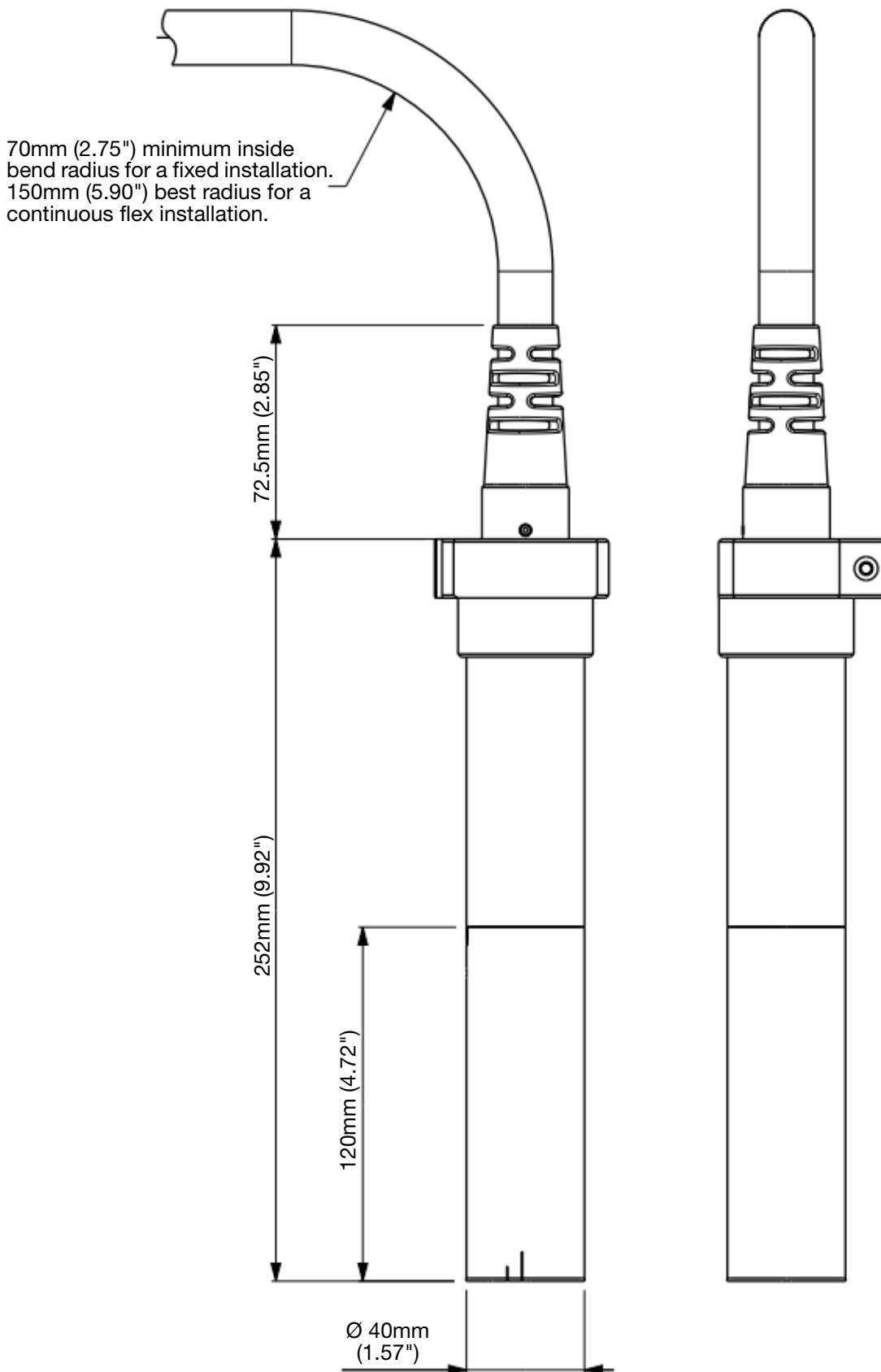
i-Pulse Duo Print Head Dimensions

i-Pulse RS Print Head with 60μ Nozzle



i-Pulse RS Print Head Dimensions

i-Pulse RS Print Head with 75 μ Nozzle



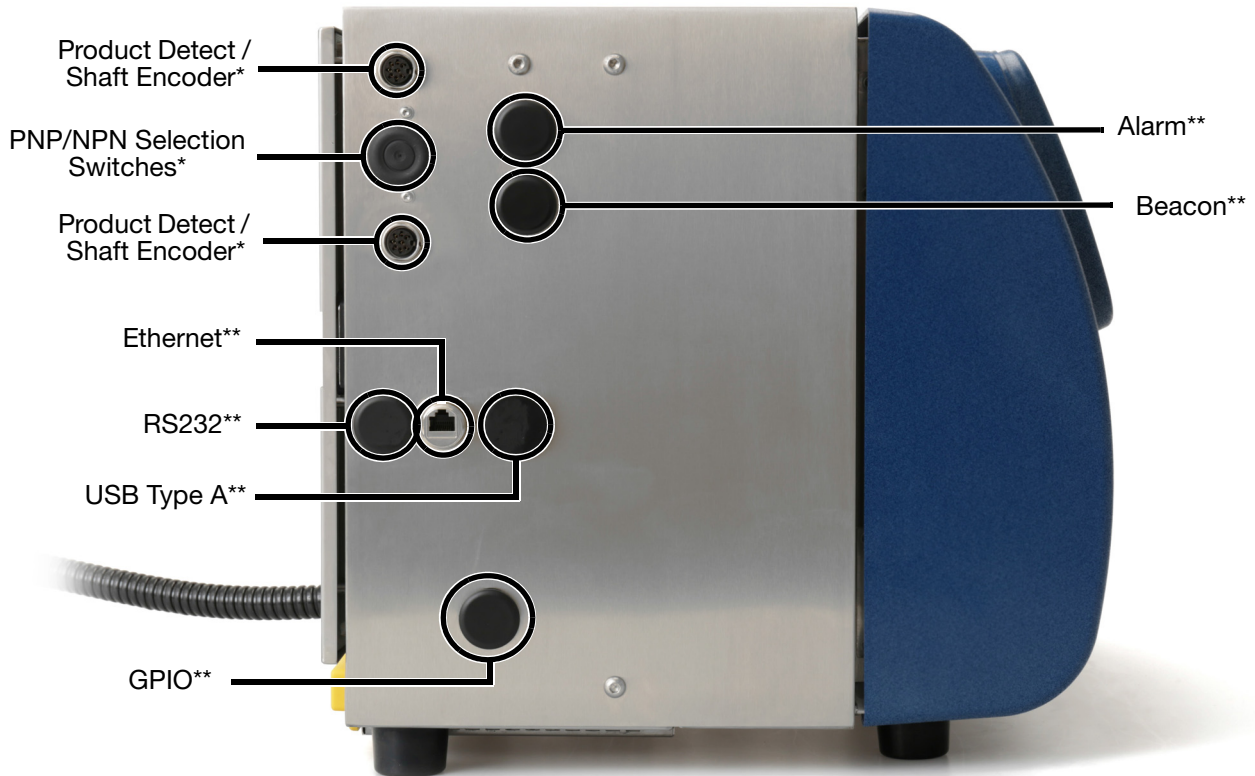
i-Pulse RS Print Head Dimensions

EXTERNAL CONNECTIONS

Ax150i

* = Standard Fit.

** = Option.



Ax150i External Connections Left View

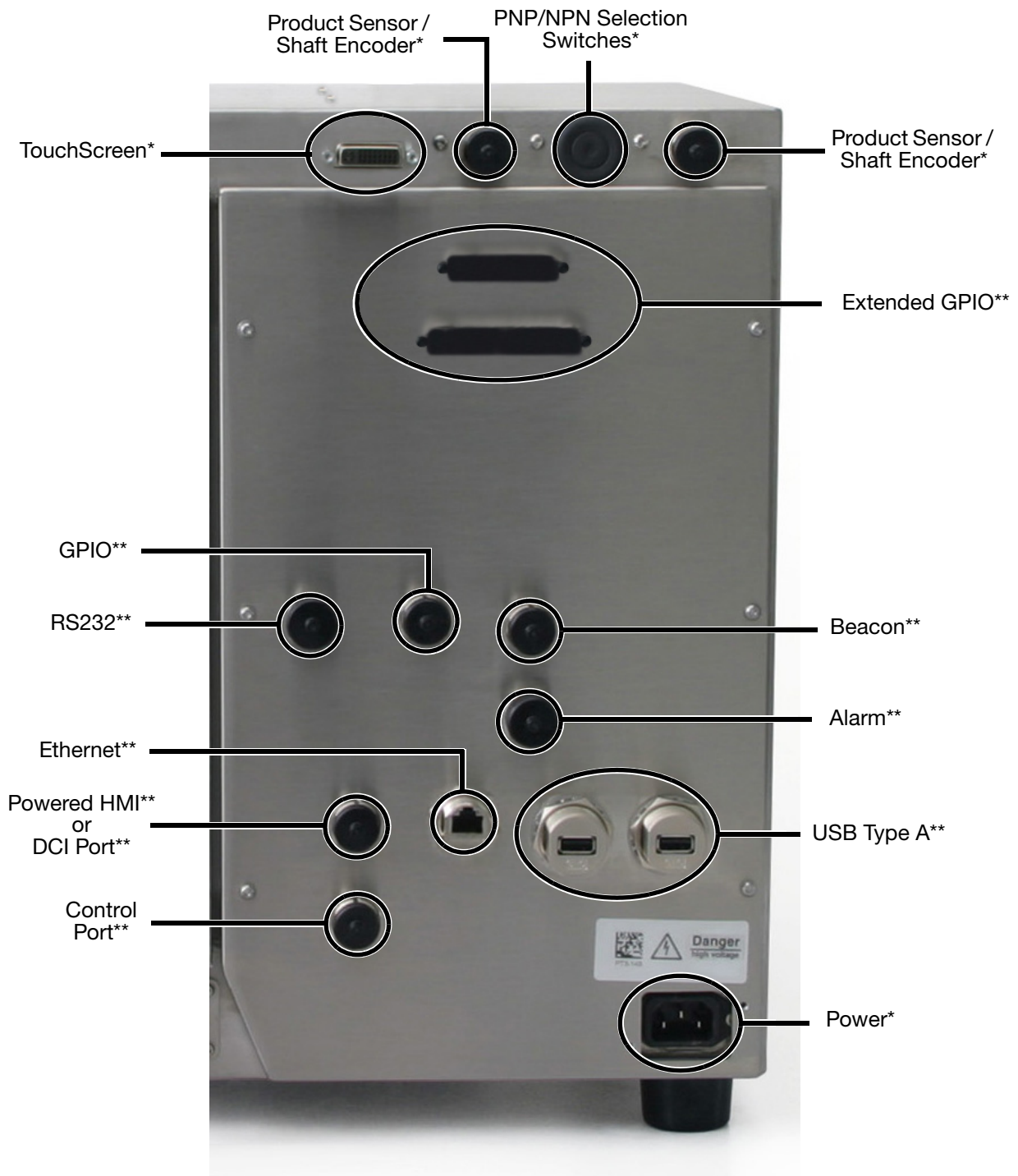


Ax150i External Connections Rear View

Ax350i/Ax550i

* = Standard Fit.

** = Pack Option.



Ax350i/Ax550i Connections

Product Detect / Shaft Encoder Socket Description



CAUTION:

Risk of equipment damage. The product detect sensor or shaft encoder can be damaged if the pins on the product detect sensor or shaft encoder plug make contact with the wrong pins on the PD/SE socket. Shut down the printer and disconnect the power lead from the rear of the printer before connecting a product detect sensor or shaft encoder to the PD/SE socket.

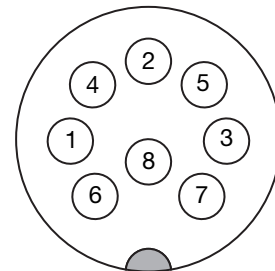
As standard, Ax-Series printers are equipped with two product detect and shaft encoder sockets. The socket locations are illustrated on [page 3-13](#) for the Ax150i and [page 3-14](#) for the Ax350i/Ax550i.

The printer provides a 24V DC supply for use by a product sensor and selectable 24V NPN (Default) or PNP input for signals from a shaft encoder. NPN or PNP can be selected by changing the micro-switches on the PCB as described on [page 3-16](#).

The diagram and table below illustrate the pin out for the product detect / shaft encoder sockets.

Note: *If the sockets are not in use, ensure the sealing caps are fitted to maintain the printer cabinet's IP rating.*

Pin	Description
1	24V Power Supply
2	Ground
3	Product Detect (+)
4	Product Detect (-)
5	Encoder Channel A (+)
6	Encoder Channel A (-)
7	Encoder Channel B (+)
8	Encoder Channel B (-)
Shell	EMC_GND via panel mounting



8-Way PD / SE Socket
(external cabinet view)

Product Detect / Shaft Encoder PNP and NPN Selection

The pictures below illustrate the PNP/NPN selection switches on the Product Detect / Shaft Encoder PCB. The configuration of these switches will set the product sensor and shaft encoder inputs to either PNP or NPN. There are LEDs on the PCB to indicate that the signal is correct.

The switches are accessed by removing the rubber grommet shown on [page 3-13](#) for the Ax150i and [page 3-13](#) for the Ax350i/Ax550i.

Note: Refit the rubber grommet to maintain the printer cabinet's IP rating.

When a single channel (non-quadrature) shaft encoder is used, only the channel SE_A can be utilised.

When a quadrature shaft encoder is used, both the Channels SE_A and SE_B can be utilised.

For information on the settings:

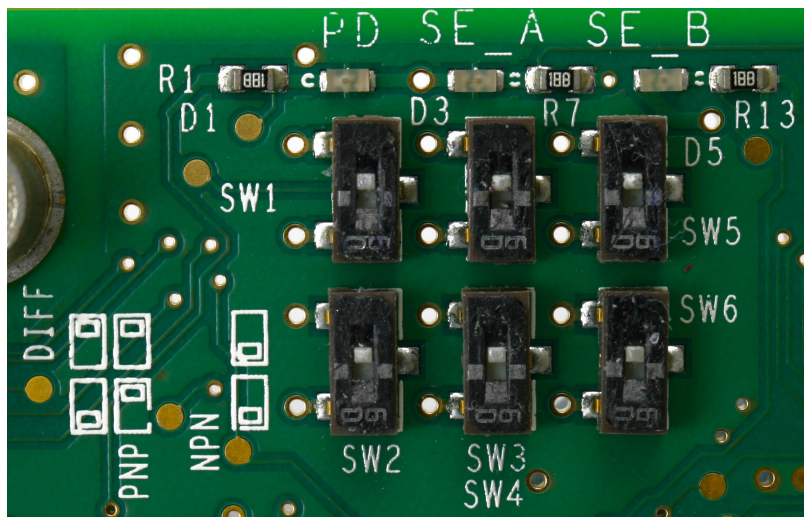
PD = Product Detect

Sinking = NPN Configuration

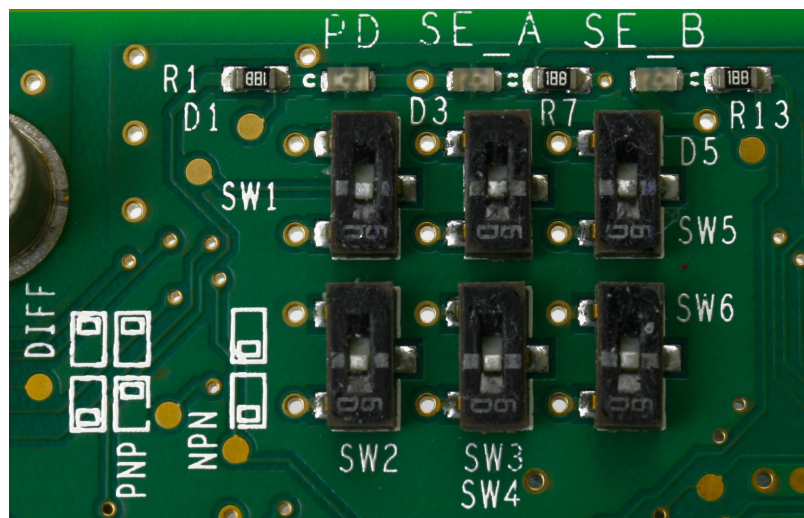
SE_A = Shaft Encoder 1 > Shaft Encoder Phase A

Sourcing = PNP Configuration

SE_B = Shaft Encoder 1 > Shaft Encoder Phase B



Switches set to PNP



Switches set to NPN

SENSOR INSTALLATION

**CAUTION:**

Risk of product detect sensor damage. The product detect sensor can be damaged if the pins on the plug make contact with the wrong pins on the PD/SE socket. To ensure the product detect sensor is not damaged when it is connected to or disconnected from the printer; shut down the printer and disconnect the power lead from the rear of the printer before connecting or disconnecting the product detect sensor.

Details of product detect sensors can be found in the Accessories Catalogue on myDomino.

Product detectors sensors are connected to the printer via one of the printer's product detect/shaft encoder sockets, see [page 3-13](#).

Sensor Positioning

Position the product detector at a location where it will detect the product before the product reaches the print head. Either the leading or trailing edge of the product detect signal can be used trigger a print.

The product detect sensor should be positioned as close as possible to the print head to avoid the next product being detected before the current print is complete.

On a very high speed production line, try to ensure the print head and product detect sensor are only one product apart. To allow a good amount of delay a minimum distance between the print head and product detect sensor of 30mm is recommended.

If it is not possible to position the product detect sensor close to the print head; the *Product queue depth* setting ([page 3-52](#)) can be set to define the number of products that come between the print head and product detect sensor.

TOUCHSCREEN INSTALLATION (Ax350i/Ax550i)

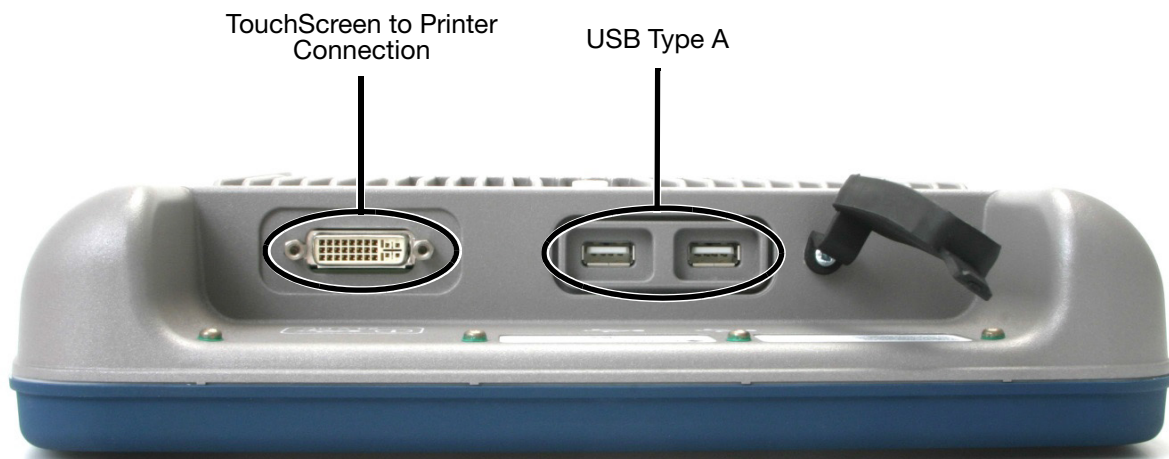
Ax350i and Ax550i printers are operated either via a TouchScreen, or via a PC.

Remote TouchScreen 10"

The Remote TouchScreen 10" is a user interface with a 10.4 inch touch screen. It has 2 USB ports for data transfer and a connector to provide control of Ax350i/Ax550i printers.



Remote TouchScreen 10" Front View



Remote TouchScreen 10" Connections

Remote TouchScreen 7"

The Remote TouchScreen 7" is a user interface with a 7 inch touch screen. It has 2 USB ports for data transfer and a connector to provide control of Ax350i/Ax550i printers.



Remote TouchScreen 7" Front View



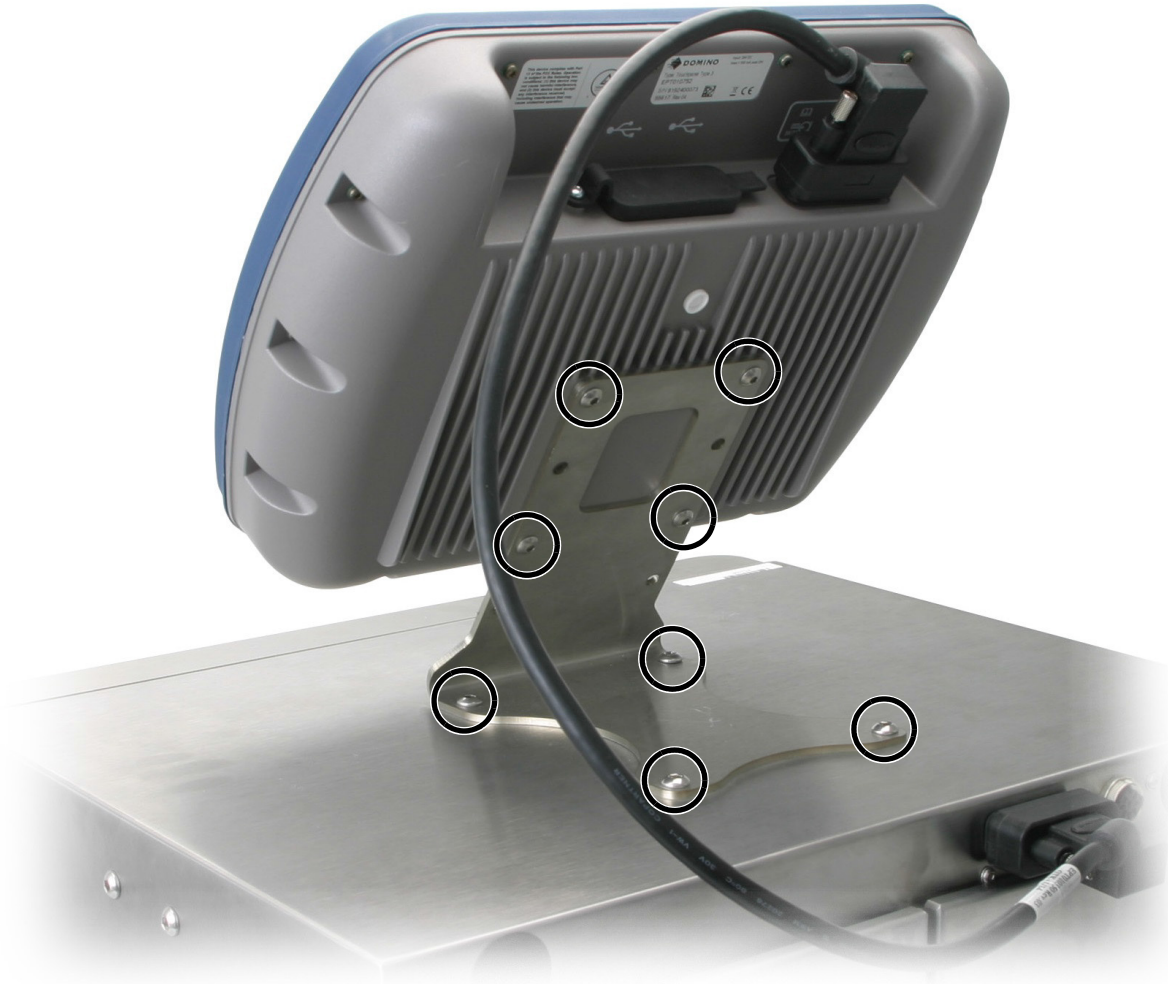
Remote TouchScreen 7" Connections

Ax350i TouchScreen Mounting

A TouchScreen can be mounted to the top of the cabinet using the angled bracket and 8 screws provided.

TouchScreens can also be mounted elsewhere on the production line using a VESA 75 bracket.

Tools required: 5mm and 4mm hex key.



Ax350i TouchScreen Bracket

Ax550i TouchScreen Mounting

A TouchScreen can be mounted to the Ax550i cabinet using the bracket and eight screws provided.

TouchScreens can also be mounted elsewhere on the production line using a VESA 75 bracket.

Tools required: 5mm and 4mm hex key.



Ax550i TouchScreen Bracket

TouchScreen Connectivity

TouchScreens are connected to the TouchScreen connector on the rear of the Ax350i/Ax550i printer. The TouchScreen will display the user interface relating to that printer when the printer is turned on.



CAUTION:

To reduce the risk of damaging cables or equipment, securely route the TouchScreen cable away from moving equipment and other hazards.



TouchScreen Connection

PREPARING THE INK SYSTEM



Ax-Series Ink System

The procedures described in this section must be followed when the ITM, ink and make-up cartridges are installed for the first time or following a long shut-down after ITM removal.

Start Up

To start up the printer:

- (1) Connect the power cable to the power socket on the rear of the printer and plug the power cable into the power supply.
- (2) Press and hold the standby button  on the printer cabinet for 2 seconds.
- (3) Wait for the printer to start up.

End of procedure.

Prepare the Print Head



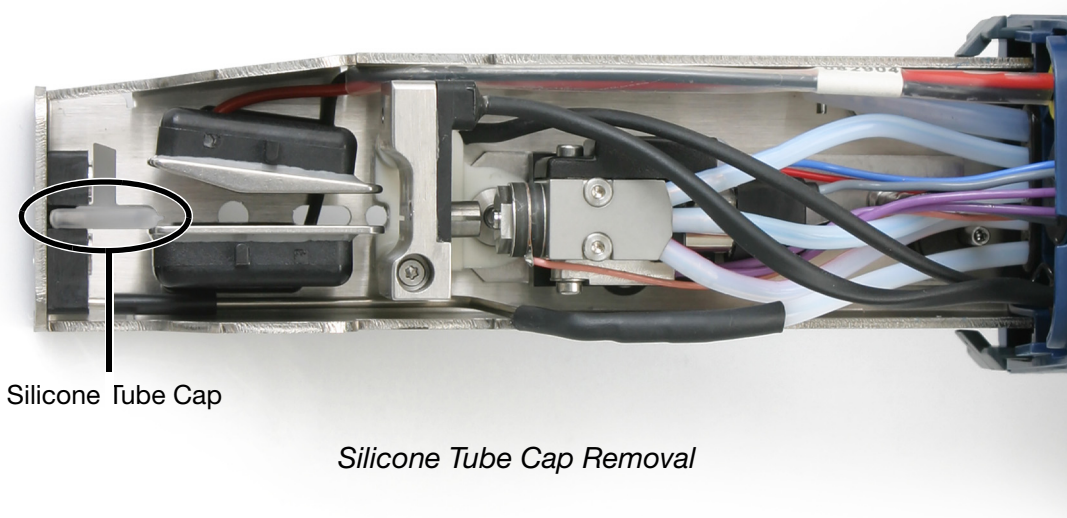
WARNING:

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with ink or make-up can cause skin or eye damage.

To prepare the print head:

- (1) Remove the print head cover.
- (2) Remove the silicone tube cap over the gutter (if fitted). Check that the print head is clean and dry.

Note: Image shows a single jet i-Pulse Print Head.



- (3) Replace the print head cover.
- (4) Fit the print head into a wash station, or place a beaker made of a conducting material and connected to earth (ground) underneath the print head.

End of procedure.

ITM Installation



- WARNINGS:**
- (1) **Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with ink or make-up can cause skin or eye damage.**
 - (2) **Pressurised ink system. Risk of ink spray. When replacing the ITM, ensure the retaining clips are fully engaged.**

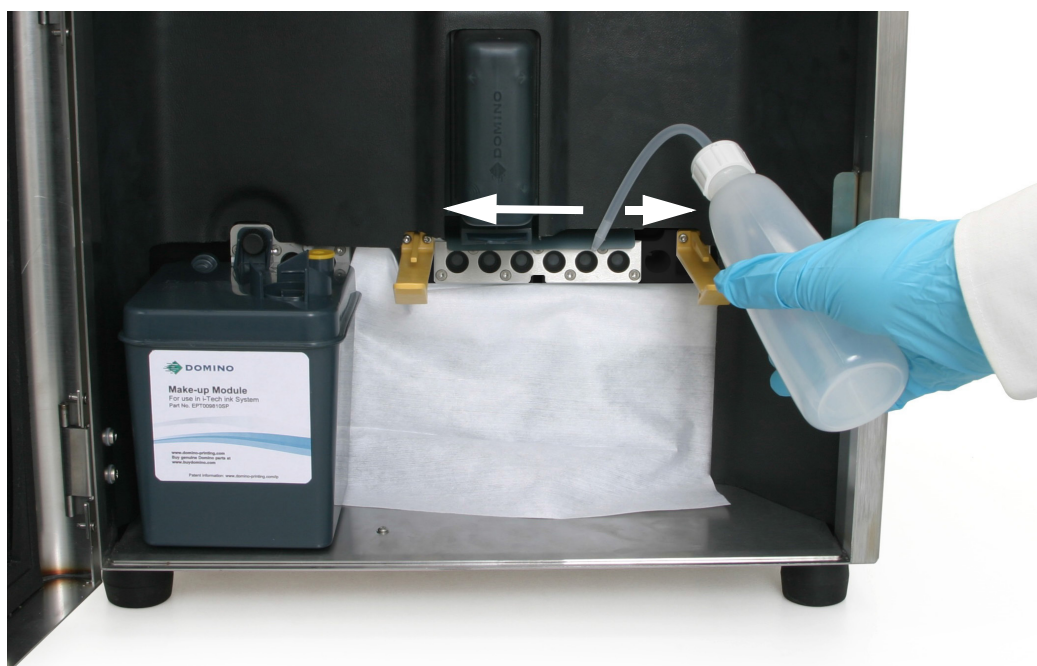
Notes: (1) Paper towels (or similar) and wash are required for this procedure.

- (2) *Waste paper towels will be contaminated with chemicals which are hazardous to the environment. Local waste disposal regulations must be followed to safely dispose of used paper towels.*

This procedure should be followed when installing the ITM for the first time or following long shut down after ITM removal. If replacing an expired ITM follow the procedure described on [page 5-10](#).

To install the ITM:

- (1) Open the printer ink cabinet.
- (2) Place paper towel (or similar) on top of the level sensor modules to catch excess fluid. Use the correct wash to remove any dried residual ink from the ink block valve face.



Washing the ink block valve face

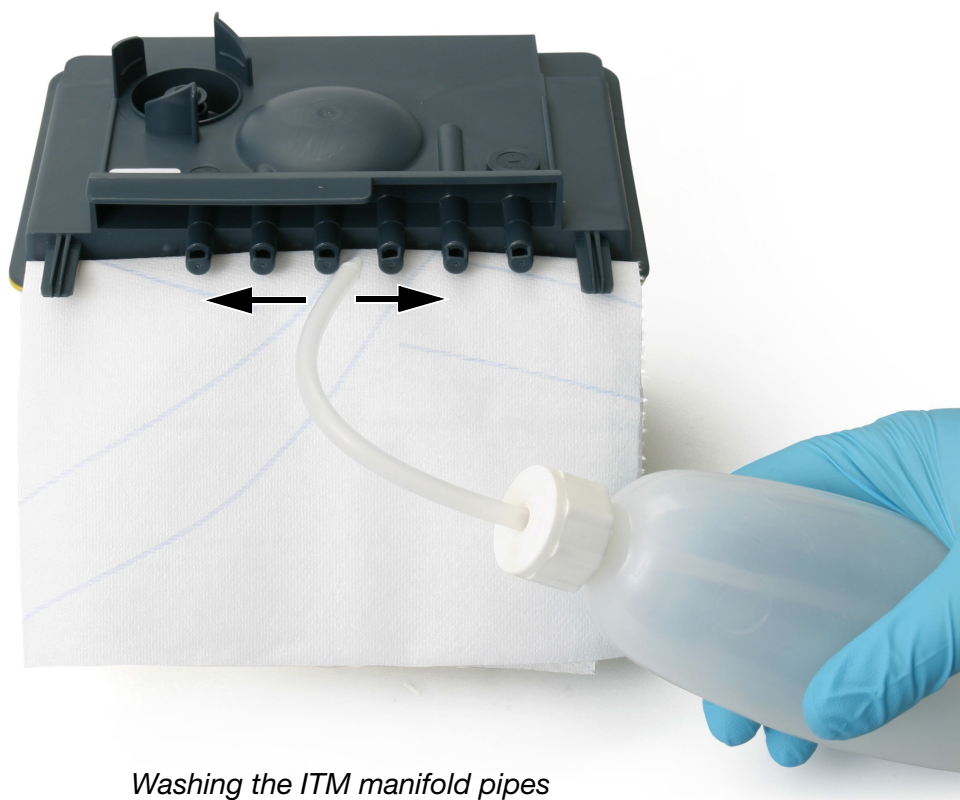
INSTALLATION

- (3) Unpack the ITM and remove the sealing strip that protects the ITM manifold pipes.



Remove ITM Sealing Strip

- (4) Place paper towel (or similar) under the ITM manifold pipes and lubricate the pipes with wash.



Washing the ITM manifold pipes

INSTALLATION

- (5) Insert the ITM between the retaining clips and firmly push the ITM manifold pipes into the ink block.



Inserting the ITM



WARNING: Pressurised ink system. Risk of ink spray. When replacing the ITM, ensure the retaining clips are fully engaged.

- (6) Continue pushing until the retaining clips engage with a click.



Retaining Clip Engaged



Retaining Clip Not Engaged

- (7) Close the printer ink cabinet.
 - (8) Dispose of used paper towels by following local waste disposal regulations.
- End of procedure.

Ink and Make-up Cartridge Installation



WARNING:

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with ink or make-up can cause skin or eye damage.

Note: Shake heavy duty ink cartridges for at least two minutes before fitting.

This procedure should be followed when installing the ink and make-up cartridge for the first time. If replacing an old or empty cartridge, follow the procedure described on [page 5-4](#).

Tools required: 6mm Hex Key.

To install the Ink and Make-up cartridges:

- (1) Before inserting the cartridges in the printer, hold the cartridges near the Quality Management Module (QMM) to check that the ink and make-up type is correct. The lights on the QMM will flash amber to indicate that the RFID tags are being read. When the RFID tags have successfully been read and validated, the lights will turn green. See [“QMM \(Quality Management Module\) Status Lights”](#) on [page 2-20](#).
- (2) Insert a 6mm hex key into the top of the cartridges and twist to break the sealing tabs. Remove the sealing tab.



Breaking the Make-up Cartridge Sealing Tab



Breaking the Ink Cartridge Sealing Tab

INSTALLATION

- (3) Push the make-up cartridge onto the make-up module manifold and push the ink cartridge onto the ITM.



Make-up Cartridge Installation



Ink Cartridge Installation

- (4) Rotate the ink cartridge clockwise and the make-up cartridge anti-clockwise, ensuring that the label is facing towards you.



Ink and Make-up Cartridge Replacement

INSTALLATION

- (5) Check inside the printer for leaks.
- (6) Shut the ink compartment access door.
- (7) The ink system must now be primed, follow the procedure on [page 3-31](#).

End of procedure.

Prime Ink System



- WARNINGS:**
- (1) Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with ink or make-up can cause skin or eye damage.
 - (2) The print head must be placed in a wash station or positioned over a beaker made of a conducting material and securely connected to earth (ground) in case the jet is misaligned out of the gutter. The electrostatic charges on the ink drops used for printing can cause a fire hazard.

The ink system must be primed before the printer can be used for the first time.

Before starting this procedure, ensure that the ink and make-up levels are at least half full. This will be indicated by yellow or green make-up and ink level icons on the information bar on the TouchScreen, see [page 4-5](#).

To run the ink priming wizard:

- (1) On the TouchScreen, select *Home > Setup > Wizards > Ink priming*.
- (2) Select *Start ink priming*.

Note: *The ink priming wizard will take 12 minutes to complete.*

- (3) Select *Complete* when the ink priming process is finished.
- (4) Check the ink jet alignment:
 - i-Pulse print head, see [page 5-22](#).
 - i-Pulse Duo print head, see [page 5-26](#).
 - i-Pulse RS print head, see [page 5-32](#).

End of procedure.

WAKE-UP MODE

Note: Wake-up mode is required for heavy duty ink types only.

Wake-up mode ensures the ink remains at a good viscosity during long periods of idle time. When enabled, the wake-up routine will run the printer's main pump for 2-3 minutes every 20 minutes to agitate the ink. The wake-up mode time interval can also be customised.

Enable/Disable Wake-up Mode

To enable or disable wake-up mode:

- (1) On the TouchScreen, select *Home > Setup > Advanced > Hardware*.
- (2) Scroll down to the *Wake-up mode* drop down setting and select one of the options described below:

Setting Name	Explanation
Factory default	Set the wake-up mode to the factory default setting.
On	Enable wake-up mode. Recommended for printers equipped with a heavy duty ink system.
Off	Disable wake-up mode. Recommended for printers equipped with a standard ink system.

End of procedure.

Set a Custom Wake-up Mode Interval

To set a custom wake-up mode time interval:

- (1) On the TouchScreen, select *Home > Setup > Advanced > Hardware*.
- (2) Scroll down to the *Wake-up mode* settings and tick the *Custom wake-up* time enabled tick box.
- (3) Adjust the *Wake-up time (min)* value to set the number of minutes in between wake-up mode cycles.

End of procedure.

INSTALLATION WIZARD

The Installation Wizard configures the language of the user interface, system pre-sets and basic settings and for the printer to operate.

Note: The correct System Date and System Time values must be entered to print accurate date or time elements within labels.

To start the printer and use the Installation Wizard:

- (1) If the printer is not already on, press and hold the standby button  for 2 seconds and wait for the printer to start up.
- (2) Select *Setup > Wizards > Installation*.
- (3) Use the drop down settings to select the required *Language, Keyboard layout* and *Primary currency*.
- (4) Select the *Next Screen* icon.



- (5) Select the *System date* and enter the current date.
- (6) Select *Save*.
- (7) Select the *System time* and enter the current time.
- (8) Select *Save*.
- (9) Select *Complete*.

End of procedure.

PRODUCTION LINE SETUP

The following settings should be configured to set the printer up on a production line.

Line Movement

Home > Production line setup > Line movement

The settings shown on the *Line movement* screen configure how the printing speed is set or measured. There are 2 different encoder input methods available which are described in the table below.

- Notes: (1) *It is recommended to use an external shaft encoder if the speed of the production line varies or there is a requirement to print barcodes or images.*
- (2) *Connect a quadrature shaft encoder instead of a single ended shaft encoder if there is a requirement to detect changes in the production line direction. A quadrature shaft encoder will also guarantee that the printer does not receive false production line movement signals when the production line is stationary.*

Encoder Input Method	Explanation
Internal	If a shaft encoder is not connected to the printer, a simulated encoder signal will be internally generated to set a fixed printing speed. To setup an internal encoder, follow the procedure described on page 3-35 .
External	Use an external shaft encoder to measure the production line speed in real time. To setup an external shaft encoder, follow the procedure described on page 3-37. If a quadrature shaft encoder is connected to the printer, the encoder mode can also be set to detect changes in the production line direction. To setup a quadrature shaft encoder, follow the procedure described on page 3-40.

Internal Encoder (Fixed Printing Speed) Setup



WARNING:

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with ink or make-up can cause skin or eye damage.

Tools required: Metric ruler.

To setup an internal encoder with a fixed printing speed:

- (1) Create a test label that includes a text element in the label design, refer to [page 4-24](#).
- (2) In the label creator, select the *Preview* tab at the top of the screen.
- (3) Make a note of the *Label width* value.
- (4) Select the *Print* icon, select *Just print* and select *OK*.



- (5) Select *Home > Production line setup > Line movement*.
- (6) Select the *START* icon and select *Sequence On*.



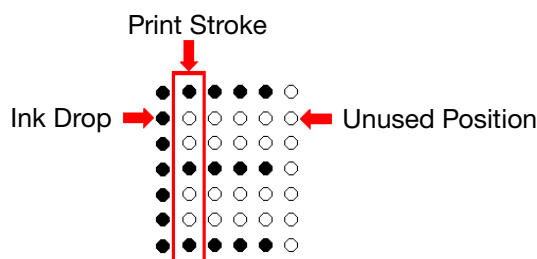
- (7) Select the *Encoder input* drop down setting and select *Internal*.
- (8) Select the *Speed (mm/s)* setting and enter the print speed in millimetres per second.
- (9) Run the production line to make a test print.
- (10) Measure the width of the test print.
- (11) If the width of the test print is not equal to the width noted at step (3), adjust the *Speed (mm/s)* setting and repeat step (9) and step (10). The equation below can also be used to calculate the correct *Speed (mm/s)* value.

$$N = O \times (P \div E)$$

New speed (mm/s) value = N
 Old speed (mm/s) value = O
 Printed label width = P
 Label width value = E

- (12) If the label width needs to be adjusted after the *Speed (mm/s)* setting has been set, adjust the *Global stroke pitch goal (mm/stroke)* setting to change the distance between print strokes (Default value: 0.43mm).

Note: A print stroke is the line of ink drops which make up each printed character as shown in the illustration below.



(13) If the Print Stretch Detected alert occurs, the bullet points below list several possible solutions to correct the alert:

- Reduce the production line speed.
- Increase the label's *Stroke pitch (mm/stroke)* value, see [page 4-25](#).
- Change the label's *Type* setting to enable the label to be printed faster, see [page 4-25](#).
- If some print stretch is acceptable, adjust the *Print stretch threshold (mm)* setting. This setting will adjust the amount of print stretch allowed before the alert is triggered.

End of procedure.

External Single Ended Shaft Encoder Setup



WARNING:

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with ink or make-up can cause skin or eye damage.

CAUTION:

Risk of equipment damage. The shaft encoder can be damaged if the pins on the shaft encoder plug make contact with the wrong pins on the PD/SE socket. Shut down the printer and disconnect the power lead from the rear of the printer before connecting a shaft encoder to the PD/SE socket.

Tools required: Metric ruler.

To setup an external single ended shaft encoder to measure the production line movement:

- (1) Create a test label that includes a text element in the label design, refer to [page 4-24](#).
- (2) In the label creator, select the *Preview* tab at the top of the screen.
- (3) Make a note of the *Label width* value.
- (4) Select the *Print* icon, select *Just print* and select *OK*.



- (5) Select *Home > Production line setup > Line movement*.
- (6) Select the *START* icon and select *Sequence On*.



- (7) Select the *Encoder input* drop down setting and select *External*.
- (8) Select the *Print optimisation* drop down setting and select *Message width*.
- (9) Scroll to the bottom of the *Line movement* menu, select the *Stroke go multiplier* setting and select *x1*.
- (10) Select the *Encoder mode* setting and select *Single mode*.
- (11) Select *Set calibration settings*.
- (12) To calibrate the shaft encoder, the production line will need to run for a measured distance. Enter the distance that the production line will run for in the *Calibration distance (mm)* setting.
- (13) Select *Start*.
- (14) Run the production line over the distance that was set at step (12).
- (15) Scroll to the bottom of the *Calibration settings* window and note the *Velocity encoder pulse count* value.
- (16) Enter the *Velocity encoder pulse count* value into the *Calibration pulse count (pulses)* setting.

INSTALLATION

(17) Adjust the *Digital gearbox multiplier* and *Digital Gearbox divider* settings to set the *Encoder stroke resolution* to a value that is close to the values described in the table below.

Notes: (1) Use the lowest possible *Digital gearbox multiplier* value to achieve a good *Encoder stroke resolution* value.

(2) The *Save* button will not be available for selection if the *Encoder stroke resolution* (pulses/stroke) value is too high. The *Encoder stroke resolution* (pulses/stroke) range is 1 - 100.

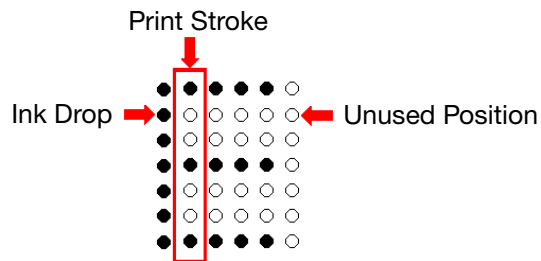
Print Application	Encoder Stroke Resolution Value
High speed printing.	10, or close to 10.
Medium and low speed printing.	Between 10 and 20.

(18) Select *Save*.

(19) Run the production line to make another test print.

(20) Measure the width of the test print and inspect the distance between each print stroke. If the width of the test print does not match the value noted at step (3), or if the spaces between print strokes varies, refer to the table below to diagnose and correct the problem.

Note: A print stroke is the line of ink drops which make up each printed character.



Problem Description	Reason	Solution
The width of the test print does not match the <i>Label width</i> value.	The <i>Calibration distance (mm)</i> value is inaccurate.	Repeat step (12) to step (20).
	The encoder has stopped at the exact position where a pulse is generated. This will cause the encoder to generate multiple pulses when the encoder is not moving and create an inaccurate <i>Velocity encoder pulse count</i> value.	Repeat step (12) to step (20). Use the equation below to calculate the correct <i>Calibration pulse count (pulses)</i> value. $N = O \times (E \div P)$ N = New pulse count (pulses) value O = Old pulse count (pulses) value E = Label width value P = Printed label width
	There is noise on the shaft encoder input signal causing false pulses.	Close the <i>Calibration settings</i> window and adjust the <i>Encoder persistence (ms)</i> value. This value sets the length of time the shaft encoder signal must be active for before the printer recognises the signal as valid (Default value: 0.001). <i>Note: Set the persistence value as low as possible. If too high a value is set, the shaft encoder signal may not be detected at high frequencies.</i>
The space between each print stroke varies across the test print.	The resolution of the shaft encoder is not accurate enough to measure a consistent stroke go pulse.	Change the <i>Stroke go multiplier</i> setting to x2 and repeat step (11) to step (19). x1 = A stroke go pulse is detected on the rising edge of channel SE_A only. x2 = A stroke go pulse is detected on the rising and falling edge of channel SE_A.

(21) If the width of the label needs to be adjusted to fit the print surface after the shaft encoder has been calibrated, adjust the *Global stroke pitch goal (mm/stroke)* setting to change the distance between print strokes. The default *Global stroke pitch goal (mm/stroke)* value is 0.43mm.

(22) If the Print Stretch Detected alert occurs, the bullet points below list several possible solutions to correct the alert:

- Reduce the production line speed.
- Increase the label's *Stroke pitch (mm/stroke)* value, see [page 4-25](#).
- Reduce the label's *Type* setting to enable the label to be printed faster, see [page 4-25](#).
- If some print stretch is acceptable, adjust the *Print stretch threshold (mm)* setting. This setting will adjust the amount of print stretch allowed before the alert is triggered.

End of procedure.

External Quadrature Shaft Encoder Setup



WARNING:

Protective equipment such as gloves and glasses must be worn when working on or near the printer. Physical contact with ink or make-up can cause skin or eye damage.

CAUTION:

Risk of equipment damage. The shaft encoder can be damaged if the pins on the shaft encoder plug make contact with the wrong pins on the PD/SE socket. Shut down the printer and disconnect the power lead from the rear of the printer before connecting a shaft encoder to the PD/SE socket.

If the production line can change direction, a quadrature shaft encoder can be used to measure the production line speed and detect the direction of travel. Four different encoder modes are available to define the printer's behaviour when the direction changes.

A quadrature shaft encoder will also guarantee that the printer does not receive false production line movement signals when the production line is stationary.

Tools required: Metric ruler.

To setup a quadrature shaft encoder:

- (1) Create a test label that includes a text element in the label design, refer to [page 4-24](#).
- (2) In the label creator, select the *Preview* tab at the top of the screen.
- (3) Make a note of the *Label width* value.
- (4) Select the *Print* icon, select *Just print* and select *OK*.



- (5) Select *Home > Production line setup > Line movement*.
- (6) Select the *START* icon and select *Sequence On*.



- (7) Select the *Encoder input* drop down setting and select *External*.
- (8) Select the *Print optimisation* drop down setting and select *Message width*.
- (9) Scroll to the bottom of the *Line movement* menu, select the *Stroke go multiplier* setting and select *x1*.
- (10) Select *Set calibration settings*.
- (11) To calibrate the shaft encoder, the production line will need to run for a measured distance. Enter the distance that the production line will run for in the *Calibration distance (mm)* setting.
- (12) Select *Start*.
- (13) Run the production line over the distance that was set at step (11).
- (14) Scroll to the bottom of the *Calibration settings* window and note the *Velocity encoder pulse count* value.
- (15) Scroll to the top of the *Calibration settings* window and enter the *Velocity encoder pulse count* value into the *Calibration pulse count (pulses)* setting.

(16) Adjust the *Digital gearbox multiplier* and *Digital Gearbox divider* settings to set the *Encoder stroke resolution* to a value that is close to the values described in the table below:

Notes: (1) Use the lowest possible *Digital gearbox multiplier* value to set the *Encoder stroke resolution*.

(2) The *Save* button will not be available for selection if the *Encoder stroke resolution (pulses/stroke)* value is too high. The *Encoder stroke resolution (pulses/stroke)* range is 1 - 100.

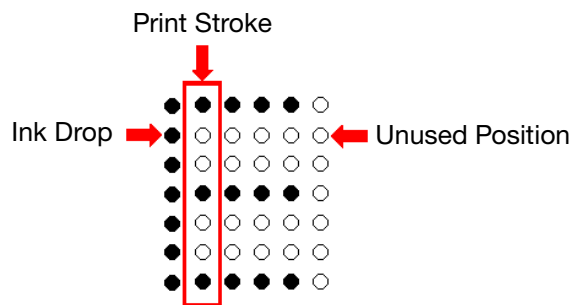
Print Application	Encoder Stroke Resolution Value
High speed printing.	10, or close to 10.
Medium and low speed printing.	Between 10 and 20.

(17) Select *Save*.

(18) Run the production line to make another test print.

(19) Measure the width of the test print and inspect the distance between each print stroke. If the width of the test print does not match the value noted at step (3), or if the space between print strokes varies, refer to the table below to diagnose and correct the fault.

Note: A print stroke is the line of ink drops which make up each printed character as shown in the illustration below.



Problem Description	Reason	Solution
The width of the test print does not match the label width value.	The <i>Calibration distance (mm)</i> value is inaccurate.	Set an encoder mode that uses both encoder channels, see page 3-44 .
		Repeat step (11) to step (19).
		Use the equation below to calculate the correct <i>Calibration pulse count (pulses)</i> value. $N = O \times (E \div P)$ N = New pulse count (pulses) value O = Old pulse count (pulses) value E = Label width value P = Printed label width
	There is noise on the shaft encoder input signal causing false pulses.	Close the <i>Calibration settings</i> window and adjust the <i>Encoder persistence (ms)</i> value. This value sets the length of time the shaft encoder signal must be active for before the printer recognises the signal as valid (Default value: 0.001). <i>Note:</i> Set the persistence value as low as possible. If too high a value is set, the shaft encoder signal may not be detected at high frequencies.
The space between each print stroke varies across the test print.	The resolution of the encoder is not accurate enough to produce a consistent stroke go signal.	Change the <i>Stroke go multiplier</i> setting to x2, or x4 if one of the backlash modes is enabled. Repeat step (11) to step (19). x1 = A stroke go pulse is detected on the rising edge of channel SE_A only. x2 = A stroke go pulse is detected on the rising and falling edge of channel SE_A. x4 = A stroke go pulse is detected on the rising and falling edge of channel SE_A and SE_B.

(20) If the width of the label needs to be adjusted to fit the print surface after the shaft encoder has been calibrated, adjust the *Global stroke pitch goal (mm/stroke)* setting to change the distance between print strokes. The default *Global stroke pitch goal (mm/stroke)* value is 0.43mm.

(21) If the Print Stretch Detected alert occurs, the bullet points below list several possible solutions to correct the alert:

- Reduce the production line speed.
- Increase the label's *Stroke pitch (mm/stroke)* value, see [page 4-25](#).
- Reduce the label's *Type* setting to enable the label to be printed faster, see [page 4-25](#).
- If some print stretch is acceptable, adjust the *Print stretch threshold (mm)* setting. This setting will adjust the amount of print stretch allowed before the alert is triggered.

End of procedure.

Encoder Mode Setup



WARNING:

Protective equipment such as gloves and glasses must be worn when working on or near the printer. Physical contact with ink or make-up can cause skin or eye damage.

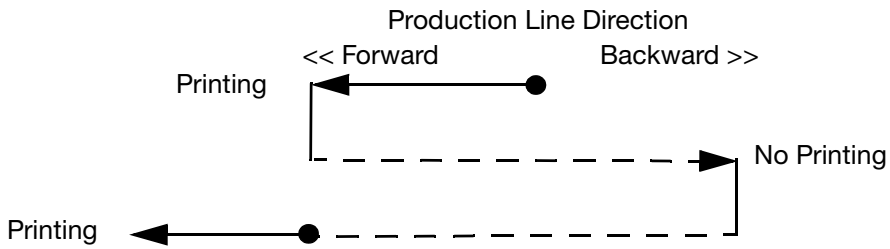
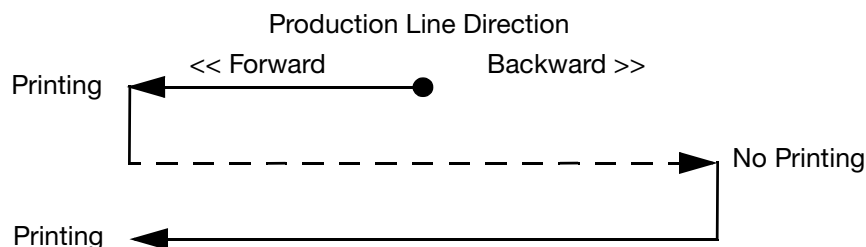
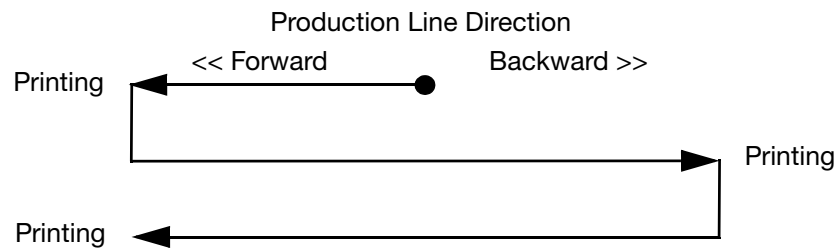
The Encoder mode setting defines the printer's behaviour when the direction of production line changes.

Note: A quadrature shaft encoder is needed to detect the production line direction. Single ended shaft encoders cannot detect the production line direction.

To set the encoder mode:

- (1) Select *Home > Setup > Production line setup > Line movement*.
- (2) Select the *Encoder mode* drop down setting.
- (3) Select one of the settings described below:

Setting Name	Explanation
Single mode	Select <i>Single mode</i> when a single ended shaft encoder is connected to the printer. The printer will be unable to detect the production line direction and printing will occur in both directions. If a quadrature shaft encoder is connected, the second input signal will be ignored.

Setting Name	Explanation
Backlash mode	Printing will only occur when the production line moves in the forward direction. When the production line moves backward, printing is paused and a count of backward encoder pulses (Current backlash count) will begin. When the production line moves forward again, the backlash count will count down until the count reaches zero. Printing will resume at this point. This enables printing to resume at the point it was paused. <i>Note: Printing will not occur in the forward direction if the current backlash count is positive. If required, select Reset to reset the backlash count to zero.</i>  The diagram illustrates the Backlash mode. It shows a horizontal line representing the production line. A dot on the line indicates the current position. Above the line, the text 'Production Line Direction' is centered. To the left of the dot, the text '<< Forward' is written, and to the right, '>> Backward' is written. Below the line, there are two horizontal arrows. The top arrow points left and is labeled 'Printing'. The bottom arrow points right and is labeled 'No Printing'. A dashed line connects the dot to the 'No Printing' arrow, indicating that printing is paused during backward movement.
Backlash suppressed	Printing will only occur when the production line moves in the forward direction. When the production line moves backward, printing is paused. Printing will resume as soon the production line moves in the forward direction again.  The diagram illustrates the Backlash suppressed mode. It shows a horizontal line representing the production line. A dot on the line indicates the current position. Above the line, the text 'Production Line Direction' is centered. To the left of the dot, the text '<< Forward' is written, and to the right, '>> Backward' is written. Below the line, there are two horizontal arrows. The top arrow points left and is labeled 'Printing'. The bottom arrow points right and is labeled 'No Printing'. A dashed line connects the dot to the 'No Printing' arrow, indicating that printing is paused during backward movement.
Backlash forward	This setting is similar to single mode, however both signals from the quadrature shaft encoder will be read. Backward movement will be treated the same as forward movement, printing will occur in both directions.  The diagram illustrates the Backlash forward mode. It shows a horizontal line representing the production line. A dot on the line indicates the current position. Above the line, the text 'Production Line Direction' is centered. To the left of the dot, the text '<< Forward' is written, and to the right, '>> Backward' is written. Below the line, there are two horizontal arrows. The top arrow points left and is labeled 'Printing'. The bottom arrow points right and is labeled 'Printing'. A dashed line connects the dot to the 'Printing' arrow, indicating that printing occurs in both directions.

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- (4) Select *Backlash direction* drop down setting and select which direction the shaft encoder will detect as forward:

Backlash Direction	Explanation
A leads B	Forward movement will be detected when the shaft encoder turns clockwise.
B leads A	Forward movement will be detected when the shaft encoder turns anti-clockwise.

End of procedure.

Stroke Release Mode

This setting defines the stroke release, or time of flight compensation mode. Do not adjust this setting unless there is a problem with print quality.

For the purpose of this setting, time of flight is defined as the time period between an ink drop breaking off from the ink jet, and an ink drop making contact with a specific point on the print surface. The time of flight changes as the production line accelerates and decelerates. The stroke release mode setting can be enabled to calculate the time of flight, so the printer can accurately position each print stroke.

Stroke Release Mode Setup



WARNING:

Protective equipment such as gloves and glasses must be worn when working on or near the printer. Physical contact with ink or make-up can cause skin or eye damage.

To set the stroke release mode:

- (1) Select *Home > Setup > Production line setup > Line movement*.
- (2) Select the *Stroke Release mode* drop down setting.
- (3) Select one of the settings described below:

Setting Name	Explanation
Off	Do not correct the print stroke release for the time of flight.
Continuous	Continuously correct each print stroke release (default setting).
First stroke only	Make a correction for the first stroke of each printed label, but do not make any corrections during the print.

End of procedure.

Setup Notes

The printer will be unable to correctly calculate the time of flight if the shaft encoder is inaccurate. Therefore, set the *Stroke release mode* to *First stroke only*. If the print quality does not improve, set the *Stroke release mode* to *Off*.

Other setup notes are listed in the table below:

Print Examples	Recommendations
	No action required.
	Set <i>Stroke release mode</i> to <i>First stroke only</i> .
	Set <i>Print optimisation mode</i> to <i>Stroke pitch</i> . Increase the <i>Encoder stroke resolution</i> value.

Print Examples	Recommendations
 Or	Set <i>Stroke release mode</i> to <i>Continuous</i>.

Print Trigger

Printing can be triggered by one of two different methods which are described in the table below.

Note: *It is recommended to use the external print trigger if the space between products varies on the production line.*

Print Trigger Method	Explanation
Internal	Generate a simulated print trigger signal internally at fixed distance intervals. To setup an internal print trigger, follow the procedure described on page 3-49 .
External	Use an external product sensor to generate a print trigger signal. To setup an external print trigger, follow the procedure described on page 3-50 .

Internal Print Trigger Setup



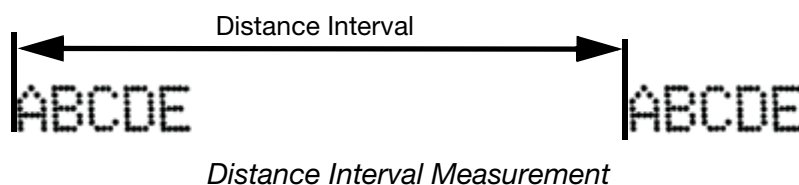
WARNING:

Protective equipment such as gloves and glasses must be worn when working on or near the printer. Physical contact with ink or make-up can cause skin or eye damage.

To setup a simulated internal print trigger signal at fixed distance intervals:

- (1) On the TouchScreen, select *Home > Setup > Production line setup > Print trigger*.
- (2) Select the *Trigger by* drop down setting and select *Internal*.
- (3) Select the *Print delay (mm)* setting and set the delay for the first label after a production run begins.
- (4) Select the *Distance interval (mm)* setting and set the distance between each printed label.

Note: *The distance interval is measured from the start of one label to the start of the next label.*



- (5) If the “Print trigger occurred while printing” alert is displayed, increase the Distance interval (mm) value so that it is larger than the width of the label.

End of procedure.

External Print Trigger Setup



WARNING:

Protective equipment such as gloves and glasses must be worn when working on or near the printer. Physical contact with ink or make-up can cause skin or eye damage.

CAUTION:

Risk of equipment damage. The product detect sensor can be damaged if the pins on the product detect sensor plug make contact with the wrong pins on the PD/SE socket. Shut down the printer and disconnect the power lead from the rear of the printer before connecting a product detect sensor to the PD/SE socket.

To setup an external print trigger:

- (1) On the TouchScreen, select *Home > Setup > Production line setup > Print trigger*.
- (2) Select the *Trigger by* drop down setting and select *External*.
- (3) Select the *Active level* drop down setting and select whether a *High* or *Low* input signal will trigger a print.

Note: Current level displays the current print trigger input signal level, either high or low.

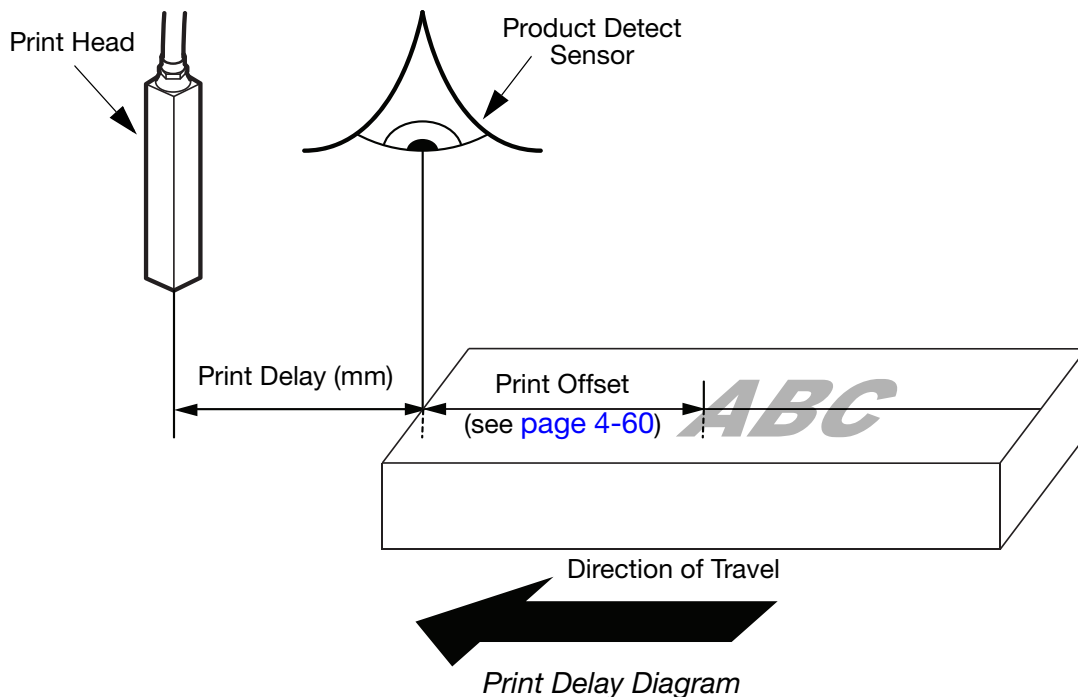
- (4) Select the *Print delay (mm)* setting and set the distance the product will move past the product detect sensor before the label is printed.

Notes: (1) Use the *Offset* settings described on [page 4-60](#) to accurately position the label on the product.

(2) The *Print delay (mm)* range depends on the printer configuration.

(3) If the “*Print delay too short*” error occurs, increase the *Print delay (mm)* value. Or move the product sensor further away from the print head.

(4) On a very high speed production line, ensure the print head and product sensor are one product apart to allow a good amount of delay. A print delay of 30mm is recommended.



- (5) If there is more than 1 product between the product detect sensor and print head, the *Product queue depth* setting should be adjusted to a value at least three more than the maximum number of products between the print head and product detect sensor to ensure reliable operation.
- (6) If required, the *Product detect persistence (mm)* setting can be set to ensure that any noise on the product detect input signal does not create false product detect triggers. The product detect signal must be active longer than the *Product detect persistence* value before the signal is recognised as valid (Default value: 1mm).

Notes: (1) Set the Product detect persistence value as low as possible. If a high value is set, then the accuracy of Print delay setting will be affected.

(2) If the print trigger signal is generated by an external machine, for example: An extrusion line set to generate a print trigger after every meter of material has been extruded. It is recommended to change the unit of measurement for the Product detect persistence value to time. The unit of measurement can be changed by a certified engineer with a USB key.

(3) If the print trigger signal is generated by a product sensor detecting physical products, it is recommended to set the unit of measurement for the Product detect persistence value by distance.

End of procedure.

Product Queue Depth

The product queue depth setting enables the printer to react to more than 1 print trigger signal. This will enable there to be more than one product between the print head and product detect sensor.

Each print trigger signal is serviced by its own independent counter that counts down the distance between the product and the print head.

Under most conditions a shaft encoder is required for acceptable performance.

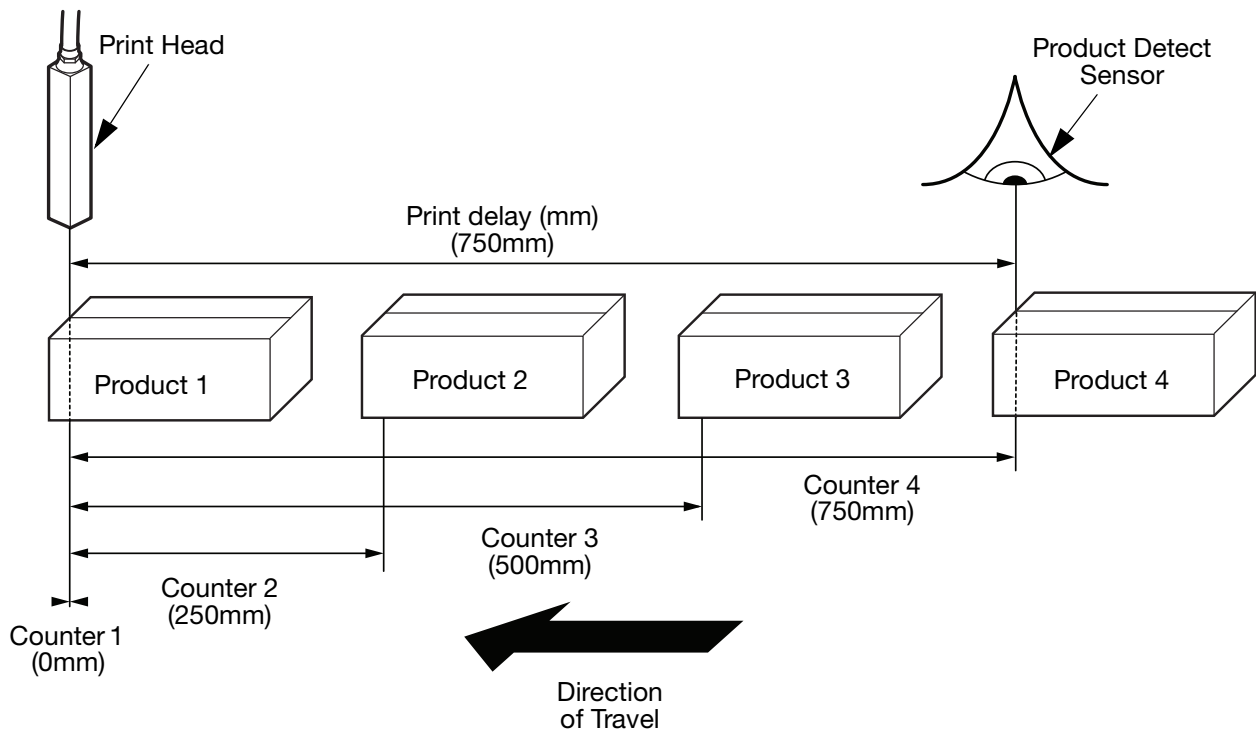
Ensure that only one print trigger signal is received for each product.

Ensure that the product queue depth value is set to at least 3 more than the maximum number of products that can come between the product detect sensor and print head.

The product queue depth setting can be found by selecting *Home > Setup > Production line setup > Print trigger*.

Example

In the example below, the *Print delay (mm)* value is set to 750mm. There are four products between the print head and product detect sensor, so the recommended *Product queue depth* value is seven.



Product Queue Depth Diagram

Ink Jet Throw Distance Setup (i-Pulse and i-Pulse RS)


WARNING:

Protective equipment such as gloves and glasses must be worn when working on or near the printer. Physical contact with ink or make-up can cause skin or eye damage.

The throw distance setting is used to define the distance between the bottom of the print head and the print surface.

Tools required: Metric ruler.

To setup the throw distance:

- (1) Measure the distance between the bottom of the print head and the print surface.
- (2) If it is possible, position the print head to the recommended distance away from the print surface as described in the table below.

Note: The nozzle diameter of the printer can be found on the TouchScreen by selecting Home > Setup > System information and viewing the Nozzle diameter (μm) value.

Nozzle Diameter	Recommended Throw Distance
75 micron	14mm
60 micron	8mm
50 micron	6mm
40 micron	4mm

- (3) On the TouchScreen, select *Home > Setup > Production line setup > Print trigger*.
- (4) Scroll down to the *Throw distance (mm)* setting and select it.
- (5) Enter the actual distance between the bottom of the print head and the print surface.
- (6) Select the green *Tick* icon.

End of Procedure.

Ink Jet Throw Distance Setup (i-Pulse Duo)



WARNING:

Protective equipment such as gloves and glasses must be worn when working on or near the printer. Physical contact with ink or make-up can cause skin or eye damage.

The throw distance setting is used to define the distance between the bottom of the print head and the print surface. Two different distances are recommended for i-Pulse Duo print heads depending on the application:

- **Stitched** - The two ink jets will be stitched together to create one large label.
- **Standard** - The two ink jets will print separately.

Tools required: Metric ruler.

To setup the throw distance:

- (1) Measure the distance between the bottom of the print head and the print surface.
- (2) If it is possible, position the print head to the recommended distance away from the print surface as described in the table below.

Note: The nozzle diameter of the printer can be found on the TouchScreen by selecting Home > Setup > System information and viewing the Nozzle diameter (µm) value.

Nozzle Diameter	Recommended Throw Distance for Standard Labels	Recommended Throw Distance for Stitched Labels
75 micron	14mm	20mm
60 micron	8mm	12mm

- (3) On the TouchScreen, select *Home > Setup > Production line setup > Print trigger*.
- (4) Scroll down to the *Throw distance (mm)* setting and select it.
- (5) Enter the actual distance between the bottom of the print head and the print surface.
- (6) Select the green *Tick* icon.

End of Procedure.

Ink Jet Velocity Setup

**WARNING:**

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with ink or make-up can cause skin or eye damage.

The Jet velocity (mm/s) setting is used to define the velocity of the ink drops between the nozzle and the print surface. However, it is recommended to leave this value at the default setting.

To setup the ink jet velocity:

- (1) On the TouchScreen, select *Home > Setup > Production line setup > Print trigger*.
- (2) Scroll down to the *Jet velocity (mm/s)* setting and select it.
- (3) Enter a jet velocity value (Range: 15000 - 25000).
- (4) Select the green *Tick* icon.

End of Procedure.

PRINT HEIGHT CALIBRATION WIZARD



WARNING:

Protective equipment such as gloves and glasses must be worn when working on or near the printer. Physical contact with ink or make-up can cause skin or eye damage.

The print height calibration wizard is a tool which is used to set the optimum default print height.

Tools required: Metric ruler.

To run the print height calibration wizard:

- (1) Check that the print head is set to the correct distance away from the print surface. The print head distance can be set by following the Ink Jet Throw Distance procedure:
 - i-Pulse and i-Pulse RS print head, see [page 3-53](#).
 - i-Pulse Duo print head, see [page 3-54](#).
- (2) If the ink jet is not already running, press the *Start/Stop* button on the printer cabinet and wait for the ink jet to sequence on.
- (3) On the TouchScreen, select *Home > Setup > Wizards > Print height calibration*.
- (4) Check that the print quality metrics listed on the TouchScreen have green ticks next to them. If any of the items are marked with a red cross, wait a few minutes to see if the ink system stabilises and a green tick appears. If the red cross remains, an engineer certified by Domino can investigate the fault.
- (5) Select the *Next Screen* icon.



- (6) Select *Testing raster* and select a suitable raster to test.
- (7) Select *Test print* and run the production line to make a test print.
- (8) The printer will print the letter H. Measure the height of the printed letter H in millimetres.
- (9) Enter the measured height of the printed letter H in the *Measured print height (mm)* setting.
- (10) If the measured print height is different from the expected height, the printer will recommend a different charge scaling (%) value. Select *Apply* to apply the recommended value and double check the print height by repeating step (7) to step (10).
- (11) Select *Complete*.

End of procedure.

NETWORK SETUP

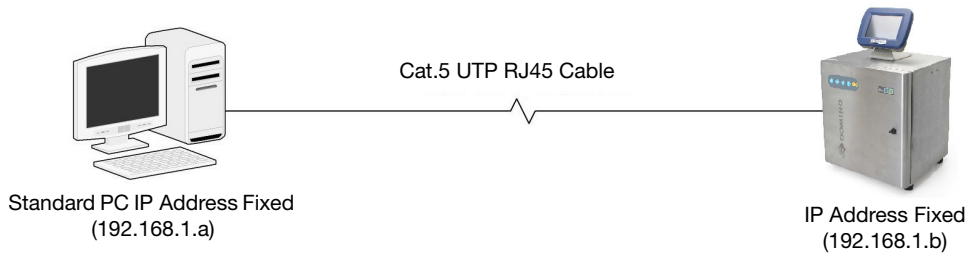
The printer can be fully controlled over a network connection using a PC which has Domino QuickStep software installed.

TouchScreens can also be used to control other printers on the same network.

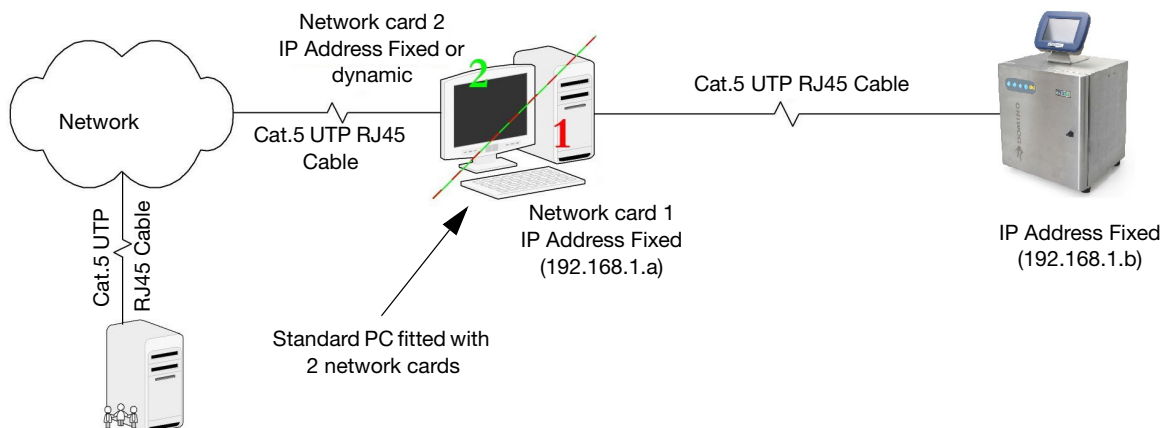
Only 1 printer can be controlled at the same time.

Network Connectivity Diagrams

Printers can be connected to a network as illustrated in the diagrams below.

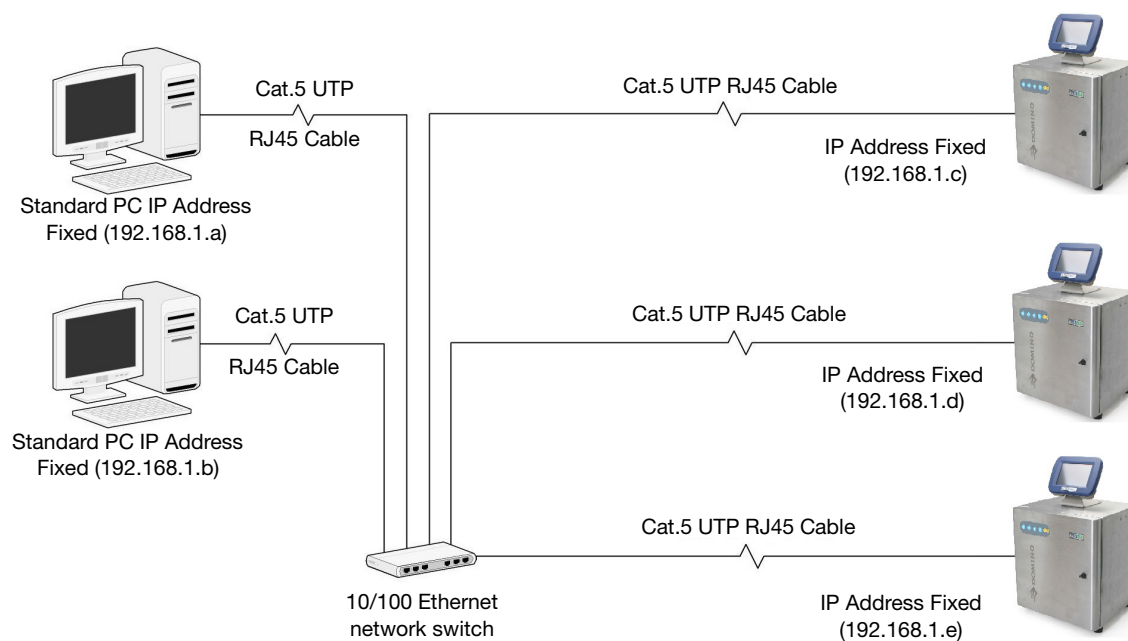


Single printer to single PC (max 1 printer) PC not connected to network

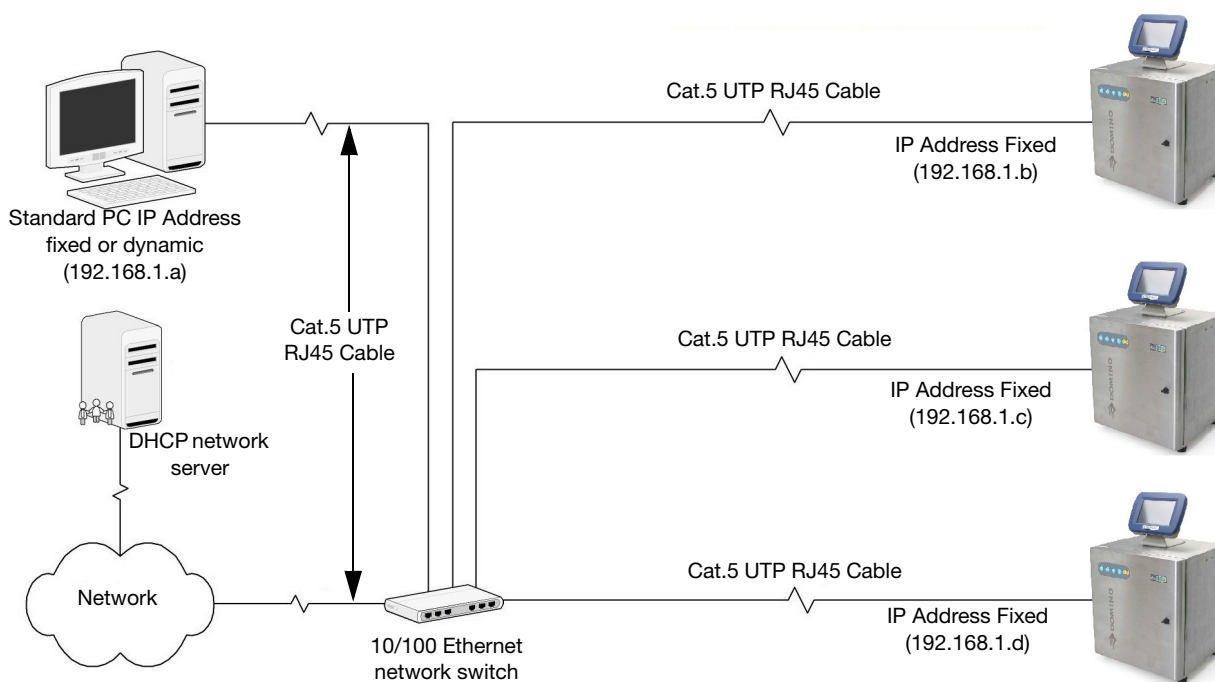


Single printer to single PC (max 1 printer) PC connected to network

INSTALLATION

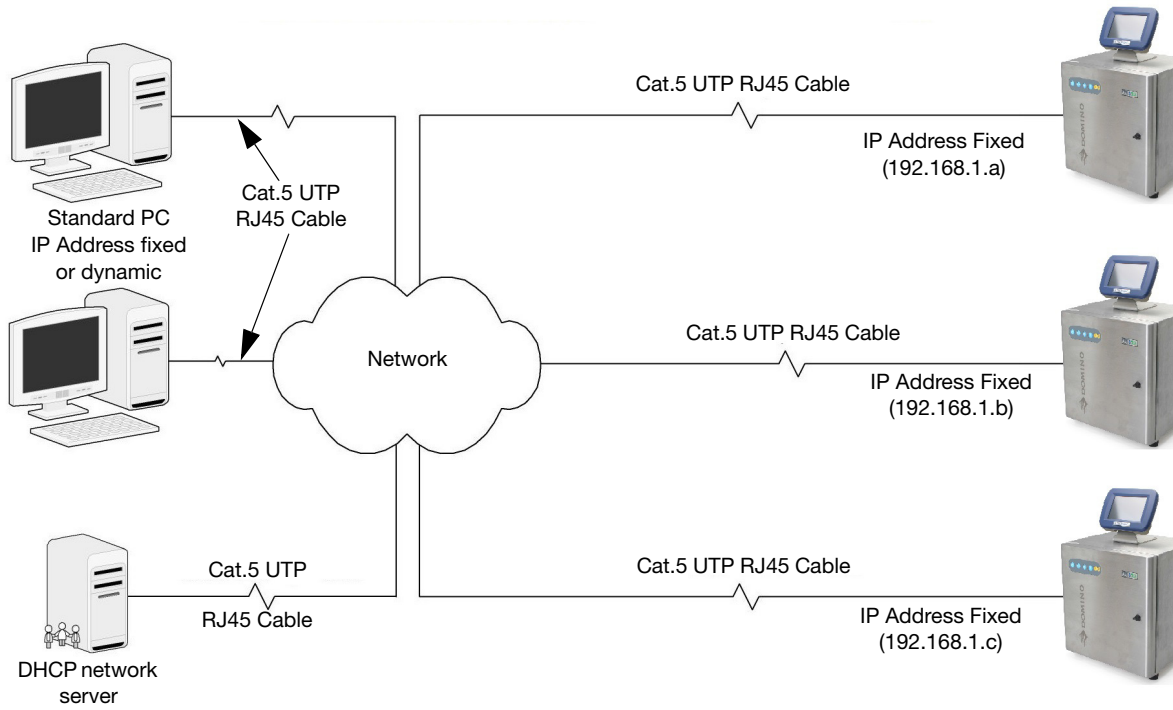


Multiple printers to PC through switch. PC not connected to network



Multiple printers to PC through switch. PC connected to network

INSTALLATION



Multiple printers to PC through network

PC to Networked Printer Settings

Note: Restart the printer after changing any network settings to ensure the new settings are saved.

To setup the PC to Printer network settings:

- (1) Install the QuickStep software on the PC which will be used to control the printer.
- (2) Ensure the printer is correctly connected to the network.
- (3) On the printer's TouchScreen, select *Home > Setup > Printer network*.
- (4) Select *Edit...*
- (5) Two different actions are available for this step depending on whether the network has a DHCP server enabled or not:

Network DHCP Server	Required Action
Enabled	Ensure that the <i>Enable DHCP</i> tick box is selected.
Not Enabled	Ensure that the <i>Enable DHCP</i> tick box is not selected. Change the IP address of the printer so that it is in the same range as the PC. For example, if the PC's IP address is 192.168.1.5 change the printers IP address to 192.168.1.55.

- (6) Make a note of the *Host name* and *IP address*.
- (7) Press and hold the standby button  for 2 seconds and restart the printer.
- (8) Start the QuickStep software on the PC.
- (9) Select the *Lock* icon.



- (10) Select *Unlock UI settings*.
- (11) Enter the UI password (Default password: QS).
- (12) Select the green *Tick* icon.
- (13) Select *OK*.
- (14) Select the *UI Settings* icon.
- (15) Select the *Connection Method* drop down setting and select one of the options described below:

Connection Method	Explanation
Broadcast	Automatically search and display a list of the printers available on the network.
Favourite	Create a list of favourite printers if there are multiple printers available on the network.
Direct	Connect to a single printer on the network.

INSTALLATION

(16) If the Broadcast connection method is used, proceed to [step \(20\)](#). If the Favourite connection method is used, select *Add favourite...* If the Direct connection method is used, select *Add direct...*

(17) Enter the following information:

Setting Name	Explanation
Name	Enter the name of the printer.
Type	Select the type of printer which the QuickStep emulator is connecting to. For an Ax-Series printer, select <i>CIJ</i> .
IP address	Enter the IP address of the printer.
Host	Enter the host name.

(18) Select the green *Tick* icon.

(19) Details of the printer will now appear on the screen.

(20) Select the *Network* icon in the bottom left corner of the PC's QuickStep programme.



(21) Select *Connect* to connect to a printer.

End of procedure.

PC Web Browser to Networked Print Setup

If the webserver is enabled in the printer's settings, the printer can be controlled from a desktop PC or a laptop using a standard web browser.

Note: Restart the printer after changing any network settings to ensure the new settings are saved.

Enable Webserver

To enable the webserver in the printer's settings:

- (1) Connect a TouchScreen to the printer and select *Home > Setup > Printer network*.
- (2) Select *Enable WebServer*.
- (3) Make a note of the printer's IP address.
- (4) Press and hold the standby button  for 2 seconds and restart the printer.

End of procedure.

Control the Printer from a PC

To control the printer using a PC based web browser:

- (1) Connect the PC to the printer directly via the Ethernet port on the rear of the printer cabinet.
Or, ensure that the PC and Printer are both setup and connected to the same network.
- (2) Power on the printer and start the PC.
- (3) Open a web browser on the PC.
- (4) Enter "http://[IP address of the printer]" into the address bar of the web browser.

End of procedure.

TouchScreen to Networked Printers

One TouchScreen can control multiple printers in the same network if required. However, the TouchScreen can only connect to 1 printer at a time.

Create a List of Favourite Printers

It is recommended to create a list of favourite printers in the TouchScreen if there is more than 1 printer on the same network which the TouchScreen can connect to.

To create a favourite list of the printers:

- (1) Select the *Lock* icon on the bottom right corner of the TouchScreen.



- (2) Select *Disconnect from printer*.
- (3) Select the *Lock* icon on the bottom right corner of the TouchScreen.
- (4) Select *Unlock UI settings*.
- (5) Enter the UI password (Default password: QS).
- (6) Select the green *Tick* icon.
- (7) Select *OK*.
- (8) Select *UI Settings*.
- (9) Select the *Connection Method* drop down setting and select *Favourite*.
- (10) Select *Add favourite...*
- (11) Enter the following information:

Setting Name	Explanation
Name	Enter the name of the printer.
Type	Select the type of printer which the TouchScreen is connecting to. For an Ax-Series printer, select <i>CIJ</i> .
IP address	Enter the IP address of the printer.
Host	Enter the host name.

- (12) Select the green *Tick* icon.
- (13) The details of the printer will now appear on the screen. Select the printer to connect to it.
- (14) Either select *Add favourite...* to add another printer or select the *Network* icon in the bottom left of the TouchScreen.



- (15) Select a printer from the favourite list to connect to.

End of procedure.

Connect to a Different Printer in the Network

Details about the printer being controlled can be viewed by selecting *Home > Setup > System Information*.

If a favourite list of printers has been set up, follow the procedure below to connect the TouchScreen to a different printer in the network.

- (1) Select the *Lock* button.
- (2) Select *Disconnect from printer*.
- (3) Select the required printer.

End of procedure.

PART 4 : OPERATION

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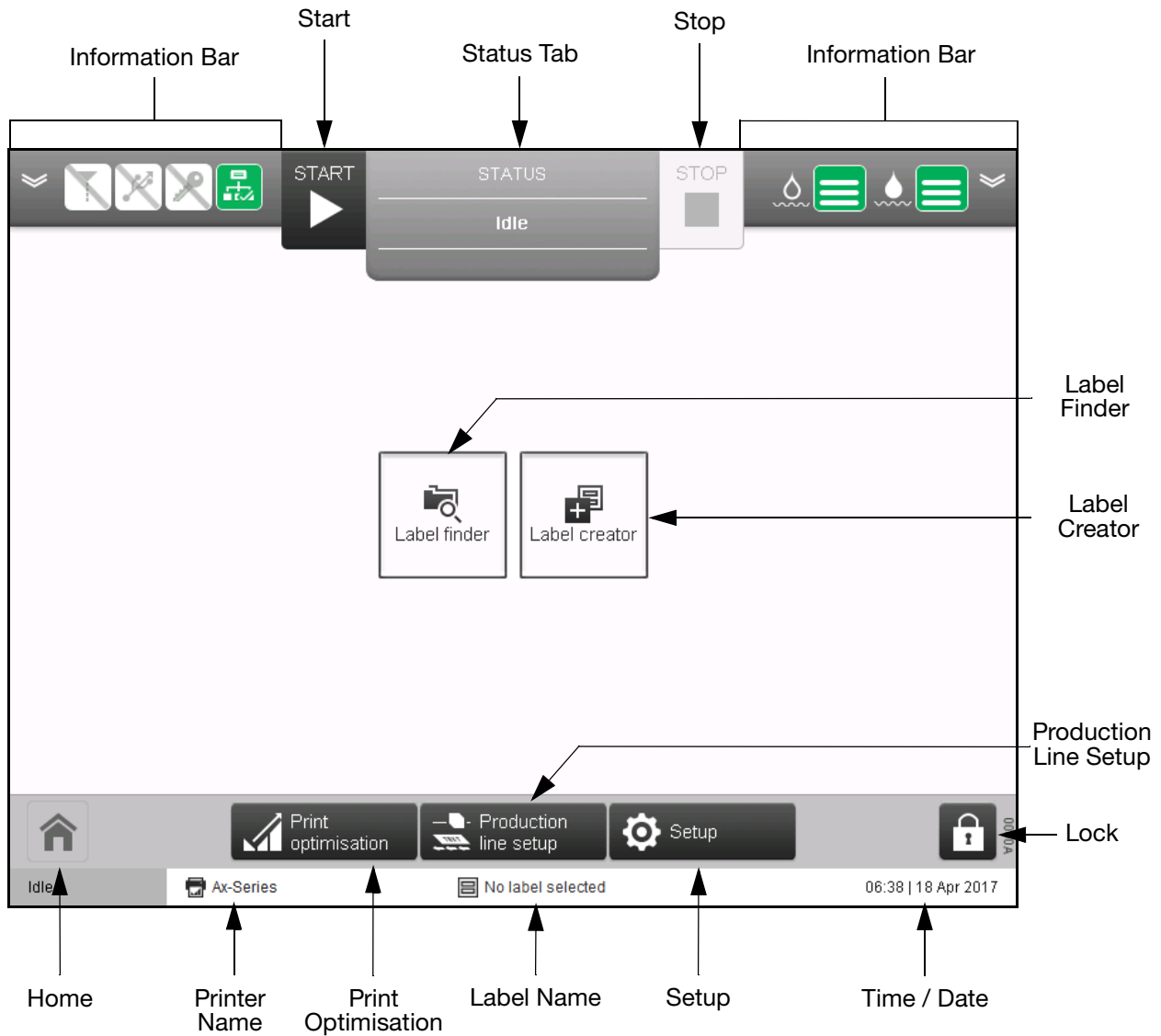
Fault Finding 4-80

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QUICKSTEP INTERFACE

Home Screen

When starting the printer, the following *Home Screen* is displayed.



QuickStep Home Screen

Setting Name	Explanation
Start	Select Start to sequence the ink jet on and enter the Ready state. When the printer is in the Ready state, the ink jet will be running and the printer will print when it receives the correct print trigger and encoder signals.
Stop	Select Stop to sequence the printer to one of the following states: <i>Sequence off</i> - The ink jet will stop running. The print head will be flushed to prevent ink from drying and blocking the nozzle or gutter. The printer will turn off. <i>Pause printing</i> - The ink jet will continue running. Printing will be paused. Power to the deflector plates will be stopped. <i>Jet running</i> - The ink jet will continue running. Printing will be paused. There will be no phasing or power to the deflector plates. <i>Phase locked</i> - The ink jet will continue running. Printing will be paused. Modulation and charging will be present, but no power to the deflector plates.
Information Bar	Displays printer information. See “PRINTER INFORMATION SCREENS” on page 4-14.
Status Tab	Displays printer status and alerts. If more than one alert is present, the highest priority alert is displayed. If an alert is displayed, select the Status Tab to show more information and to clear the alert. See “STATUSES, ALERTS AND FAULT FINDING” on page 4-68.
Label Finder	Find a saved label from the printer’s label store to preview, print or edit.
Label Creator	Open the Label Creator to create a new label.
Lock	Log the user out from the printer. Disconnect the TouchScreen from the printer. Change the user’s password. Lock the screen to prevent accidental changes.
Date / Time	Displays the current system date / time.
Setup	Open the Setup screen. See “Setup Screen” on page 4-8.
Production Line Setup	Open the Production Line Setup screen. See “Production Line Setup Screen” on page 4-9.

OPERATION

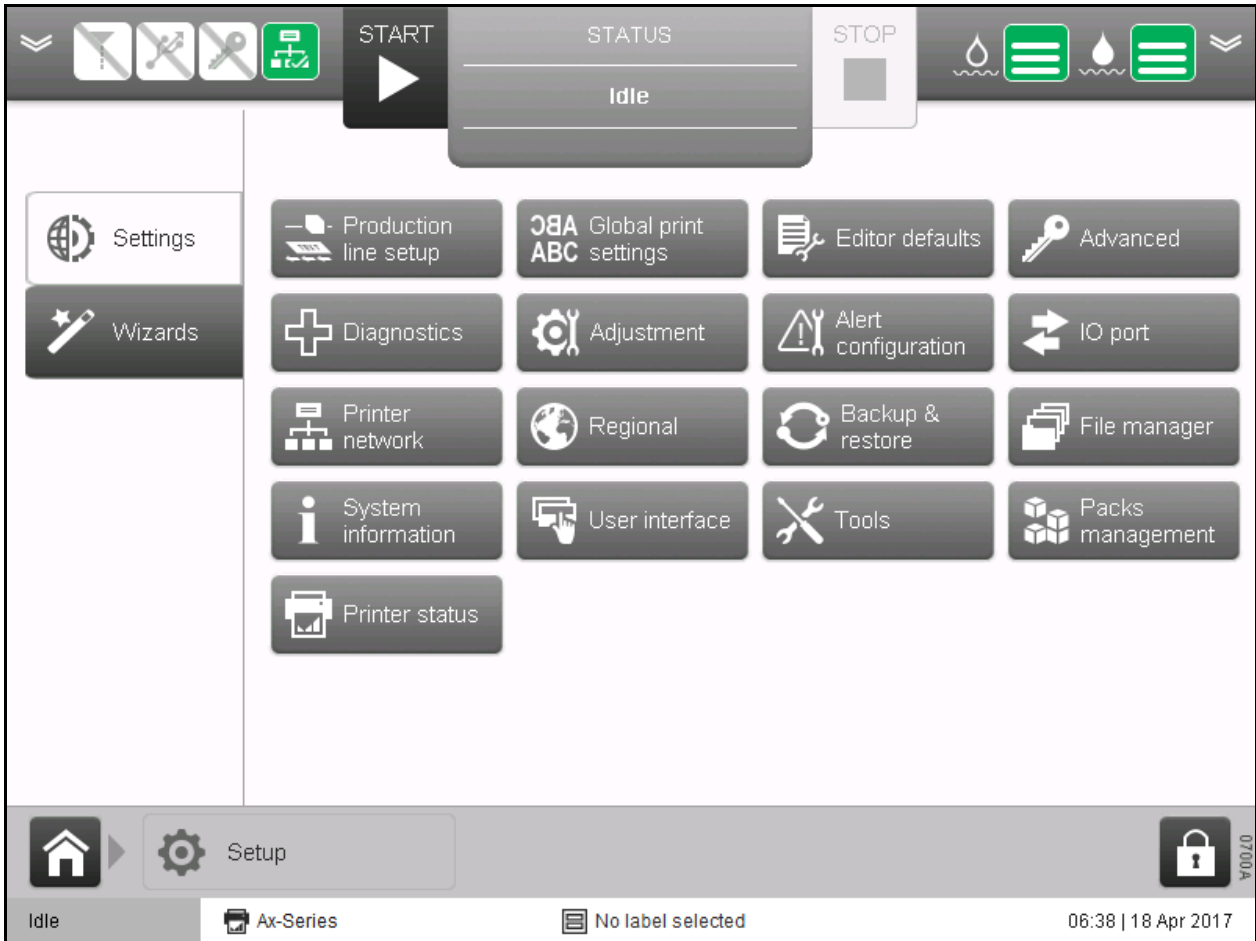
Setting Name	Explanation
Print Optimisation	Open the Print Optimisation screen. See “Print Optimisation Screen” on page 4-10.
Label Name	Displays the name of the label which is currently loaded.
Printer Name	Displays the name of the printer that the user interface is connected to. The name can be changed by selecting <i>Home > Setup > System information</i> and changing the <i>Printer name</i> setting.
Home	Press Home at any time to return to this Home Screen.

Setup Screen

The Setup screen contains advanced printer settings, diagnostic tools and wizards.

This screen is accessed by selecting *Setup* on the Home screen.

The following illustration shows the Setup screen.



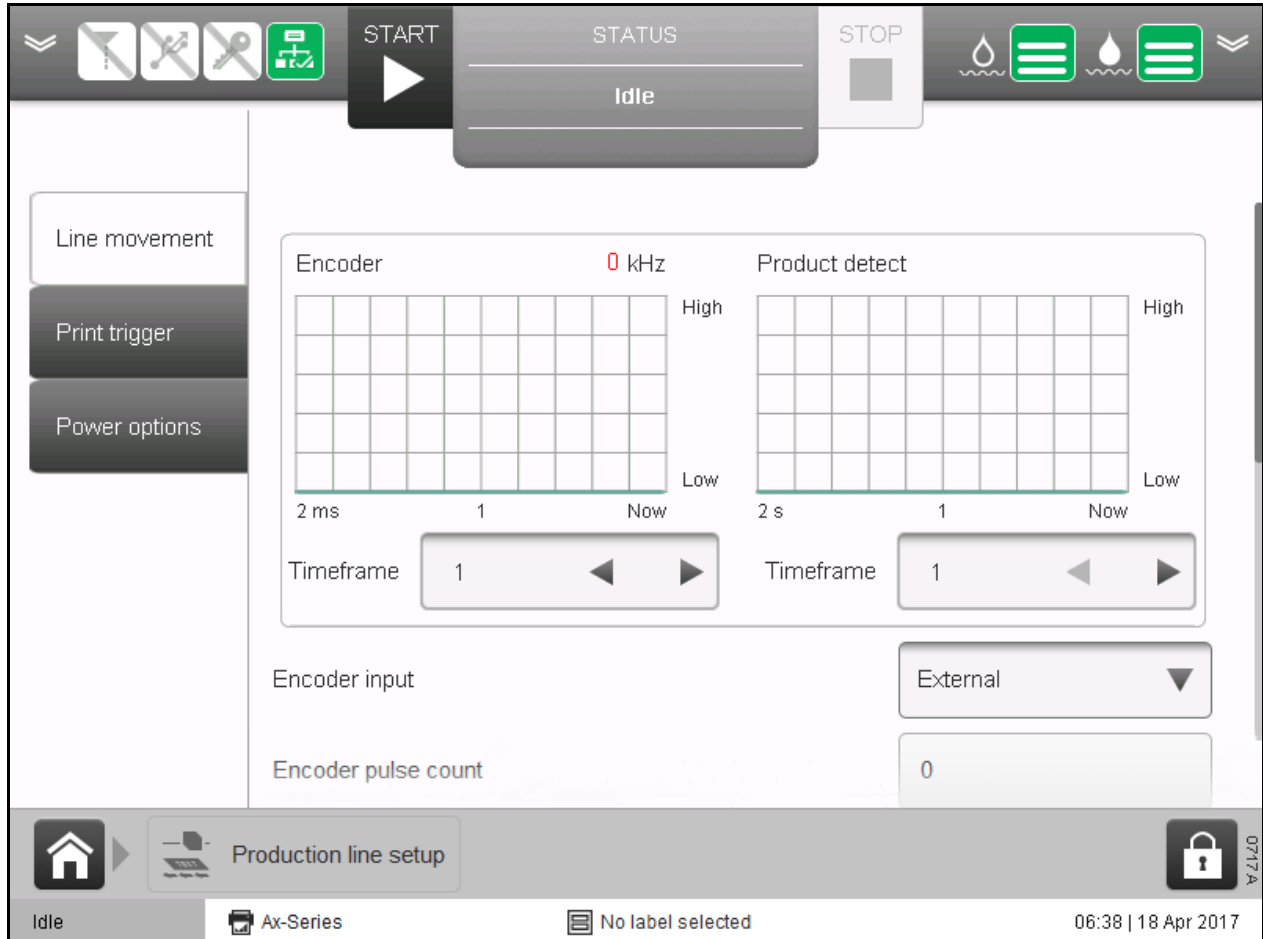
Setup Screen

Production Line Setup Screen

The Production Line Setup screen contains print trigger, encoder and power settings which should be configured during printer installation. See [“PRODUCTION LINE SETUP”](#) on page 3-34.

This screen is accessed by selecting *Production Line Setup* on the Home screen.

The following illustration shows the Production Line Setup screen.



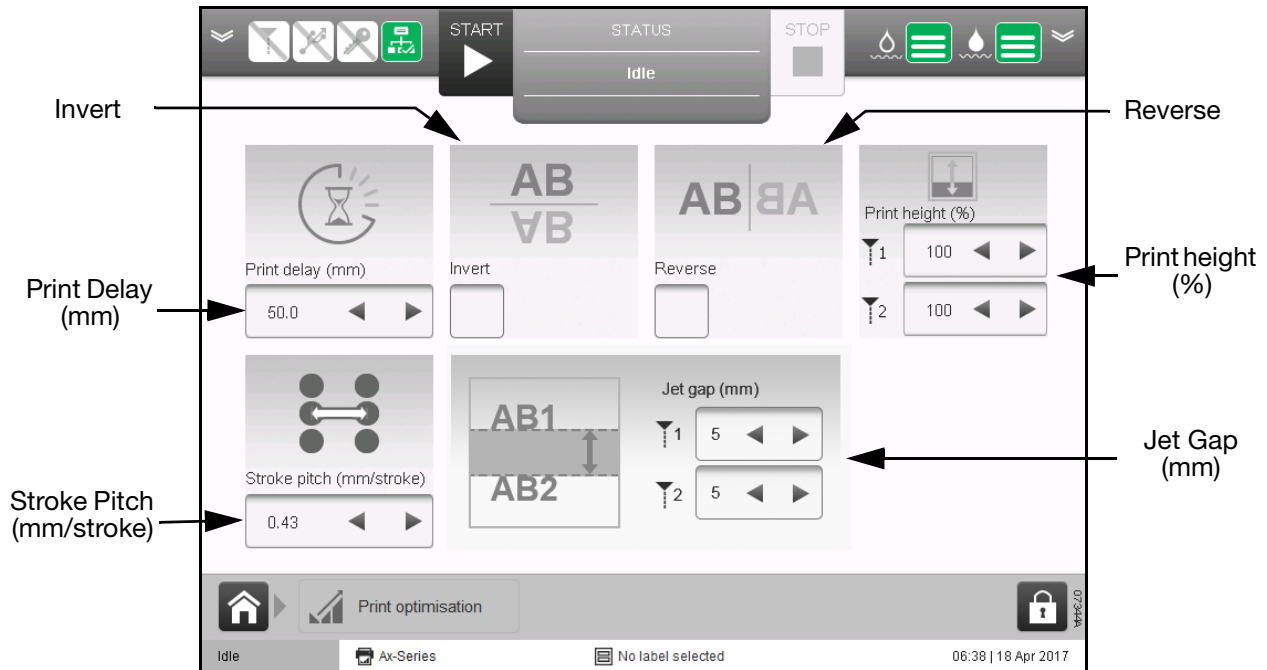
Production Line Setup Screen

Print Optimisation Screen

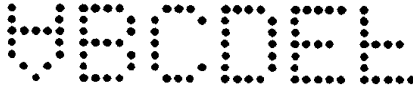
The Print Optimisation screen contains settings which are used to make everyday adjustments to label appearance.

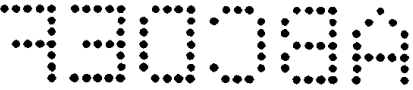
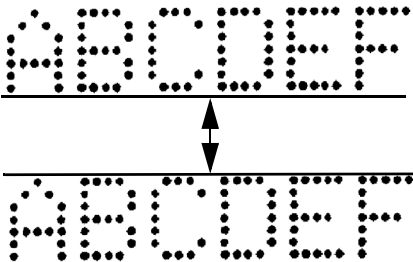
This screen is accessed by selecting *Print Optimisation* on the *Home Screen*.

The following illustration shows the Print Optimisation screen.



Print Optimisation Screen

Setting Name	Explanation
Invert	<i>Note:</i> Not available if the Continuous Printing Pack is installed. Invert the label. 
Print Delay (mm)	<i>Note:</i> Not available if the Continuous Printing Pack is installed. Set the delay between a product sensor detecting a product and the label being printed.
Stroke Pitch (mm/stroke)	Adjust the width of the label by increasing or decreasing the distance between each line of ink drops that form each printed character. The default value is 0.43.

Setting Name	Explanation
Reverse	<i>Note: Not available if the Continuous Printing Pack is installed.</i> Reverse the label. 
Print Height (%)	<i>Notes: (1) Jet 1 and Jet 2 settings are available for the Duo Print Head only.</i> <i>(2) Adjustment of this setting on a printer equipped with a Duo Print Head will change the accuracy of the Jet Gap (mm) setting.</i> Adjust the print height percentage. 
Jet Gap (mm)	<i>Notes: (1) For Duo Print Head only.</i> <i>(2) If the Print Height (%) is also adjusted, the accuracy of the Jet Gap value will be effected.</i> Adjust the vertical gap between the two ink jet printing areas. 

Label Creator Screen

The Label Creator screen is used in label creation and label editing.

Refer to [“CREATING AND EDITING A LABEL”](#) on page 4-24.



Label Creator Screen

Setting Name	Explanation
Label View	The Label Creator contains three different viewing options: • Standard - View, add and edit label elements. • Segment - View, add and edit label segments (Visible only when the Professional Printing Pack is installed). • Preview - Preview how the label will look when printed.
Label Zoom Options	Zoom in and out of the label design, or fit the label design to the canvas area.
Max print speed	Displays the maximum print speed of the current label design.
Save/Save As/Print	Save the label design or send it to print.
Side Menu	The Side Menu contains two tabs: • Element - Add and edit label elements such as text, barcode and graphics. • Label - Edit the label layout. Edit label print settings such as the label offset and label repeat. Manage label elements by locking them.

OPERATION

Setting Name	Explanation
Element Moving Options	Choose to move elements in the label design area by dragging them. Or, choose to display arrow buttons to move elements more precisely.
Undo/Redo	Undo or redo the previous action.

PRINTER INFORMATION SCREENS

The Information Screens display detailed live information about the printer's performance.

To open the Information Screens:

- (1) Press the  icon on the Information Bar.
- (2) Swipe the screen left, or right to go to the next or previous screen.
- (3) Press the  icon to close the Information Screen.

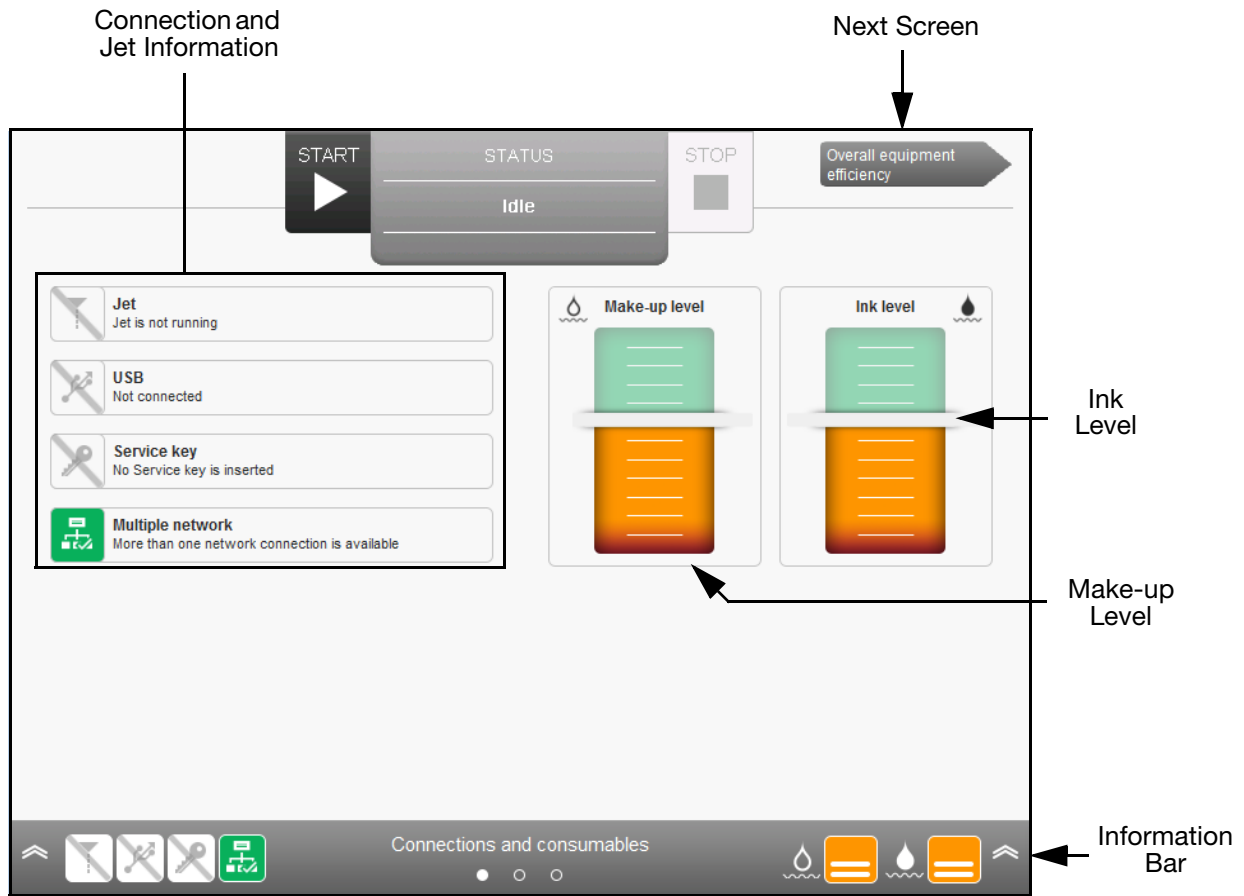
Note: The Information Bar on the Home Screen will display summarised information from the Information Screen which was closed.

The following Information Screens are available:

- Connection and Consumable, see [page 4-15](#).
- Live Status, see [page 4-16](#).
- Overall Equipment Efficiency, see [page 4-16](#).

Connections and Consumables Screen

The following illustration shows the Connections and Consumables screen.

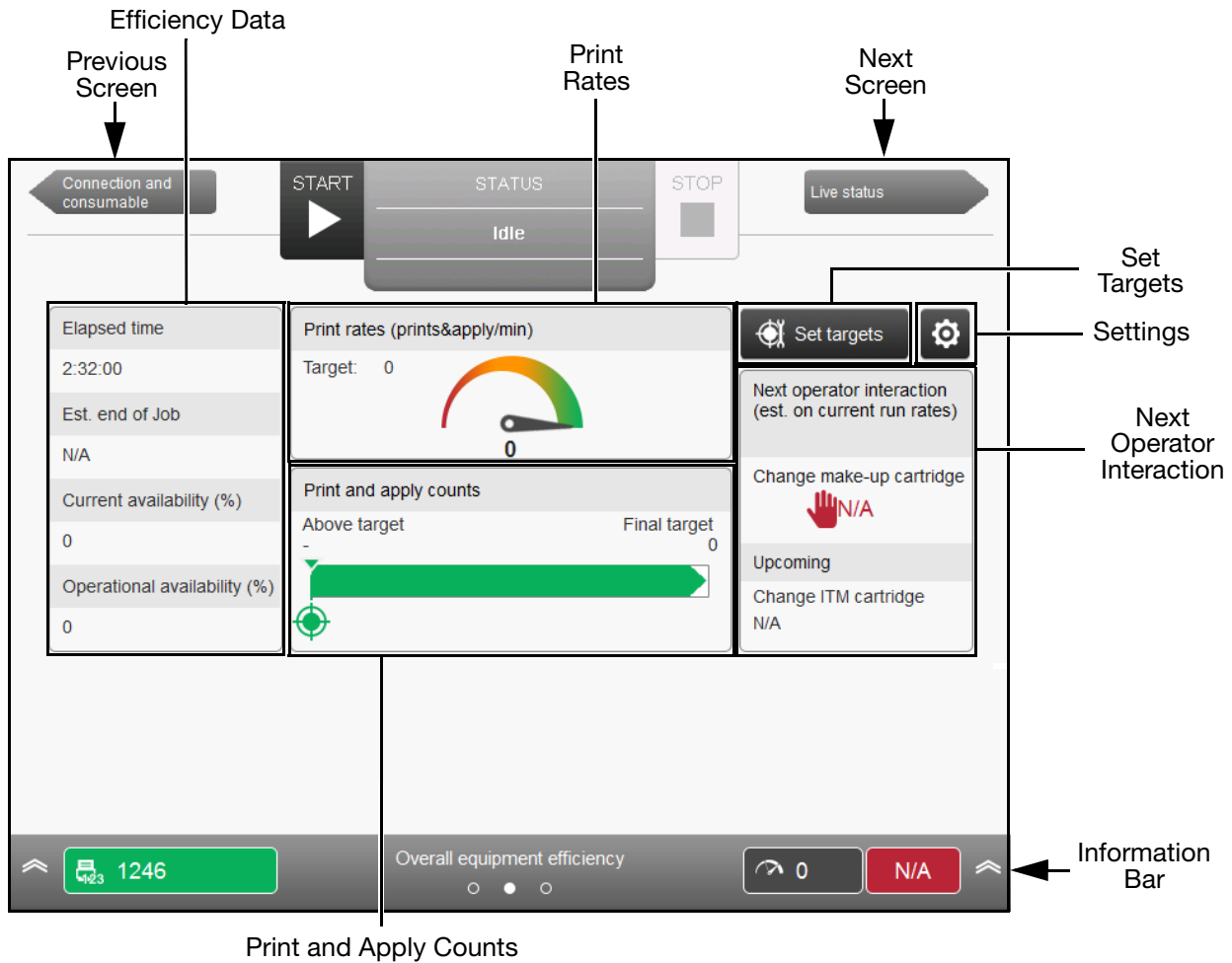


Connections and Consumables Screen

Setting Name	Explanation
Next Screen	Proceed to the Overall equipment efficiency screen.
Connection and Consumable Information	Shows connection and jet information about the printer: • Jet running/stopped. • USB device connected. • Service key connected. • Multiple network connected.
Ink Level	Shows the fluid level in the ink cartridge and ITM. The indicator moves to amber when the ink cartridge is empty and the printer is using only the fluid in the ITM.
Make-up Level	Shows the fluid level in the make-up cartridge and MUM. The indicator moves to amber when the make-up cartridge is empty and the printer is using only the fluid in the MUM.
Information Bar	Shows status icons which will be displayed on the Home Screen when the Connections and Consumables screen is closed.

Overall Equipment Efficiency Screen

The following illustration shows the Overall Equipment and Efficiency screen.



Overall Equipment Efficiency Screen

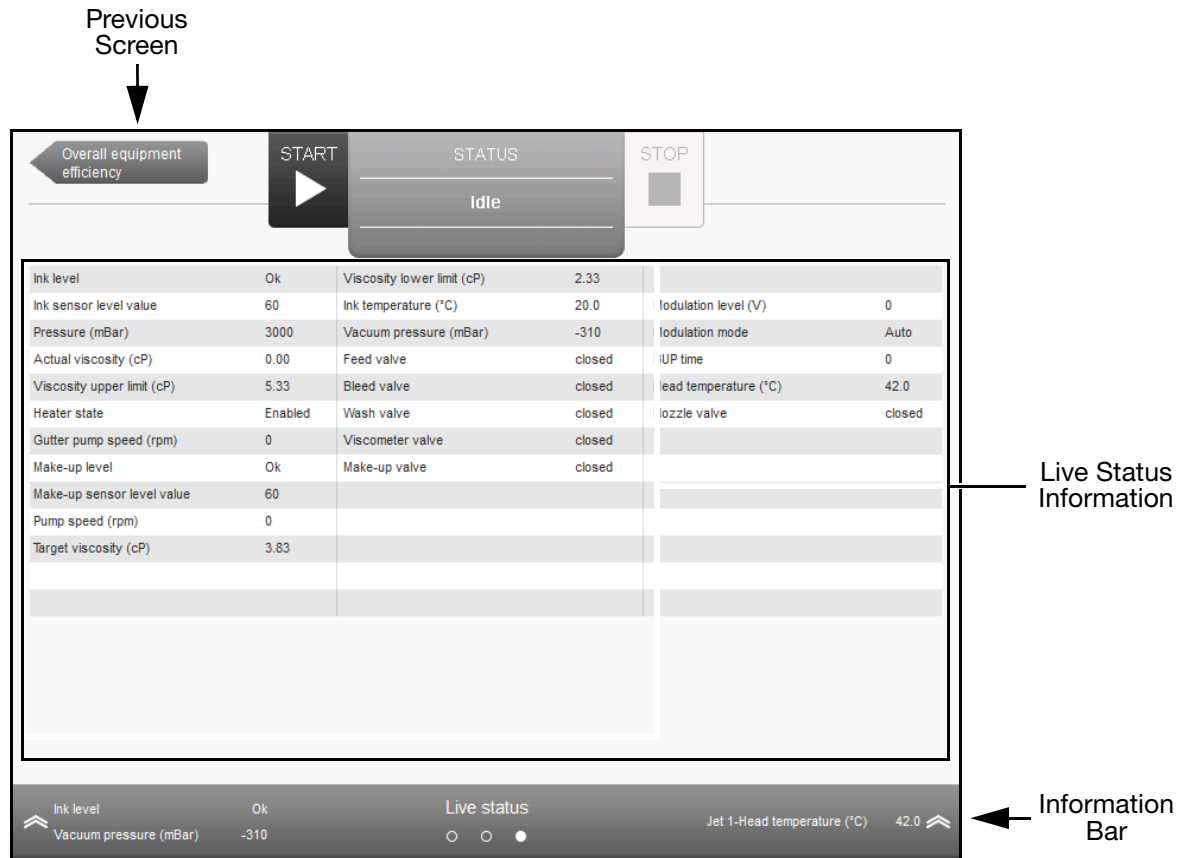
Setting Name	Explanation
Next Screen	Proceed to the Live status screen.
Previous Screen	Go back to the Connections and consumables screen.
Set Targets	Set print rate and print count targets.
Settings	Choose when to reset the OEE targets.
Print Rates	Shows the targeted number of prints per minute against the current number of prints per minute.
Next Operator Interaction	Shows the estimated time of future operator interactions.
Efficiency Data	Printer efficiency data including elapsed time, estimated time to end of job etc.
Print and Apply Counts	The targeted number of prints shown against the actual number of prints.

OPERATION

Setting Name	Explanation
Information Bar	Shows status icons which will be displayed on the Home Screen when the Overall Equipment Efficiency screen is closed.

Live Status Screen

The following illustration shows the Live Status screen.



Live Status Screen

Setting Name	Explanation
Previous Screen	Go back to the Overall equipment efficiency screen.
Live Status Information	Displays live status information about the printer.
Information Bar	Shows status icons which will be displayed on the Home Screen when the Live Status screen is closed.

START-UP

Start-up to the Ready State

When the printer is in the Ready state, the ink jet will be running and the printer will print when it receives the correct print trigger and encoder signals.

To start-up the printer and enter the Ready state:

- (1) Ensure the power connector at the rear of the printer is connected to a power source.

Note: *The red status light on the standby button will illuminate to indicate that the printer is connected to a power source.*

- (2) Press and hold the *Start/Stop* button  on the printer cabinet for 2 seconds.
- (3) The printer will now sequence on in the following order:
 - (a) The green indicator lights on the standby button and start/stop button will begin flashing to indicate that the printer is starting up.
 - (b) The TouchScreen will display the QuickStep start-up progress log.
 - (c) The red indicator light on the standby button will turn off.
 - (d) The status tab will display the printer status and any faults that require attention.
 - (e) The printer is now ready to print labels.

End of procedure.

Start-up to the Idle State

When the printer is in the Idle state, the ink jet will not be running and the printer will not print. Changes can be made to printer settings and labels can be created or edited in the label creator.

To start-up the printer and enter the Idle state:

- (1) Ensure the power connector at the rear of the printer is connected to a power source.

Note: *The red status light on the standby button will illuminate to indicate that the printer is connected to a power source.*

- (2) Press and hold the standby button  on the printer cabinet for 2 seconds.
- (3) The printer will now sequence on in the following order:
 - (a) The green indicator light on the standby button will begin flashing to indicate that the printer is starting up.
 - (b) The TouchScreen will display the QuickStep start-up progress log.
 - (c) The red indicator light on the standby button will turn off
 - (d) The status tab will display the printer status and any faults that require attention.
 - (e) The printer is now ready to perform non-printing functions such as creating a message.

End of procedure.

Switch Between the Ready State and Idle State

To switch the printer between the Ready state and Idle state:

- (1) Press the *Start/Stop* button .
- (2) Depending on the current printer status, the printer will enter one of the following two states:
 - From the Ready state, the printer will enter the Idle state.
 - From the Idle state, the printer will enter the Ready state.

End of procedure.

SHUT DOWN

**CAUTION:**

Do not disconnect the power supply if wake-up mode is enabled. Risk of ink degradation. Wake-up mode keeps the ink agitated during long periods of idle time. Wake-up mode will be disabled if power is disconnected from the printer, see [page 3-32](#).

To shut down a printer:

- (1) Press and hold the standby button  for 2 seconds.
- (2) The printer will now begin shutting down in the following sequence:
 - (a) A progress bar will be displayed on the TouchScreen.
 - (b) The green indicator light on the standby button will begin flashing.
 - (c) The print head will run an auto-flush cycle to prevent ink from drying and blocking inside the ink jet nozzle and print head gutter.
 - (d) The printer will shut down.
 - (e) The red status light on the standby button will illuminate to indicate that the printer is connected to a power source.

End of procedure.

RECOMMENDATIONS FOR A LONG SHUT DOWN

Less Than 14 Days

**WARNING:**

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with ink or make-up can cause skin or eye damage.

CAUTION:

Do not disconnect the power supply if wake-up mode is enabled. Risk of ink degradation. Wake-up mode keeps the ink agitated during long periods of idle time. Wake-up mode will be disabled if power is disconnected from the printer, see [page 3-32](#).

Depending on the ink type and the environment that the printer will be stored in, it's generally ok to shut down and store the printer for up to 14 days by following the procedure below.

However, some heavier ink types may need to be flushed out of the printer sooner than 14 days by an engineer certified by Domino. Also, if the printer will be stored in an environment outside of the recommended storage environment for the ink type, the ink will need to be flushed out of the printer by an engineer certified by Domino. This prevents ink from degrading in the printer's ink system and causing faults when the printer is re-started. Contact your local support office for more information.

To shut down and store the printer for less than 14 days:

(1) Clean the print head:

- i-Pulse print head, see [page 5-19](#).
- i-Pulse Duo print head, see [page 5-23](#).
- i-Pulse RS print head, see [page 5-27](#).

(2) Press and hold the standby button  for 2 seconds.

(3) Flush the gutter with the correct wash during the gutter clearing cycle to ensure the gutter is completely clean.

(4) Remove any factory air supply.

End of procedure.

Longer Than 14 Days

If the printer is to be shut down for longer than 14 days or stored in an environment that could degrade the ink, the ink must be flushed out of the printer by an engineer certified by Domino. This prevents ink from degrading in the printer's ink system and prevents faults when the printer is restarted. Some heavier ink types may need to be flushed out of the printer sooner than 14 days after shut down. Contact your local support office for more information.

ESSENTIAL SECURITY MODE

Note: This feature was added in software version 30.

Essential security mode is a security feature that is available on all Ax-Series printers. The Advanced Security Pack is available separately for applications that require more advanced and customisable security features, see [page 6-14](#).

Enable/Disable Essential Security

When essential security mode is active, only users with a valid password will be able to operate the printer through the user interface.

- Notes:*
- (1) *You will be immediately logged out of the printer when the security mode is enabled. Make sure you have a valid password to log into the printer before enabling essential security mode.*
 - (2) *Either a logged in admin level user, or an engineer with an engineer's USB key can disable essential security.*

To enable/disable essential security mode:

- (1) Select *Home > Setup > Security*.
- (2) Select the *Security mode* drop down setting.
- (3) Select *Essential* or *Off*.

End of procedure.

Log In

To log in:

- (1) Select the *Padlock* icon in the bottom right corner of the user interface.
- (2) Select *Log in*.
- (3) Enter the password for the required user group. The default password for each user group is defined in the table below:

Note: Passwords are case sensitive.

User Group	Password	Explanation
operator	op	Operator access to the basic settings required for normal printer operation.
admin	admin	Administrator access to all printer settings.

End of procedure.

Log Out

To log out:

- (1) Select the *Padlock* icon in the bottom right corner of the user interface.
- (2) Select *Log out*.

End of procedure.

Enable Auto Log-in

To enable auto log-in:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security*.
- (3) Tick the *Enable auto log-in* tick box.
- (4) Select the *Auto login user* drop down setting.
- (5) Select the user that will automatically be logged in when the printer is turned on.

End of procedure.

Disable Auto Log-in

To disable auto log-in:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security*.
- (3) Untick the *Enable auto log-in* tick box.

End of procedure.

Change User Password

To change a user password:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security > Users*.
- (3) Select a user.
- (4) Select *Change password*.
- (5) Enter the following information:

Setting Name	Explanation
Current password	<i>Note: Required to change admin password only.</i> Enter the current user password.
New password	Enter the new password.
Retype password	Retype the new password.

- (6) Select *Save*.

End of procedure.

CREATING AND EDITING A LABEL

Create a New Label

To create a new label:

- (1) From the TouchScreen's Home Screen, select *Label creator*.
- (2) Select *Blank*.
- (3) Select the *Label name* text box.
- (4) Use the on screen keyboard to enter a name which the label design will be saved as.
- (5) Select the green *Tick* icon.
- (6) Select the  icon to configure the label layout as described in the table below.

Note: *Printers equipped with an i-Pulse Duo print head will display label layout settings for both ink jets.*

Setting Name	Explanation
Number of lines	Select the required number of lines.
Line height (drops)	Select the required height for each line in ink drops.
Type	Select the required label quality.
Overall height (mm)	Select the height of the label.
Use default height (%)	Tick the tick box to use the default print height setting. Untick the tick box to display the <i>Height (%)</i> setting.
Height (%)	<i>Note: This setting is only valid if the Use default height (%) tick box is not ticked.</i> Set the print height percentage.
Use default stroke pitch	Tick the tick box to use the default stroke pitch setting. Untick the tick box to display the <i>stroke pitch (mm/stroke)</i> setting.
Stroke pitch (mm/stroke)	<i>Note: This setting is only valid if the Use default stroke pitch tick box is not ticked.</i> Set the distance between print strokes. A stroke is the line of ink drops which is used to make up each printed character.

- (7) Select *Create label*.

End of procedure.

Edit a Label Layout

To edit a label layout:

- (1) Open the label which requires editing.
- (2) In the side menu, select the *Label* tab.
- (3) Select the *Edit* menu.
- (4) The layout settings as described in the table below can now be edited.

Setting Name	Explanation
Number of lines	Select the required number of lines.
Line height (drops)	Select the required height for each line in ink drops.
Type	Select the required label quality.
Overall height (mm)	Select the height of the label.
Use default height (%)	Tick the tick box to use the default print height setting. Untick the tick box to display the <i>Height (%)</i> setting.
Height (%)	<i>Note: This setting is only valid if the Use default height (%) tick box is not ticked.</i> Set the print height percentage.
Use default stroke pitch	Tick the tick box to use the default stroke pitch setting. Untick the tick box to display the <i>stroke pitch (mm/stroke)</i> setting.
Stroke pitch (mm/stroke)	<i>Note: This setting is only valid if the Use default stroke pitch tick box is not ticked.</i> Set the distance between print strokes. A stroke is the line of ink drops which is used to make up each printed character.

- (5) Select the Save icon to save the label layout.



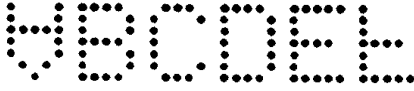
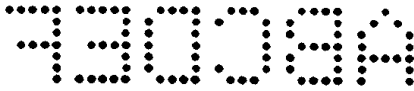
End of procedure.

Label Settings

Each individual label design can be created and saved with its own custom print settings.

To edit the label settings:

- (1) Open the label which requires editing.
- (2) In the side menu, select the *Label* tab.
- (3) Select the *Settings* menu.
- (4) The settings described in the table below can now be edited:

Setting Name	Explanation
Forward and Reverse offset	Adjust the position of the label on the print surface in millimetres, see page 4-27 .
Invert mode	Invert the label design, see page 4-28 . 
Reverse mode	Reverse the label design, see page 4-29 . 
Repeat mode	Set the label design to print repeatedly, see page 4-30 .
Max. print speed	Set the unit of measurement that the maximum print speed will be displayed in.
Name	Display the name the label is saved as. This value cannot be changed.
Shift codes	Create a shift code table that can be inserted into the label design, see page 4-32 .
Script variables	Edit variables in LUA script elements that have been inserted into the label. <i>Note: LUA scripts can only be used if the Professional Printing Pack or Continuous Printing Pack are installed.</i>

- (5) Select the Save icon to save the settings.

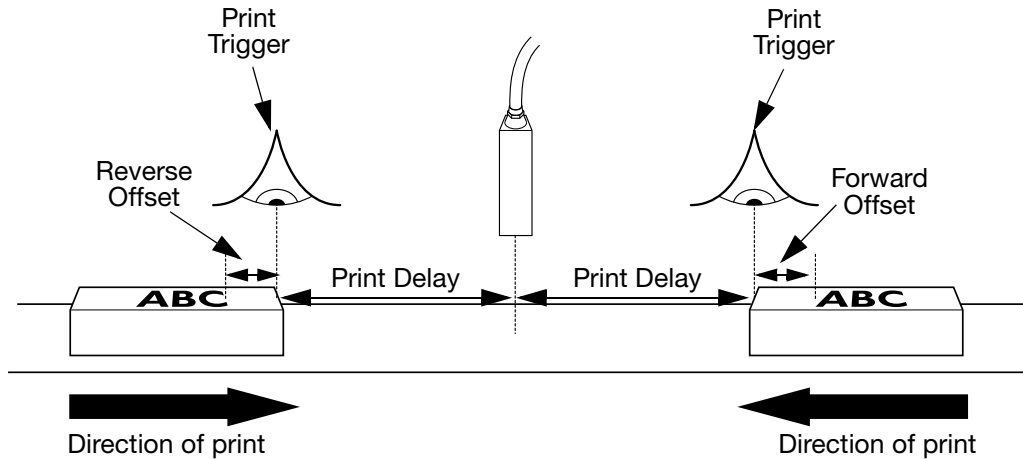


End of procedure.

Forward and Reverse Offset

Forward offset is used to adjust the position of the label on the print surface. If the print head is on a traversing line the Reverse offset can also be set.

Note: A global offset setting is also available to adjust the offset for all label designs, see [page 4-60](#).



Print Offset Diagram

To adjust the offset for an individual label design:

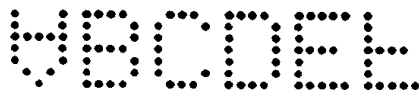
- (1) In the label creator's side menu, select the *Label* tab.
- (2) Select the *Settings* menu.
- (3) Adjust the *Forward offset (mm)* or *Reverse offset (mm)* setting.
- (4) Select the *Save* icon to save the setting.



End of procedure.

Invert Mode

Invert mode is used to invert an individual label design as shown in the illustration below.



This setting can also be used on a traversing line to invert the label design for a specified number of times before inverting the label back. If the printer is fitted with a GPIO Pack or Extended GPIO Pack, this setting can be configured to invert the label when an assigned input pin is activated.

Notes: (1) A global invert setting is available to invert all label designs. However, the global setting does not allow the number of inverted prints to be counted. Also, the global setting does not allow inverted prints to be triggered by a user port, see [page 4-66](#).

(2) If required, the label can also be inverted on the Print optimisation screen, see [page 4-10](#).

To setup invert mode for an individual label design:

- (1) In the label creator's side menu, select the *Label* tab.
- (2) Select the *Settings* menu.
- (3) Select the *Invert mode* drop down setting.
- (4) Select and configure one of the options described in the table below:

Setting Name	Explanation
On - Always	The printed label will be inverted at all times.
On - Counted	The printed label will be inverted for a specified number of prints. When this option is selected, the settings described below will also be displayed: Invert: Initial orientation - Select the orientation of the first label print. Invert repeat count - Select the number of times the label will be printed in an inverted orientation.
On - GPI / User port	<i>Note: Only valid if a GPIO Pack or Extended GPIO Pack is installed.</i> The printed label will be inverted when an assigned input pin is activated.
Off	The printed will not be inverted.

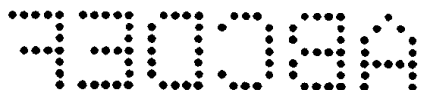
- (5) Select the *Save* icon to save the setting.



End of procedure.

Reverse Mode

Reverse mode is used to reverse an individual label design as shown in the illustration below.



This setting can be configured to reverse the label design for a specified number of times, before reversing the label back for the same number of times. If the printer is fitted with a GPIO Pack or Extended GPIO Pack, this setting can be configured to reverse the label when an assigned input pin is activated.

- Notes: (1) A global reverse setting is available to reverse all label designs. However, the global setting does not allow the number of reversed prints to be counted. Also, the global setting does not allow reversed prints to be triggered by a user port, see [page 4-66](#).*
- (2) If required, the label can also be inverted on the Print optimisation screen, see [page 4-10](#).*

To setup reverse mode for an individual label design:

- (1) In the label creator's side menu, select the *Label* tab.
- (2) Select the *Settings* menu.
- (3) Select the *Reverse mode* drop down setting.
- (4) Select and configure one of the options described in the table below:

Setting Name	Explanation
On - Always	The printed label will be reversed at all times.
On - Counted	The printed label will be reversed for a specified number of prints. When this option is selected, the settings described below will also be displayed: Reverse: Initial orientation - Select the orientation of the first label print. Reverse repeat count - Select the number of times the label will be printed in each orientation.
On - GPI / User port	<i>Note: Only valid if a GPIO Pack or Extended GPIO Pack is installed.</i> The label design will be reversed when an assigned input pin is activated.
Off	The printed label will not be reversed.

- (5) Select the *Save* icon to save the setting.



End of procedure.

Repeat Mode

Repeat mode can be configured to repeatedly print an individual label design for a specified number of times after a single print trigger signal is received. Repeat mode can also be configured to repeatedly print the label design when a continuous print trigger signal is received.

Note: A global repeat setting is available to repeat print all label designs, see [page 4-61](#).

To setup repeat mode for an individual label design:

- (1) In the label creator's side menu, select the *Label* tab.
- (2) Select the *Settings* menu.
- (3) Select the *Repeat mode* drop down setting.
- (4) Select and configure one of the options described in the table below:

Setting Name	Explanation
Off	The printed label will not be repeated.
Counted	The printed label will be repeated for a specified number of extra prints. When this option is selected, the settings described below will also be displayed: Repeat count - Enter the number of times to repeat the current label. The number of repeat counts is extra to the initial print, i.e. 6 will print 6 extra prints (7 in total). Repeat spacing mode - Select how the distance between each print is defined. Select, <i>Space by pitch</i> to define the distance between the start of one print and the start of the next print in mm. Or, select <i>Space by gap</i> to define the distance between the end of one print to the start of the next print in mm. Repeat spacing (mm) - Define the distance between prints. Repeat: Update each print - Tick to update clock and counter information for each print. Or, untick to keep the same information on each print. Repeat: End of product - Define when and how to stop repeat printing. Select <i>Ignore</i> to ignore the print trigger signal and continue printing. Select <i>Complete</i> to complete the next print. Select <i>Cancel</i> to stop before the next print.

Setting Name	Explanation
Continuous	Continuously repeat print the label when a constant print trigger signal is received by the printer. The printer will stop printing when the print trigger signal stops. When this option is selected, the settings described below will also be displayed: Repeat spacing mode - Select how the distance between each print is defined. Select <i>Space by pitch</i> to define the distance between the start of one print and the start of the next print in mm. Or, select <i>Space by gap</i> to define the distance between the end of one print to the start of the next print in mm. Repeat spacing (mm) - Define the distance between prints. Repeat: Update each print - Tick to update clock and counter information each print. Or, Untick to keep the same information on each print. Repeat: End of product - Define when and how to stop continuous print. Select <i>Complete</i> to complete the next print. Select <i>Cancel</i> to stop before the next print.

(5) Select the Save icon to save the setting.



End of procedure.

Shift Codes

Shift codes is used to create a shift code table which can be inserted into the label design.

To create a shift code table:

- (1) In the label creator's side menu, select the *Label* tab.
- (2) Select the *Settings* menu.
- (3) Select *Shift codes*.
- (4) Select the *Add New* icon.
- (5) Select the *Name* setting and enter the name of the shift code table.
- (6) Select *View...*
- (7) Select *+Add row...*
- (8) Enter a *Start time* and a *Code* for the first shift.
- (9) To add more shifts select *+Add row...*
- (10) Select *Save* to confirm the shift codes.
- (11) Select the green *Tick* icon to save the shift code table.
- (12) To add the shift code table into the label design, see [page 4-46](#).

End of procedure.

Label Elements

The text, barcode and graphic items which make up label designs are known as elements.

The following pages explain how to add and edit different label elements.

Add a Text Element

To add a text element into a label design:

- (1) In the label creator's side menu select the *Element* tab.
- (2) Select the *Add* menu.
- (3) Select *Text*.
- (4) Enter the required text using the on screen keyboard.

Note: The keyboard type and language can be changed by selecting the appropriate icon at the bottom of the text entry screen.

- (5) Select the green *Tick* icon to add the text element to the label design.
- (6) Select the *Save* icon to save the label design.



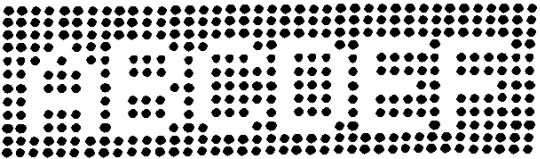
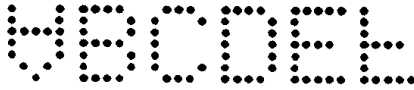
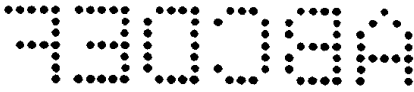
End of procedure.

Edit a Text Element

To edit a text element in the label design:

- (1) Select the text element that requires editing.
- (2) In the label creator's side menu select the *Element* tab.
- (3) Select the *Edit* menu.
- (4) The following settings can now be used to edit the text element:

Setting Name	Explanation
Name	Edit the name of the element.
Font type	<i>Tower</i> - Each character will be rotated 90° anti clockwise. <i>Fixed</i> - Each character will take up an equal amount of horizontal space. <i>Proportional</i> - Each character will take up the amount of horizontal space required for that specific character only.
Font	Change the height of the text in ink drops.
Bold	Print text in bold. Range: 0-3 Set to 1 in the example below.

Setting Name	Explanation
Inter-character gap (strokes)	Vary the gap between text characters. Range: 1-50 
Invert black/white	Invert the printed and unprinted parts of the text element. 
Invert	Invert the text element. 
Reverse	Reverse the text element. 
	Open the keyboard to edit the text. <i>Note: The keyboard type and language can be changed by selecting the appropriate icon at the bottom of the text entry screen.</i>
	Copy the text element.
	Delete the text element.

(5) Select the Save icon to save the change.



End of procedure.

Unicode

To add a Unicode character into a text element:

- (1) When editing or creating a new text element, select the *Unicode* icon in the text entry screen.



- (2) Enter the Unicode value:

Common Unicode Characters			
00A3	Pound (Sterling)	20AA	Shekel (Israel)
0024	Dollar (US)	20AB	Dong (Vietnam)
00A2	Cent (US)	20A2	Cruzeiro (Brazil)
00A5	Yen (Japan)	20A6	Naira (Nigeria)
20AC	Euro	20A8	Rupee
20A1	Colon (Costa Rica)	20A9	Won (South Korea)

- (3) Select the *Tick* icon in the top right of the screen to confirm the Unicode character.
- (4) Select the green *Tick* icon to enter the Unicode character into the text element.
- (5) Select the *Save* icon to save the label design.



End of procedure.

Input Method Editor (IME)

Entered in a similar manner to entering Unicode characters.

- (1) When editing or creating a new text element, select the *IME* icon in the text entry screen.



- (2) Enter the required characters.
- (3) Select the green *Tick* icon to enter the IME character into the text element.
- (4) Select the *Save* icon to save the label design.



End of procedure.

Add a Barcode Element

Notes: (1) The printer can currently print 10x10, 12x12, 14x14, 16x16, 18x18, 20x20, 22x22, 24x24, 26x26, 32x32, 8x18, 8x32, 12x26, 12x36, 16x36 and 16x48 data matrix symbols. The amount of data that can be contained within a data matrix symbol is dependent on the data matrix size.

(2) The printer applies a limit of 64 characters to unrestricted codes.

To add a barcode element into the label design:

- (1) In the label creator's side menu, select the *Element* tab.
- (2) Select the *Add* menu.
- (3) Select *Barcode*.
- (4) Select the barcode type.
- (5) Define any other barcode type specific settings.
- (6) Select *Add*.
- (7) Enter the required barcode data using the keyboard.

Note: The keyboard type can be changed by selecting the appropriate icon at the bottom of the data entry screen.

- (8) Select the green *Tick* icon to add the barcode data to the label design.
- (9) In the label design area, select the barcode.
- (10) In the side menu, select the *Element* tab.
- (11) Select the *Edit* menu.
- (12) The following barcode settings can now be defined:

Setting Name	Explanation
Name	Enter the name of the barcode element.
Type	Select the barcode type.
Height (drops)	Specify the height of the barcode in ink drops.
Module width	Specify the barcode module width. Range: 1-10
Show human readable code	Tick the <i>Show human readable code</i> tick box to display the barcode data in human readable text underneath the barcode.
Bearer bars	Tick the <i>Bearer bars</i> tick box to add bearer bars above and below the barcode.
Bearer box	Tick the <i>Bearer box</i> tick box to add a bearer box around the barcode.
Bar width ratios	Specify the width ratio between the widest and thinnest bar in the barcode.
Space width ratios	Specify the width ratio between the widest and thinnest space in the barcode.
Invert black/white	Invert the printed and unprinted parts of the barcode element.
Invert	Flip the barcode upside down.

Setting Name	Explanation
Reverse	Reverse the barcode.
Rotation	Rotate the barcode by 0° 90° 180° or 270°.
Check digit algorithm	If required, select a check digit algorithm. Default setting: None
Quiet zone: Left	Add an empty area to the left of the barcode.
Quiet zone: Right	Add an empty area to the right of the barcode.

(13) Select the Save icon to save the change.



End of procedure.

Add a Graphic Element

Notes: (1) To create and save an image file in the printer, see [page 4-51](#).

(2) To import and save an image file in the printer from a USB flash drive, see [page 4-52](#).

To insert a graphic element into the label:

- (1) In the label creator's side menu, select the *Element* tab.
- (2) Select the *Add* menu.
- (3) Select *Graphics*.
- (4) Select the required graphic from the folder.
- (5) Select the Save icon to save the change.



End of procedure.

Variable Label Elements

Variable label elements are elements which contain variable data such as clocks, counters and shift codes.

The following pages explain how to add and edit different variable label elements.

Add a Clock Element

To add a clock element into the label design:

- (1) In the Side-menu, select the *Element* tab.
- (2) Select the *Add* menu.
- (3) Select *Text*.
- (4) Select *+Variable* at the top of the screen.
- (5) Select *Insert new...*
- (6) Select *Clock*.
- (7) The following clock settings can now be defined:

Setting Name	Explanation
Format	Input the required date format: HH - Hour with leading zero, 24 hour clock (00-23) hh - Hour with leading zero, 12 hour clock (01-12) H - Hour code QQ - Quarter hour code (00-95) mm - Minute with leading zero (00-59) ss - Second with leading zero (00-59) AMPM - AM or PM AP - A or P DD - Day of the month with leading zero (01-31) JJJ - Julian day of the year (001-366) Y - Julian year of the decade (0-9) YY - 2 digit year (00-99) YYYY - 4 digit year (0000-9999) MM - Month number with leading zero (01-12) PGMM - Month name (January-December) M - Mandatory alpha character (upper case only (A-Z) Q - Quarter year code (five year cycle) QD - Day in the quarter PGDDD - Day name (Monday-Sunday) WW - Calendar week number (01-53) D - Day of the week (1-7) A7 - Day of week code A31 - Day of month code
Calendar	Select the calendar type: • Gregorian • Jalali • Hijri

Setting Name	Explanation
Language	Select the language: • Western • Arabic • Farsi
Offset	If required, an offset value can be set in years, months, weeks, days, hours and minutes.
Roll over time	This setting is a delay that adjusts time when the calendar date will roll over to the next day. Only a positive value can be set. For example, a value of 03:30:00 will retain the previous calendar date past midnight until 03:30.

(8) Select the green *Tick* icon to confirm the clock settings.

(9) Select the green *Tick* icon to add the clock element to the label design.

(10) Select the *Save* icon to save the change.



End of procedure.

Add a Counter Element

To add a counter element to the label design:

- (1) In the label creator's side menu, select the *Element* tab.
- (2) Select the *Add* menu.
- (3) Select *Text*.
- (4) Select *+Variable* at the top of the screen.
- (5) Select *Insert new...*
- (6) Select *Counter*.
- (7) The following counter settings can now be defined:

Setting Name	Explanation
Name	Enter the name of the counter element.
Format	Input the required counter format, for example: 0000 = Four mandatory numeric characters (0-9) LLLL = Four mandatory alpha character (A-Z or a-z)
Start value	Input the counter start value.
Repeat count	Enter the number of times the same counter value will be repeated.
Show additional properties	Tick the <i>Show additional properties</i> tick box to show more counter settings.
Step	Enter the number of steps and the direction that the counter will count in, for example: If the <i>Start value</i> is set to 00 and <i>Step</i> is set to 5, the counter will count upwards from 00 in the following sequence: 00, 05, 10, 15, etc. The counter will reset when it reaches the <i>Upper limit</i> value. If the <i>Start value</i> is set to 20 and <i>Step</i> is set to -5, the counter will count downwards from 20 in steps of 5 in the following sequence: 20, 15, 10, 05, etc. The counter will reset when it reaches the <i>Lower limit</i> value.
Lower limit	Enter the minimum counter value. When the counter reaches this value it will reset.
Upper limit	Enter the maximum counter value. When the counter reaches this value it will reset.
Leading zeros	Tick the <i>Leading zeros</i> tick box to add padding characters to the counter.
Padding character	If the <i>Leading zeros</i> tick box has been ticked, this setting is used to define the padding character that will be printed.

Setting Name	Explanation
Trigger	Specify what method will trigger the counter value to update: Product Detect - The counter value will update each time a product detect signal is received. Another Counter - The counter value will update when another selected counter element reaches its <i>Upper limit</i> or <i>Lower limit</i> value.
Language	Select the language: • Western • Arabic • Farsi
Store persistent counter value	Select when the counter value is saved: • never • on print start • on print complete

(8) Select the green *Tick* icon to confirm the counter settings.

(9) Select the green *Tick* icon to add the counter element to the label design.

(10) Select the *Save* icon to save the change.



End of procedure.

Add a Prompted Field Element

When a label containing a prompted field element is sent to print, the operator will be prompted to update data in the prompted field element before printing can start.

The type of data that can be entered into a prompted field element can be Time, Date or Text.

To add a prompted field element into the label design:

- (1) In the *Side* menu select the *Element* tab.
- (2) Select the *Add* menu.
- (3) Select *Text*.
- (4) Select *+Variable* at the top of the screen.
- (5) Select *Insert new...*
- (6) Select *Prompted field*.
- (7) The following prompted field settings can now be defined:

Setting Name	Explanation
Name	Enter the name of the prompted field element.
Type	Select the type of prompted field: None - The user will be prompted to enter standard text. Clock - The user will be prompted to enter a date or time.
Prompt	Enter a prompt message which will be displayed to the operator to prompt when the label is sent to print.
Input mask	If Type is set to None, enter the number and type of characters which the operator must input. If Type is set to Clock or Time Conditional, enter the format the operator must use to input the date or time.
Output format	Enter the format which the clock or time will be printed in. <i>Note: This setting is only valid if Type is set to Clock.</i>
Language	Select the clock or date language. <i>Note: This setting is only valid if Type is set to Clock.</i>

- (8) Select the green *Tick* icon to confirm the prompted field settings.
- (9) Select the green *Tick* icon to add the prompted field element to the label design.
- (10) Select the *Save* icon to save the change.



End of procedure.

Add a Link Element

A link element can be used to print data contained within a barcode as human readable text elsewhere in the same label design.

Note: A link element can only link to data from barcode elements.

To add a link element into the label design:

- (1) In the *Side* menu select the *Element* tab.
- (2) Select the *Add* menu.
- (3) Select *Text*.
- (4) Select *+Variable* at the top of the screen.
- (5) Select *Insert new...*
- (6) Select *Link*.
- (7) The following link settings can now be defined:

Setting Name	Explanation
Source ID	Select the name of the barcode element that will be linked.
Source length	Set the maximum length of the linked data.
Source offset	Add an offset to the linked data.

- (8) Select the green *Tick* icon to confirm the link settings.
- (9) Select the green *Tick* icon to add the link element to the label design.
- (10) Select the *Save* icon to save the change.



End of procedure.

Add a Script Element

Note: The Professional Printing Pack is required to use this feature.

Script elements written in Lua programming language can be imported into the printer using the file manager, see [page 6-8](#). Once imported, the script element can then be added into the label design.

To add a script element into the label design:

- (1) In the *Side* menu select the *Element* tab.
- (2) Select the *Add* menu.
- (3) Select *Text*.
- (4) Select *+Variable* at the top of the screen.
- (5) Select *Insert new...*
- (6) Select *Script*.
- (7) Select *Source*.
- (8) Navigate to the location of the script file and select it.
- (9) Select the green *Tick* icon to confirm the correct script file has been selected.
- (10) Select the *Placeholder* setting.
- (11) Enter text that will act as a place holder for the script element on the label design area and select the green *Tick* icon.
- (12) Select the green *Tick* icon to add the script element to the label design.
- (13) Select the *Save* icon to save the change.



End of procedure.

Add an External Data Element

An external data element is a field that allows data to be printed from an external source. Data can be transmitted to the printer via Ethernet TCP or Serial connection.

To add an external data element into the label design:

- (1) In the *Side* menu select the *Element* tab.
- (2) Select the *Add* menu.
- (3) Select *Text*.
- (4) Select *+Variable* at the top of the screen.
- (5) Select *Insert new...*
- (6) Select *External data*.
- (7) The external data settings can now be defined:

Setting Name	Explanation
Source	Select the external data communication source.
Length	Set the maximum length of the printed data. For example, if the data received by the printer is 01234 and the length is set to 3, the printed data will be 012.
Offset	The offset value specifies which part of the data will be printed. For example, if the data received by the printer is 01234, the length is set to 3 and the offset is set to 2, the printed data will be 234.
Delimited	Tick the delimited box to display the Delimiter and Index settings.
Delimiter	If data sent to the printer is made up of multiple data blocks, the delimiter is the character that separates each data block. The delimiter setting specifies the delimiter character used in the data stream that is sent to the printer. The delimiter can be any character.
Index	The index value selects a specific block in the data stream to print. For example, if the data is 123,456,789,132,425 and the index is set to 4, the 4th data block will be printed which is 132.
Example data	Displays an illustration of the format that the data will be printed in.

- (8) Select the green *Tick* icon to confirm the external data settings.
- (9) Select the green *Tick* icon to add the external data element to the label design.
- (10) Select the *Save* icon to save the change.



End of procedure.

Add a Shift Code Element

A shift code element can be used to print data from a shift code table.

Note: A shift code table must be created before a shift code element can be added to the label design. To create a shift code table, see [page 4-32](#).

To add a shift code element into the label design:

- (1) In the *Side* menu select the *Element* tab.
- (2) Select the *Add* menu.
- (3) Select *Text*.
- (4) Select *+Variable* at the top of the screen.
- (5) Select *Insert new...*
- (6) Select *Shift code*.
- (7) Select the *Shift-code table* drop down menu.
- (8) Select the required shift code table.
- (9) Select the green *Tick* icon to confirm the shift code table selection.
- (10) Select the green *Tick* icon to add the shift code element to the label design.
- (11) Select the *Save* icon to save the change.



End of procedure.

Lock Label Elements

Label elements can be locked to stop them from being accidentally edited or moved within the label design.

To lock a label element:

- (1) In the label creator's side menu, select the *Element* tab.
- (2) Select the *Manage* menu.
- (3) Select the element to be locked from the list of elements, or select the element in the label design area.
- (4) Select the Lock icon in the side menu to lock the selected element.
- (5) Select the Save icon to save the setting.



End of procedure.

Preview Label

Select *Preview* at the top of the Label Creator to view a preview image of the label design.

Open and Edit a Label

To edit an existing label:

- (1) From the *Home Screen*, select *Label finder*.
- (2) Open the label store.
- (3) Select *Edit* next to the label that requires editing.

The label will open in the Label Creator. To edit the content within the Label Creator:

- (4) Select an element to edit within the label design.
- (5) Edit the item using the available settings in the side menu.

End of procedure.

Copy a Label Element

To copy an element within the label design:

- (1) Select the element to copy.
- (2) In the side menu, select the *Element* tab.
- (3) Select the *Edit* menu.
- (4) Select the *Copy* icon.



- (5) Drag the new element to a suitable position in the label design area.
- (6) Select the *Save* icon to save the change.



End of procedure.

Copy Multiple Label Elements

To copy multiple elements within the label design:

- (1) In the side menu, select the *Element* tab.
- (2) Select the *Manage* menu.
- (3) Select the tick box next to the name of each element which will be copied.

Note: A red square will be displayed around each selected element in the label design area.

- (4) Select the *Element* tab.
- (5) Select the *Edit* menu.
- (6) Select the *Copy* icon.



- (7) Drag the new elements to suitable positions in the label design area.
- (8) Select the *Save* icon to save the change.



End of procedure.

Delete a Label Element

To delete an element within the label design:

- (1) Select the element to delete.
- (2) In the side menu, select the *Element* tab.
- (3) Select the *Edit* menu.
- (4) Select the *Delete* icon.



- (5) Select the *Save* icon to save the change.



End of procedure.

Delete Multiple Label Elements

To delete multiple label elements within the label design:

- (1) In the side menu, select the *Element* tab.
- (2) Select the *Manage* menu.
- (3) Select the tick box next to the name of each element which will be deleted.

Note: A red square will be displayed around each selected element in the label design area.

- (4) Select the *Element* tab.
- (5) Select the *Edit* menu.
- (6) Select the *Delete* icon.



- (7) Select the *Save* icon to save the change.



End of procedure.

Move a Label Element

To move an element within the label design:

- (1) Select the element to move.
- (2) Either:
 - (a) Select the *Drag* icon and drag the element to move it.



- (b) Select the *Arrows* icon and use the arrows to move the selected element precisely.



- (3) Select the *Save* icon to save the change.



End of procedure.

Zoom

Select the *Fit to Screen*, *Zoom In* or *Zoom Out* icon to change the magnification of the label area.



Fit to Screen



Zoom In



Zoom Out

Save Label

Once the label has been created:

- (1) Select the *Save as* icon.



- (2) Give the label a name and select a location.
- (3) Select the *Save* icon.



End of procedure.

IMAGES

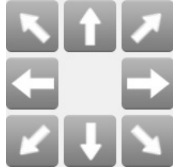
Create an Image

To create an image:

- (1) Select *Home > Setup > Tools*.
- (2) Use the *Width* and *Height* settings to specify the width and height of the design area.

Note: This will define the size occupied by the image even if the content is smaller than this size.

- (3) Use the *Arrow* icons to place the cursor in the required position.



- (4) Click on the *Paint Brush* icon and draw the logo by moving the brush using the *Arrow* icons.



- (5) Click on the *Eraser* icon to remove pixels where required.



- (6) Select the *Save As* icon to save the image to the Images folder.



- (7) Enter a filename for the logo in the *Name* text box.
- (8) Select *Save*.

End of procedure.

Edit an Image

To edit an image:

- (1) Select *Home > Setup > Tools*.
- (2) Select the *Folder* icon.



- (3) Select the image which requires editing.
- (4) Edit the logo as required.
- (5) Save the logo using the *Save* or *Save As* icons.



End of procedure.

Import an Image

To import an image into the printer from a USB flash drive:

- (1) Insert the USB flash drive containing the image files into the USB port on the TouchScreen.
- (2) Wait until the USB icon on the information bar turns green.



- (3) Select *Home > Setup > File Manager*.
- (4) Open the *USB* folder.
- (5) Open the folder that contains the image file.
- (6) Select the tick box next to the image to be imported.



- (7) Select either the *Copy* or *Cut* icon.



- (8) Navigate to and open the *Labels* folder.



- (9) Open the *IMAGE* folder.



(10) Select the *Paste* icon.



(11) Remove the USB flash drive from the TouchScreen.

End of procedure.

LABEL STORE AND FILE MANAGEMENT

Selecting an Existing Label

Note: Where no label is selected, "No label selected" will be displayed on the Home screen.

To open an existing label:

- (1) On the Home Screen, select *Label finder*.
- (2) Open the *Labels* folder.
- (3) If the label name or part of the label name is known, select *Filter labels...* and enter the label's name. If the label name is not known, swipe the screen vertically to scroll through the available the labels.

Note: Stored labels can be viewed by name, or by name and a preview image:



View labels by name.



View labels by name and preview image.



Refresh the label list.

- (4) When the required label has been found, select either *Print*, *Edit* or *Preview*.

End of procedure.

File Manager

The File Manager is a useful tool for reviewing, organising and editing stored labels, images, fonts, rasters, scripts and other file types. The File Manager can also copy files from USB memory devices.

To use the file manager:

- (1) Select *Home > Setup > File Manager*.
- (2) Open a folder to view the files saved in that folder.
- (3) Select the tick box next to a file to enable the settings listed below:
 - Paste
 - Copy
 - Cut
 - Rename
 - Detail
 - Delete

End of procedure.

Import a Label(s)

To import a label into the printer from a USB flash drive:

- (1) Insert the USB flash drive containing the label files into the USB port on the TouchScreen.
- (2) Wait until the USB icon on the information bar turns green.



- (3) Select *Home > Setup > File Manager*.
- (4) Open the *USB* folder.
- (5) Open the folder that contains the label file.
- (6) Select the tick box next to the label to be imported.



- (7) Select either the *Copy* or *Cut* icon.



- (8) Navigate to and open the *Labels* folder.



- (9) Select the *Paste* icon.



- (10) Remove the USB flash drive from the TouchScreen.

End of procedure.

Import Single Jet Labels into Duo Printers

Single jet labels can be imported into a Duo printer in the normal way as described above. However, when the label is opened in the Editor a prompt will appear to specify loading it in the upper, lower or both regions.

Export a Label(s)

To export a label from the printer to a USB flash drive:

- (1) Insert the USB flash drive into the USB port on the TouchScreen.
- (2) Wait until the USB icon on the information bar turns green.



- (3) Select *Home > Setup > File Manager*.
- (4) Open the *Labels* folder.

 \[DefaultStore]\Labels

Open

- (5) Select the tick box next to the label to be exported.

 ABCDE EXAMPLE LABEL.lbl



- (6) Select either the *Copy* or *Cut* icon.



- (7) Navigate to and open the USB folder.

 \USB

Open

- (8) Select the *Paste* icon.



- (9) Remove the USB flash drive from the TouchScreen.

End of procedure.

Delete a Label

To delete a label from the Labels folder:

- (1) Select *Home > Setup > File Manager*.
- (2) Open the *Labels* folder.



- (3) Select the tick box next to the label to be deleted.



- (4) Select the *Delete* icon.



End of procedure.

BACKUP PRINTER

There are 2 types of file information that can be backed up to a USB device.

A label backup will copy all of the labels in the printer's label store. A label backup can also be restored to other printers or retained to restore the original printer's labels.

A service backup will copy information that is useful to a service engineer such as log files, labels, screen shots, diagnostic data, and jet profile information. A service backup cannot be copied back to the printer as the files are for information only.

Create a Backup

To create a backup:

- (1) Insert a USB flash drive into the USB port on the TouchScreen.
- (2) Wait until the USB icon on the information bar turns green.



- (3) Select, *Home > Setup > Backup & Restore > Backup*.

Notes: (1) The "Select folder for backup" setting can be used to select a specific folder on the USB flash drive to save the backup file to.

- (2) *The "Optional reference string" setting can be used to enter information that will help identify the backup file after it is saved on the USB flash drive.*

- (4) Select the *Select backup type* drop down setting.
- (4) Select *Label* or *Service*.
- (5) Select *Backup*.
- (6) A prompt will display when the backup is complete, select *OK* to clear the prompt.
- (7) Remove the USB flash drive from the TouchScreen.

End of procedure.

Restore Labels from a Backup

Only label backup files can be restored to the printer.

Note: It is necessary to restart the printer as part of the restore procedure.

To restore labels from a backup file:

- (1) Insert the USB flash drive containing the backup file into the USB port on the TouchScreen.
- (2) Wait until the USB icon on the information bar turns green.



- (3) Select, *Home > Setup > Backup & Restore > Restore*.
- (4) Select the *Select backup file* setting.
- (5) Select the label backup file.
- (6) Select *Restore*.
- (7) A prompt will display when the restore is complete, select *OK* to clear the prompt.
- (8) Remove the USB flash drive from the TouchScreen.
- (9) Press and hold the standby button  for 2 seconds and restart the printer.

End of procedure.

EDITOR DEFAULTS

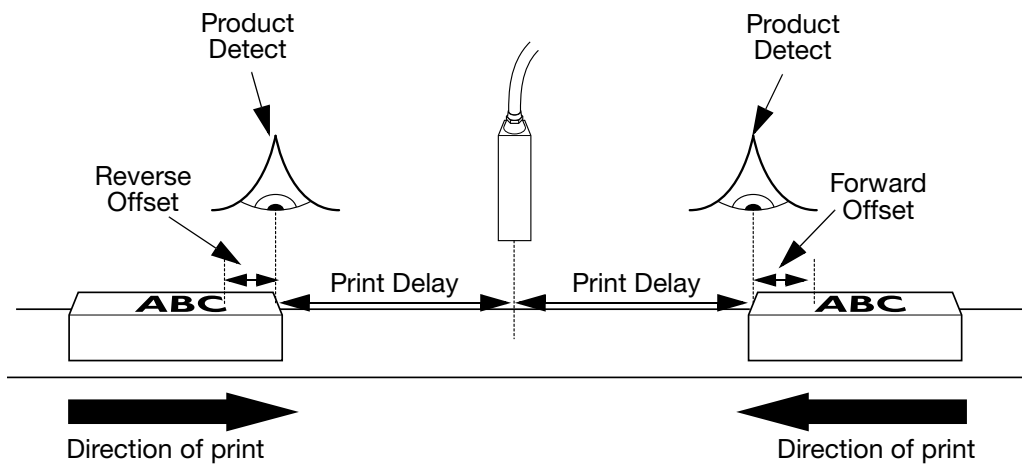
Set the default label settings. New labels will use these defaults.

Print Offset

Home > Setup > Editor Defaults > Properties

The Forward offset is used to adjust the position of the label on the print surface. If the print head is on a traversing line the Reverse offset can also be set.

Note: This is a global print setting, label designs can also be saved with their own individual print offset, see [page 4-26](#).



Print Offset Diagram

Repeat Print

Home > Setup > Editor Defaults > Properties

Labels can be repeat printed using one of two different methods. Or, repeat printing can be turned off as described in the table below.

Note: This is a global setting which will affect all labels sent to print. To configure an individual label to print repeatedly, see [page 4-30](#).

Repeat Method	Explanation
Off	Repeat printing will not be enabled.
Counted	Repeatedly print the same label for a specified number of times after a single pulsed print trigger signal is received. See page 4-62 . Typical application: Printing on lengths of extrusions of different specification each requiring a different label.
Continuous	Continuously print the same label when a constant print trigger signal is received by the printer. The printer will stop printing when the print trigger signal stops. See page 4-63 . Typical application: Printing on cable.

Counted

To set Counted repeat printing:

- (1) Select *Home > Setup > Editor Defaults > Properties*.
- (2) Configure the settings described in the table below:

Setting Name	Explanation
Repeat	Select <i>Counted</i> .
Repeat count	Enter the number of times to repeat the current label. <i>Note: The number of repeat counts is extra to the initial print, i.e. 6 will print 6 extra prints (7 in total).</i>
Repeat spacing type	Select how the distance between prints is defined: <i>Space by pitch</i> - The distance between the start of one print and the start of the next print defined in mm. <i>Space by gap</i> - The distance between the start of one print and the start of the next print defined using the gap between characters in mm.
Repeat Spacing (mm)	Set the distance from the start of one print to the start of the next.
Repeat: Update each print	Tick the <i>Update each print</i> tick box to update any clock and counter information contained in the label with each print. Alternatively, Untick the <i>Update each print</i> tick box to keep the same information on each print.
Repeat: End of product	Use the print trigger signal to set the behaviour of repeat printing: <i>Ignore</i> - End of detected product will have no effect on repeat printing. <i>Complete</i> - If the end of product is detected between prints, one more repeat label will be printed. If the end of product is detected while printing a repeat, the print will complete and no further repeats will be printed. <i>Cancel</i> - If the end of product is detected between prints, no further repeats will be printed (and the pending print will be cancelled). If end of product is detected while printing a repeat, the print will complete.

End of procedure.

Continuous

To set Continuous Repeat Printing:

- (1) Select *Home > Setup > Editor Defaults > Properties*.
- (2) Configure the settings described in the table below:

Setting Name	Explanation
Repeat	Select <i>Continuous</i> .
Repeat Spacing Type	Select how the distance between prints is defined: <i>Space by pitch</i> - The distance between the start of one print and the start of the next print defined in mm. <i>Space by gap</i> - The distance between the start of one print and the start of the next print defined using the gap between characters in mm.
Repeat Spacing (mm)	Set the distance from the start of one print to the start of the next.
Repeat: Update each print	Tick the <i>Update each print</i> tick box to update any clock and counter information contained in the label with each print. Alternatively, untick the <i>Update each print</i> tick box to keep the same information on each print.
Repeat: End of product	Use the print trigger signal to set the behaviour of repeat printing: <i>Ignore</i> - End of detected product will have no effect on repeat printing. <i>Complete</i> - If end of product is detected between prints, one more repeat label will be printed. If the end of product is detected while printing a repeat, the print will complete and no further repeats will be printed. <i>Cancel</i> - If the end of product is detected between prints, no further repeats will be printed (and the pending print will be cancelled). If end of product is detected while printing a repeat, the print will complete.

End of procedure.

Default Font Size

To set the default font size for label text elements:

- (1) Select *Home > Setup > Editor Defaults > Properties*.
- (2) Select the *Default font size for text elements (drops)* drop down setting.
- (3) Select the a new default font size option.

End of procedure.

Prevent Duplicate Prompts

If a label containing a prompted field is sent to print on a printer equipped with a duo print head, the prompt will be displayed for each ink jet. This setting will stop the prompt being displayed twice when the setting is enabled.

To enable or disable duplicate prompts:

- (1) Select *Home > Setup > Editor Defaults > Properties*.
- (2) Select or deselect the *Prevent duplicate prompts for prompted fields* tick box.

End of procedure.

Clocks/Dates

Home > Setup > Editor defaults > Clocks / dates

View and change:

- Hour codes.
- Day of week codes.
- Day of month codes.
- Month codes.
- Day names.
- Month names.

REGIONAL SETTINGS

Set the Language and Keyboard

Home > Setup > Regional > Language & keyboard

Set:

- Language
- Keyboard layout
- IME Scheme
- Primary currency.

Set the Master Clock

Home > Setup > Regional > Date & time

Note: The correct System Date and System Time values must be entered to print accurate date or time elements within label designs.

Set:

- System date
- System time (24 hour clock).

GLOBAL PRINT SETTINGS

Position / Orientation

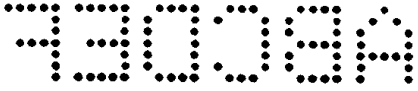
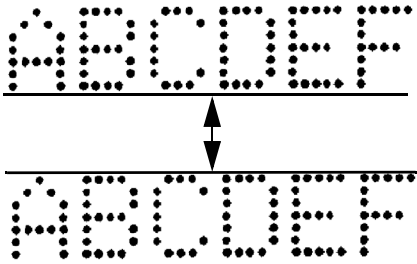
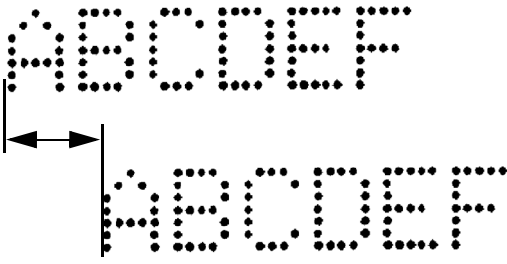
Notes: (1) Everyday adjustments to print height, print delay, invert and reverse should be made on the Print Optimisation screen, see [page 4-10](#).

(2) These are global settings which will affect all label designs when printed. The Invert and Reverse settings can also be configured and saved with individual label designs, see [page 4-26](#).

To adjust the label print position and orientation:

- (1) Select *Home > Setup > Global print settings > Position / orientation*.
- (2) View and change the settings listed below:

Setting Name	Explanation
Calibrate	Use the print height calibration wizard to set the print height, see page 3-56 .
Print delay (mm)	Set the distance between a product sensor detecting a product and the label being printed.
Internal stroke rate (mm/s)	Note: This setting is only available if the Encoder input is set to internal, see page 3-35. If an internal encoder is used to set the print speed this setting will define the print speed in millimetres per second. This setting is the same as the Speed (mm/s) setting on the Line movement screen, see page 3-35.
Print height (%)	Notes: (1) Jet 1 and Jet 2 settings are available for the Duo Print Head only. (2) Adjustment of this setting on a printer equipped with a Duo Print Head will change the accuracy of the Jet Gap (mm) setting. Adjust the print height percentage. 
Invert	Invert the label. 

Setting Name	Explanation
Reverse	Reverse the label. 
Stroke pitch (mm/stroke)	Set the distance between print strokes. A stroke is the line of ink drops which is used to make up each printed ink character.
Jet gap (mm)	Notes: (1) For i-Pulse Duo Print Head only. (2) If the Print Height (%) is also adjusted, the accuracy of the Jet Gap value will be effected. Adjust the vertical gap between the two ink jet printing areas. 
Region offset	Note: For i-Pulse Duo Print Head only. Adjust the horizontal offset between the two ink jet printing areas. 

End of procedure.

STATUSES, ALERTS AND FAULT FINDING

Statuses

Shown below are some printer statuses and their explanations. Printer statuses are displayed on the Status tab at the top of the QuickStep interface.

Status Name	Explanation
Idle	The ink jet is off, and the printer will not print, but the user interface can still be used.
Sequencing On	The printer is getting ready to print. Once completed, the printer will be in a Ready state.
Printhead warming up	The printer is heating the print head to the optimum temperature for the ink.
Ready	The printer is ready to print. If a label is on-line, it will print on receiving a product detect signal.
Sequencing Off	The printer is changing from the Ready state to the Idle state.
Gutter clearing in progress	The print head gutter is being cleared to stop ink from drying and blocking the gutter.

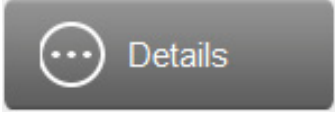
Alerts

Alerts are displayed on the Status tab at the top of the QuickStep interface. When multiple alerts occur only the highest alert will be displayed. If multiple alerts occur, the full list of alerts can be viewed by selecting the status tab.

Common alert ID codes, their causes and remedies are listed on [page 4-70](#).

To clear an alert, or view information about how to solve the alert:

- (1) Select the *Status* tab.
- (2) The icons illustrated below will become available to select.

Icon	Explanation
	Select the <i>Details</i> icon to display the following information: • Date and time the alert occurred. • Alert ID number. • Alert level. • Detailed information about the alert. • A solve icon which can be pressed to display recommendations for fixing the problem. • A QR code which can be scanned by a mobile phone or tablet device to show a video or further details about how to solve the alert. • A clear button to acknowledge and clear the alert.
	Select this icon to acknowledge and clear the alert. If this icon is greyed out, then an action must be performed before the alert can be cleared.

End of procedure.

Alert ID Codes

The table below describes some alerts which the printer might encounter. The table also lists possible causes and remedies.

If an alert occurs that is not listed in the table:

- Select the *Solve* icon to view options for solving the alert.
- Scan the QR code with a mobile phone or tablet device to show a video or further details about how to solve the alert.
- Contact the local support office for further help.

Alert ID	Description	Possible Cause and Remedy
10	Print Speed too high - Print max speed limit in force	Incorrect print configuration for the production line speed. There are three solutions available: • Reduce the production line speed. • Increase the <i>Stroke pitch (mm/stroke)</i> value, see page 4-25. • Reduce the print quality to allow the label to be printed faster, see page 4-25.

Alert ID	Description	Possible Cause and Remedy
15	Deflection EHT has tripped.	Spark across the EHT plates. Run the print height calibration wizard and slightly reduce the <i>EHT (%)</i> value, see page 3-56. <i>Note: The EHT (%) value must only be adjusted in small steps.</i>
		Ink build up on charge electrode. Clean the print head: • i-Pulse print head, see page 5-19. • i-Pulse Duo print head, see page 5-23. • i-Pulse RS print head, see page 5-27.
		Partially blocked nozzle. Run the nozzle unblocking wizard, see page 5-17.
		Ink jet clipping the gutter. Check the print height calibration, page 3-56. Also, check the ink jet alignment: • i-Pulse print head, see page 5-22. • i-Pulse Duo print head, see page 5-26. • i-Pulse RS print head, see page 5-32.
		Excessive print height. Run the print height calibration wizard, see page 3-56.
20	Ink Level Below Minimum	Ink leak at nozzle. Check for ink drops around the nozzle and call an engineer certified by Domino to repair or replace the nozzle.
		The ink level is running low. Fit a new ink cartridge, see page 5-4.

Alert ID	Description	Possible Cause and Remedy
24	Gutter Dry at Start-up	The printer was not shut down correctly. Clean the print head: • i-Pulse print head, see page 5-19. • i-Pulse Duo print head, see page 5-23. • i-Pulse RS print head, see page 5-27. And restart the printer.
		Blocked nozzle. Run the nozzle unblocking wizard, see page 5-17.
		Dried ink in the gutter. Clean the print head: • i-Pulse print head, see page 5-19. • i-Pulse Duo print head, see page 5-23. • i-Pulse RS print head, see page 5-27. And restart the printer.
		Ink jet incorrectly aligned. Check the ink jet alignment: • i-Pulse print head, see page 5-22. • i-Pulse Duo print head, see page 5-26. • i-Pulse RS print head, see page 5-32.
25	Add make-up cartridge	A new make-up cartridge is required. Fit a new make-up cartridge, see page 5-4.

Alert ID	Description	Possible Cause and Remedy
41	Ink level too high	The ink viscosity has increased, and extra make-up has been added to the ink causing the level to increase. Replace the ITM, see page 5-10. Or; Remove the ITM, empty some of the ink into a suitable container and replace the ITM, see page 3-49. Follow local waste disposal regulations to correctly dispose of waste ink. Frequent flushing has occurred, either due to repeated start/stop, repeated flushing or cleaning of the print head. Avoid frequent start/stop and shut down the printer before cleaning the print head. The printer has been moved to a different location with the ink and make-up cartridges fitted. Shut down the printer and remove the ink and make-up cartridges before moving the printer.
53	Make-up Level Below Minimum	The make-up level is running low. Fit a new make-up cartridge, see page 5-4.
134	Unexpected shutdown	The printer was not powered off using the correct method. Follow the correct shut down procedure, see page 4-20.
139	Print trigger occurred while printing	The printer received a false trigger signal before the label finished printing. Increase the persistence level for the print trigger, see page 3-50. The product sensor is faulty. Replace the product sensor. Label too long for the product. Check the printed label and reduce the label length.
213	Inspection Required	An inspection is required by an engineer certified by Domino. Contact the local support office to organise an inspection.

Alert ID	Description	Possible Cause and Remedy
234	Gutter Stall	The operator has cleaned the print head whilst the printer is running. This has caused excess solvent to enter the gutter causing it to stall. The printer will self recover from this fault. Follow the correct print head cleaning procedure: • i-Pulse print head, see page 5-19. • i-Pulse Duo print head, see page 5-23. • i-Pulse RS print head, see page 5-27.
		The gutter has stalled. Restart the printer following the correct procedure, see page 4-19. If the fault reoccurs, switch off the printer and contact the local support office.
235	Gutter Pump not Rotating	The gutter pump has stopped. Restart the printer following the correct procedure, see page 4-19. If the fault reoccurs, switch off the printer and contact the local support office.
236	Gutter Blocked Attempting to Clear	The printer is attempting to clear an obstruction in the gutter.

Alert ID	Description	Possible Cause and Remedy
237	Gutter Blocked	Dried ink in the gutter. Clean the print head: • i-Pulse print head, see page 5-19. • i-Pulse Duo print head, see page 5-23. • i-Pulse RS print head, see page 5-27. And restart the printer.
		Incorrect shut down procedure followed. Clean the print head: • i-Pulse print head, see page 5-19. • i-Pulse Duo print head, see page 5-23. • i-Pulse RS print head, see page 5-27. And restart the printer.
		Blocked nozzle. Run the nozzle unblocking wizard, see page 5-17.
		Ink jet incorrectly aligned. Check the ink jet alignment: • i-Pulse print head, see page 5-22. • i-Pulse Duo print head, see page 5-26. • i-Pulse RS print head, see page 5-32.
238	Insufficient gutter vacuum - Retrying	The printer is attempting to increase the vacuum within the gutter. Follow any instructions given on the TouchScreen.
517	ITM life expired (currently in grace period)	The ITM has reached the end of its life. The printer will continue to operate for a 72 hour grace period. If the ink jet is sequenced off during the grace period a password is required to continue operation. • Replace the ITM, see page 5-10.
520	Change ITM - ITM Ink Type Incorrect	The ITM which has been fitted is the wrong type. Shut down the printer, remove the ITM and replace it with the correct type, see page 5-10.
521	Change ITM - ITM Tag Dead	The ITM has reached the end of its life and can no longer be used. Replace the ITM, see page 5-10.

Alert ID	Description	Possible Cause and Remedy
523	Change ITM - Expired	The ITM has reached the end of its life and can no longer be used. Replace the ITM, see page 5-10.
524	Insert ITM - No Valid ITM present	No ITM fitted.
		Fit an ITM, see page 5-10 .
		ITM not fitted correctly Check that the ITM is correctly fitted in the printer cabinet, see page 5-10.
528	Add ink cartridge	The RFID tag on the ITM is not working correctly.
		Replace the ITM, see page 5-10 .
528	Add ink cartridge	A new ink cartridge is required. Fit a new ink cartridge, see page 5-4.
530	Change Ink Cartridge - Ink Type Incorrect	The printer has detected that the ink cartridge fitted (or offered up to the QMM) is not the correct type. If the cartridge has been fitted, immediately switch off the printer and contact the local support office.
531	Change Ink Cartridge - Ink Cartridge Expired	The ink cartridge has expired, and a new cartridge is required. Fit a new ink cartridge, see page 5-4.
532	Change Ink Cartridge - Empty	The ink cartridge is empty. Fit a new ink cartridge, see page 5-4.
534	Insert Ink Cartridge - No Valid Ink Cartridge Present	No ink cartridge fitted.
		Fit an ink cartridge, see page 5-4 .
		Ink cartridge not fitted correctly Check that the ink cartridge is correctly fitted in the printer cabinet, see page 5-4.
539	Insert Make-up Cartridge	The RFID tag on the ink cartridge is not working correctly.
		Replace the ink cartridge, see page 5-4 .
539	Insert Make-up Cartridge	A new make-up cartridge is required. Fit a new make-up cartridge, see page 5-4.

Alert ID	Description	Possible Cause and Remedy
540	Make-up low	The make-up level in the make-up module (MUM) is low. Fit a new make-up cartridge, see page 5-4.
542	Change Make-up Cartridge - Cartridge Empty	A new make-up cartridge is required. Fit a new make-up cartridge, see page 5-4.
543	Change Make-up Cartridge - Cartridge Dead	The make-up cartridge has reached the end of its life and can no longer be used. Fit a new make-up cartridge, see page 5-4.
544	Insert Make-up Cartridge - No valid Cartridge present	No make-up cartridge fitted. Fit a make-up cartridge, see page 5-4.
		Make-up cartridge not fitted correctly Check that the make-up cartridge is correctly fitted in the printer cabinet, see page 5-4.
		The RFID tag on the make-up cartridge is not working correctly. Replace the make-up cartridge, see page 5-4.
556	Print head warming up	The print head is warming up. If this status persists, contact the local support office.
557	Misaligned jet at start-up	The gutter is blocked. Clean the print head: • i-Pulse print head, see page 5-19. • i-Pulse Duo print head, see page 5-23. • i-Pulse RS print head, see page 5-27.
		The nozzle is blocked. Run the nozzle unblocking wizard, see page 5-17.
		The ink jet is misaligned. Check the ink jet: • i-Pulse print head, see page 5-22. • i-Pulse Duo print head, see page 5-26. • i-Pulse RS print head, see page 5-32.

Alert ID	Description	Possible Cause and Remedy
561	The label print height is out of range	The <i>Height (%)</i> setting in the label has been set too high. The printer will not print at the value that has been set but will print at the highest possible value. Reduce the <i>Height (%)</i> to clear the alert, see page 4-25.
572	The printed label is too long, speed too high or the shaft encoder has stopped mid-label.	Labels are being printed too close together meaning there is not enough phasing time. There are two solutions are available: • Increase the gap between printed labels. • Lower the production line speed.
1017	Print delay too short	The product detect persistence is set too long. Reduce the product detect persistence, see page 3-49.
		The product sensor and print head are too close together. Physically increase the distance between the product sensor and print head, then reset the print trigger settings, page 3-49.
		The print delay setting is too short. Increase the print delay setting, see page 3-49.

Alert ID	Description	Possible Cause and Remedy
1312	Charge Detection Error	The printer cannot detect ink drops within the charge electrode due to a build of ink of the charge electrode. Clean the print head: • i-Pulse print head, see page 5-19. • i-Pulse Duo print head, see page 5-23. • i-Pulse RS print head, see page 5-27. And restart the printer.
		The ink jet is not modulating correctly. Check the ink jet modulation and break-up within the charge electrode. If this is not visible contact the local support office.
		Ink jet incorrectly aligned. Check the ink jet alignment: • i-Pulse print head, see page 5-22. • i-Pulse Duo print head, see page 5-26. • i-Pulse RS print head, see page 5-32.
1313	Recovered Successfully	The printer has recovered from a previous fault or alert, no user action should be required. If this alert repeats and the print quality is reduced, contact the local support office.
1370	Print head cover removed	The i-Pulse RS print head contains a sensor to detect when the print head cover is removed. Replace the print head cover to clear the alert and resume printing.

Fault Finding

Issue	Possible Cause	Remedy
Not Printing, error Gutter Dry is displayed	Blocked Nozzle.	<i>Home > Setup > Wizards > Nozzle unblocking.</i>
Ink on deflector plates and/or charge electrode, possible faults could be: • Ink detected on Charge Electrode. • Charge detection has failed. • Fall back jet modulation in use. • Deflection EHT has tripped.	Blocked Nozzle or dirty print head.	Clean the print head: • i-Pulse, see page 5-19. • i-Pulse Duo, see page 5-23. • i-Pulse RS, see page 5-27.
No print occurs but message has been sent.	Faulty sensor.	Check the sensor and its position.
	Faulty encoder.	Check the encoder is outputting signals (can be seen via the Ext I/ F screen).
	Incorrect setting of sensor.	Check that the active level is set correctly.
	Print delay and/or offset set incorrectly.	Check the delay and offset for this message are suitable, adjust as required.
	Enable/Disable (Send to Print) button has been pressed when in the Home screen	This button will toggle printing within the Home screen, so only necessary action is to press button again.

PART 5 : MAINTENANCE

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GENERAL MAINTENANCE



WARNINGS:

- (1) Protective equipment such as gloves and glasses must be worn when working on or near the printer. Physical contact with ink or make-up can cause skin or eye damage.
- (2) If the printer is ever operated in a way that allows it to print into a beaker, the beaker must be made of conducting material and be securely connected to earth (ground), as the electrostatic charges on the ink drops used for printing can cause a fire hazard.
- (3) Do not smoke or allow naked flames (or other sources of ignition) in the vicinity of any inks or solvents as this is highly dangerous.

Ink and Make-up Cartridge Replacement



WARNING:

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.

CAUTIONS:

- (1) *If the make-up cartridge is not replaced when needed, the make-up module will empty, and the ink viscosity will go outside the printer's operating limits.*
- (2) *If the make-up cartridge is not replaced when needed, head flushing will not be carried out when the printer is shut down. Ink may be deposited on the print head components and dry causing a blockage.*
- (3) *Do not replace a cartridge until prompted. Risk of ink leak. If the make-up or ink cartridge is replaced when it is not needed, the ink level in the ITM could overfill. This will cause ink to leak out of the printer cabinet. Only replace the ink or make-up cartridge when prompted by the user interface or status lights.*

Notes: (1) The used ink and make-up cartridges contain residual chemicals which are hazardous to the environment. Local waste disposal regulations must be followed to safely dispose of the used cartridges.

(2) Shake heavy duty ink cartridges for at least two minutes before fitting.

When the ink or make-up cartridges need to be replaced, the messages 'Add Ink Cartridge' or 'Add Make-up Cartridge' will appear on the TouchScreen status tab. If these messages are ignored, the messages 'Ink Level Below Minimum' and 'Make-up Level Below Minimum' will appear. Also, the amber alert light and the ink or make-up level alert light on the printer cabinet will illuminate.

When the ink or make-up cartridges require replacing immediately, the messages 'Change Ink Cartridge - Empty' or 'Change Make-up Cartridge - Cartridge Empty' will appear on the TouchScreen status tab. Also, the red alert light will illuminate and the ink or make-up light on the printer cabinet will start flashing. If the ink or make-up cartridge is not replaced at this stage, the printer will stop working and will not continue printing until a new cartridge is fitted.

Tools required: 6mm Hex Key.

To replace the ink or make-up cartridge:

- (1) Open the access door to the printer's ink compartment.
- (2) Remove the old cartridges by rotating the ink cartridge anti-clockwise and the make-up cartridge clockwise to free them before lifting the cartridge away.



Removal of Ink and Make-up Cartridges

- (3) Before breaking the tab on the new cartridge and inserting it, hold the cartridge near the Quality Management Module (QMM) to check that the ink or make-up type is correct. The lights on the QMM will flash amber to indicate that the RFID tags are being read. When the RFID tags have successfully been read and validated, the lights will turn green. See [“QMM \(Quality Management Module\) Status Lights”](#) on page 2-20.

Notes: (1) If a fault is detected, the QMM lights will turn red and an alert will be displayed on the Status tab.

- (2) If the RFID tag cannot be read, or an RFID tag is not present, the QMM lights will turn solid amber.

- (4) Insert a 6mm hex key into the top of the new cartridge, twist to break the sealing tab and remove the sealing tab.



*Breaking the Make-up
Cartridge Sealing Tab*



*Breaking the Ink Cartridge
Sealing Tab*

- (5) Push the ink cartridge onto the ITM or push the make-up cartridge onto the make-up module.



*Make-up Cartridge
Replacement*



*Ink Cartridge
Replacement*

MAINTENANCE

- (6) Rotate the ink cartridge clockwise and the make-up cartridge anti-clockwise. Ensure that the label is facing towards you.



Ink and Make-up Cartridge Replacement

- (7) Check for leaks inside the printer.
 - (8) Close the access door to the printer ink compartment.
 - (9) Dispose of empty cartridges by following local waste disposal regulations.
- End of procedure.

Make-Up Module Filter Replacement



WARNING:

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.

CAUTIONS:

- (1) Cleanliness is of extreme importance. Ensure debris does not enter the make-up module during this process or the printer may become damaged. Observe good cleanliness procedures at all times.
- (2) The used make-up module filter contains residual chemicals which are hazardous to the environment. Local waste disposal regulations must be followed to safely dispose of the filter.

Tools required: 6mm hex key.

The make-up filter is situated in the make-up module. As a user replaceable part, it is coloured yellow.

To replace the make-up filter:

- (1) If the printer is on, press and hold the standby button  for 2 seconds and wait for the printer to shut down.
- (2) Open the access door to the printer ink compartment.
- (3) If a make-up cartridge is attached to the make-up module, remove it by twisting the cartridge in a clockwise direction and lifting it out of the printer (refer to [“Ink and Make-up Cartridge Replacement” on page 5-4](#)).
- (4) Remove the old make-up filter from the module by inserting a 6mm hex key into the yellow filter top and unscrewing the filter from the module.



Make-Up Module Filter Replacement

MAINTENANCE

- (5) Use the 6mm hex key to screw the new make-up filter into the make-up module.
 - (6) Replace the make-up cartridge (if fitted).
 - (7) Press and hold the standby button  for 2 seconds and wait for the printer to start up.
 - (8) Check inside the printer cabinet for leaks.
 - (9) Shut the ink compartment access door.
 - (10) On the TouchScreen, select *Home > Setup > Wizards > Ink priming*.
 - (11) Select *Start* and follow the on screen instructions to prime the ink system.
 - (12) Dispose of the old make-up filter by following local waste disposal regulations.
- End of procedure.

ITM Replacement



- WARNINGS:**
- (1) The printer must be shut down and the power cable must be disconnected before the ITM can be removed. The ink system is pressurised. If the printer is not shut down ink will spray out of the ITM manifold over the person removing the ITM.
 - (2) Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.
 - (3) Pressurised ink system. Risk of ink spray. When replacing the ITM, ensure the retaining clips are fully engaged.

CAUTION: *Cleanliness is of extreme importance. Ensure debris does not enter the new ITM or ink block during this process or the printer may become damaged. Observe good cleanliness procedures at all times.*

- Notes:**
- (1) Paper towels (or similar) and wash are required for this procedure.
 - (2) Waste paper towels will be contaminated with chemicals which are hazardous to the environment. Local waste disposal regulations must be followed to safely dispose of used paper towels.
 - (3) The used ITM contains residual chemicals which are hazardous to the environment. Local waste disposal regulations must be followed to safely dispose of the used ITM.

The main ink and gutter filters are situated in the ITM and are an integral part of the printer's ink system. Replacement of these filters therefore occurs automatically during scheduled ITM replacement.

The messages, 'The ITM life is soon to expire. Ensure a replacement ITM is available' and 'Intelligent Ink System is in depletion mode to prepare for ITM change' will appear near the end of the ITM life. Printing will not continue beyond the expiry of the ITM life.

The ink system will automatically run down the level of ink in the ink cartridge to a minimum level, so it is usual to replace the ink cartridge at the same time.

To replace the ITM:

- (1) If the printer is on, press and hold the standby button  for 2 seconds and wait for the printer to shut down.
- (2) Disconnect the power lead from the rear of the printer.
- (3) Open the access door to the printer ink compartment.
- (4) If an ink cartridge is attached to the ITM, remove it by twisting the cartridge anti-clockwise and lifting it out of the printer.

MAINTENANCE

- (5) Pull the two ITM retaining clips upwards away from the ITM.



Removing the ITM

- (6) Pull the ITM backwards, withdrawing the connecting manifold from the ink block and remove the old ITM.

Notes: (1) If the ITM does not withdraw with moderate pressure, push the ITM fully forward (which will break any seal caused by dry ink) and try again.

(2) The manifold pipes will contain residual ink. Take care when removing the ITM to avoid spillage.

- (7) Unpack the new ITM and remove the sealing strip that protects the ITM manifold.



Remove ITM Sealing Strip

- (8) Re-use the sealing strip to seal the manifold on the old ITM.

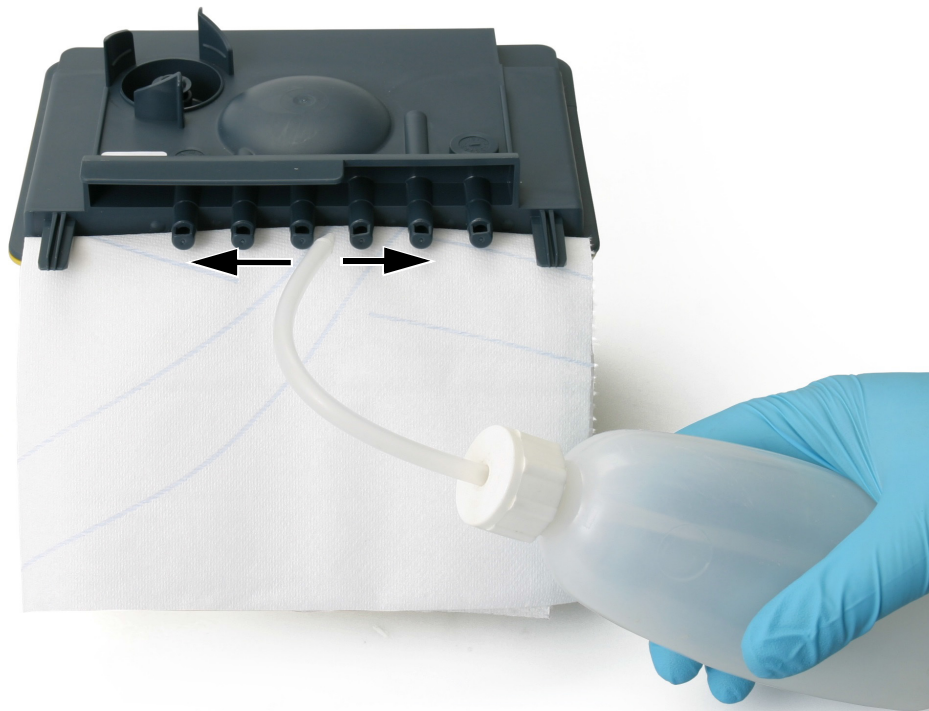
MAINTENANCE

- (9) Place paper towel or similar on top of the level sensor modules to catch excess fluid and using wash, remove any dried residual ink from the ink block valve face.



Washing the ink block valve face

- (10) Place paper towel or similar under the ITM manifold pipes and lubricate the pipes with wash.



Washing the ITM manifold pipes

(11) Insert the ITM between the retaining clips and push the manifold pipes into the ink block.



Inserting the ITM



WARNING:

Pressurised ink system. Risk of ink spray. When replacing the ITM, ensure the retaining clips are fully engaged.

(12) Continue pushing until the retaining clips engage with a click.



Retaining Clip Engaged



Retaining Clip Not Engaged

MAINTENANCE

(13) Re-connect the power lead to the rear of the printer.

(14) Press and hold the standby button  for 2 seconds and wait for the printer to start.

(15) Attach a new ink cartridge to the ITM or re-use the existing ink cartridge. See, [“Ink and Make-up Cartridge Replacement” on page 5-4.](#)

(16) Check the ink compartment for leaks.

(17) The status lights on the Quality Management Module (QMM) will flash amber to indicate that the RFID tags on the ITM, Ink Cartridge and Make-up Cartridge are being read. When the RFID tags have successfully been read and validated, the status lights will turn green.

Notes: (1) If a fault is detected, the QMM status lights will turn red and an alert will be displayed on the Status tab.

(2) If the RFID tag cannot be read, or an RFID tag is not present, the QMM status lights will turn solid amber.



Quality Management Module Status Lights

(18) Shut the ink compartment access door.

(19) Dispose of contaminated paper towels and the old ITM by following local waste disposal regulations.

End of procedure.

Air Filter Replacement

**WARNING:**

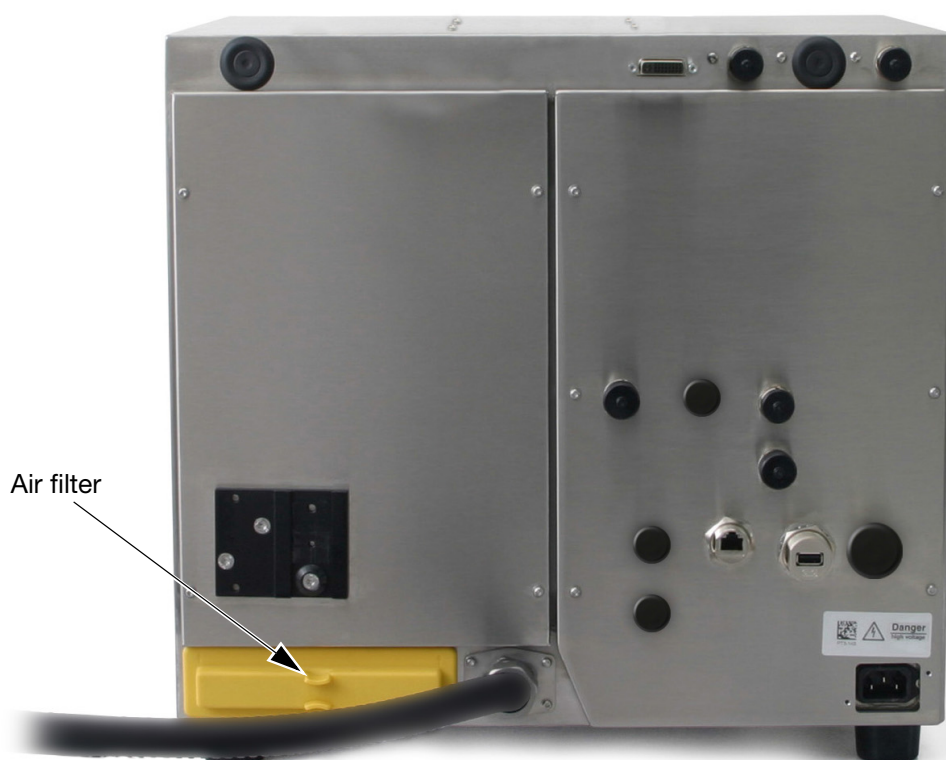
Protective equipment such as gloves and glasses must be worn when this procedure is carried out. The air filter may contain hazardous particles.

Air filter replacement is recommended after every 2000 hours of operation but may need to be replaced sooner depending on the operating environment.

Note: The air filter is not designed to be cleaned, it must be replaced.

A new air filter can be ordered through www.buydomino.com

The air filter is located in a slot at the back of the printer. As a user replaceable part, it is coloured yellow.



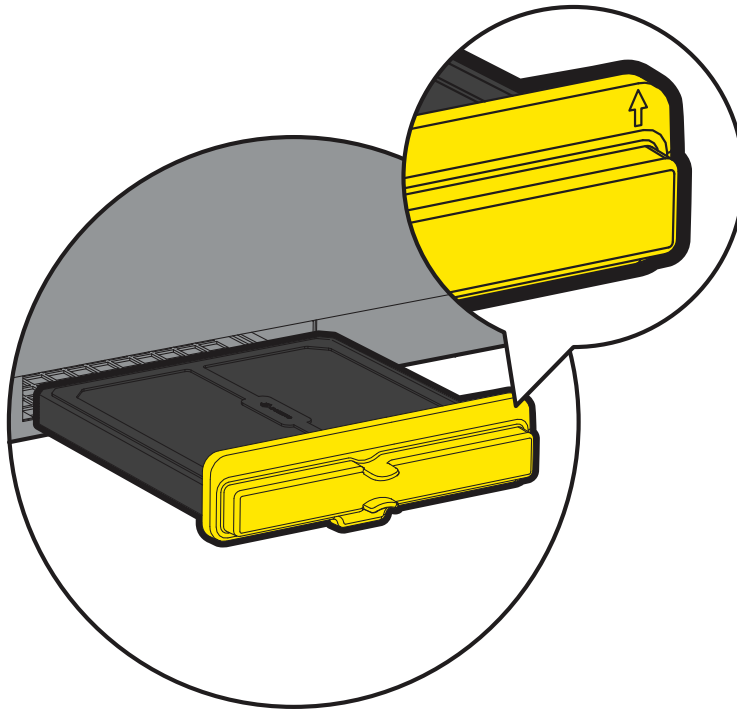
Air Filter Location

To replace the air filter:

- (1) If the printer is on, press and hold the standby button  for 2 seconds and wait for the printer to shut down.
- (2) Remove the air filter by holding the yellow exposed part of the filter and pulling it gently out of the filter housing.

- (3) Replace the air filter with a new unit.

Note: Ensure the new air filter is correctly orientated by confirming the arrow symbols on the yellow filter moulding are pointing up. If the filter is inserted upside-down it will get stuck in the printer cabinet.



Filter Orientation

- (4) Press and hold the standby button  for 2 seconds to start up the printer.
- (5) Select *Home > Setup > Adjustment > Inspection > Add...*
- (6) Select the *Replaced part* drop down menu and select *Air filter*.
- (7) Select *Operator name* and enter your name.
- (8) Select *Notes* and enter any notes which may be useful.
- (9) Select *Save*.

End of procedure.

Clearing a Blocked Nozzle

**WARNING:**

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.

CAUTIONS:

- (1) *Do not run the nozzle unblocking wizard more than 2 times. Each time the wizard is run, the printer adds make-up to the ink. If the wizard is run too many times the ink viscosity will thin and go out of the printer's operating range. If the nozzle is not clear after 2 attempts, contact the local support office to organise a visit from an engineer certified by Domino.*
- (2) *Unless trained, do not attempt to remove the nozzle assembly. Removing the nozzle assembly can cause contamination to enter the ink system. The nozzle assembly is also tuned to the print head and will require re-tuning if removed.*

Clear a blocked nozzle as follows:

- (1) Ensure the printer is in the Idle state, see [page 4-19](#).
- (2) On the TouchScreen, select *Home > Setup > Wizards > Nozzle unblocking*.
- (3) Select *Start nozzle unblocking*.
- (4) Follow the on screen instructions.

End of procedure.

Gutter Airflow Adjustment

Note: Increasing the gutter airflow will increase make-up consumption.

The gutter in the print head captures unused drops of ink and returns them to the ink system.

Airflow through the gutter is automatically controlled by the printer's software. The printer's software adjusts the gutter pump speed to reach a predicted airflow target.

If the gutter is not clearing ink, gutter stall alerts are occurring or the gutter block is messy, the gutter airflow can be increased manually.

To adjust the gutter airflow:

- (1) Select *Home > Setup > Advanced > Gutter*.
- (2) Increase the *Airflow adjustment (ml/min)* value until the issue is resolved.

End of procedure.

i-PULSE PRINT HEAD MAINTENANCE

Print Head Cleaning



WARNING:

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.

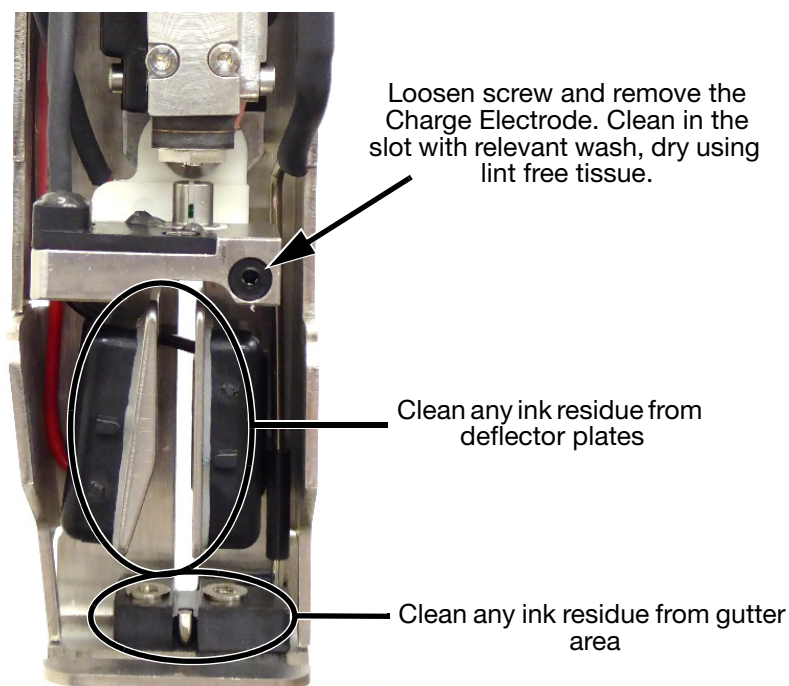
CAUTION:

The printer must be switched off before starting this procedure. If the printer is running whilst this procedure is carried out, wash will enter the gutter and may cause the viscosity of the ink to thin.

Required tools: T6 Torx Driver.

Clean the print head as follows:

- (1) If the printer is on, press and hold the standby button  for 2 seconds and wait for the printer to shut down.
- (2) Disconnect the power lead from the rear of the printer.
- (3) At the print head, remove the holster by pressing in the two holster retaining clips on the sides of the print head.
- (4) Remove the internal cover by gently squeezing in the sides to release the clips and lifting the lower part away from the print head and gently pulling downwards.
- (5) Follow the annotated instructions in the following illustration.



i-Pulse Print Head Cleaning

- (6) Replace the print head holster.

End of procedure.

Gutter Cleaning



WARNING:

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.

To clean the gutter:

- (1) Ensure the printer is on and in the Idle state.
- (2) At the print head, press in the two retaining clips on the sides of the print head and remove the holster.
- (3) On the TouchPanel, select *Home > Setup > Advanced > Gutter*.
- (4) Tick the *Gutter pump at maximum speed* tick box.
- (5) Select the  icon on the Information Bar.
- (6) Swipe the screen from right to left twice to navigate to the *Live status* screen.
- (7) Observe the *Vacuum pressure (mbar)* value.

Overall equipment efficiency

START

STATUS

STOP

Idle

Ink level	Ok	Viscosity lower limit (cP)	2.33		
Ink sensor level value	60	Ink temperature (°C)	20.0	Modulation level (V)	0
Pressure (mBar)	3000	Vacuum pressure (mBar)	-310	Modulation mode	Auto
Actual viscosity (cP)	0.00	Feed valve	closed	BUP time	0
Viscosity upper limit (cP)	5.33	Bleed valve	closed	Head temperature (°C)	42.0
Heater state	Enabled	Wash valve	closed	Nozzle valve	closed
Gutter pump speed (rpm)	0	Viscometer valve	closed		
Make-up level	Ok	Make-up valve	closed		
Make-up sensor level value	60				
Pump speed (rpm)	0				
Target viscosity (cP)	3.83				

Ink level

Vacuum pressure (mBar)

Ok

-310

Live status

○ ○ ●

Jet 1-Head temperature (°C)

42.0

Vacuum Pressure (mbar) Value

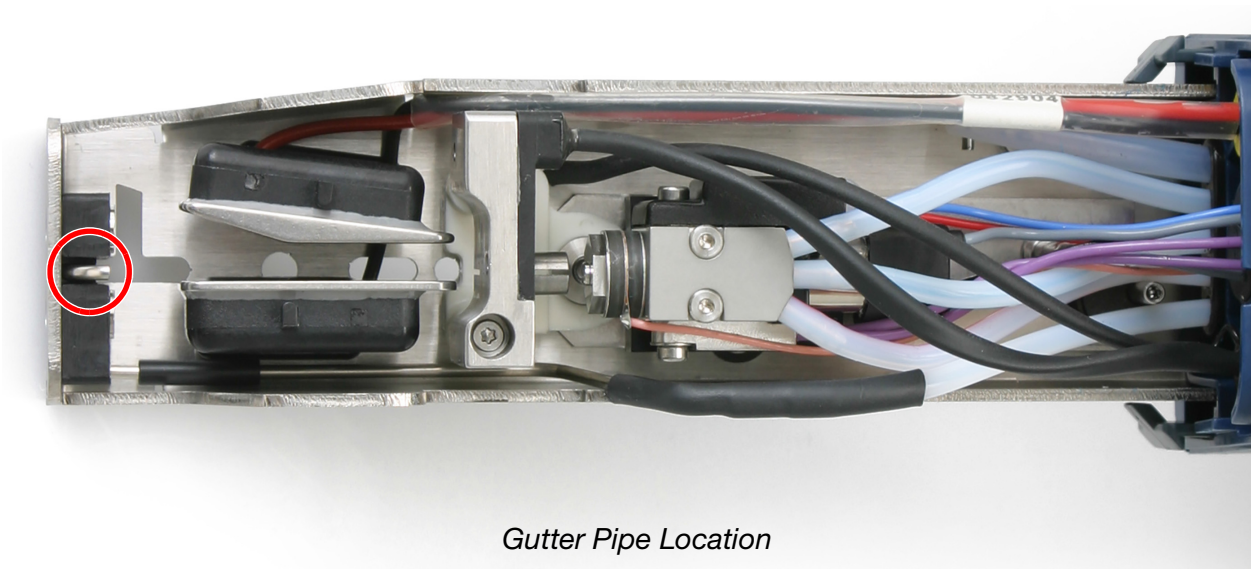
MAINTENANCE

- (8) If the *Vacuum pressure (mbar)* value is greater than -180 it is likely that the gutter pipe is partially blocked with dried ink. Squirt a small amount of solvent into the gutter pipe and wait for the *Vacuum pressure (mbar)* value to stabilise.



CAUTION:

Risk of sending ink viscosity outside of printer operating limit. The ink viscosity will thin and go out of the printers operating range if a large amount of solvent enters the gutter.



Gutter Pipe Location

- (9) If the *Vacuum pressure (mbar)* value is now lower than -180 continue to the next step. If the *Vacuum pressure (mbar)* value is still greater than -180 repeat step (8).
- (10) Select the  icon at the bottom of the screen.
- (11) Select *Home > Setup > Advanced > Gutter*.
- (12) Untick the *Gutter pump at maximum speed* tick box.
- (13) Replace the print head holster.
- End of procedure.

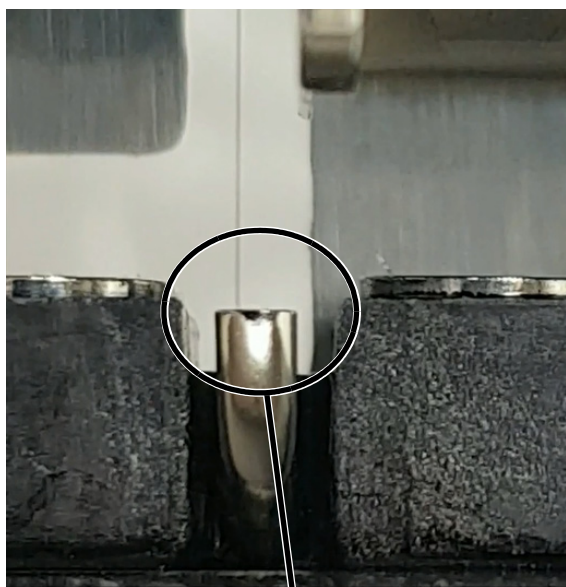
Ink Jet Alignment Check



- WARNINGS:**
- (1) Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.
 - (2) The print head must be placed in a wash station, or positioned over a beaker made of a conducting material and securely connected to earth (ground) in case the jet is misaligned out of the gutter. The electrostatic charges on the ink drops used for printing can cause a fire hazard.

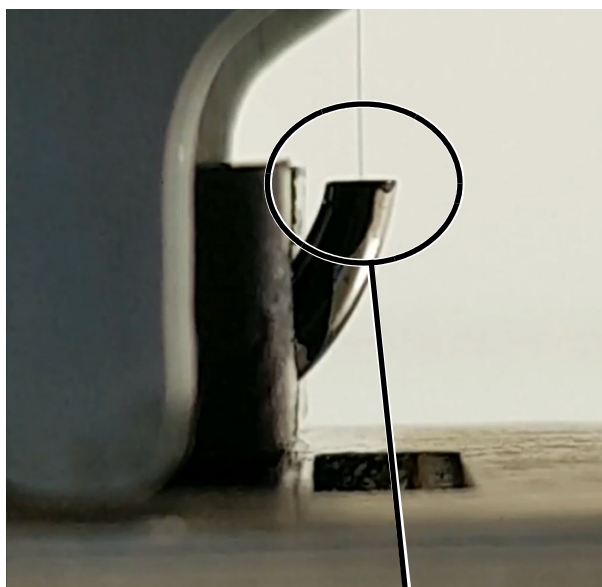
Check the ink jet alignment as follows:

- (1) Remove the holster from the print head by pressing in the two holster retaining clips and withdrawing the holster.
- (2) Fit the print head into a wash station, or place a beaker made of a conducting material and connected to earth (ground) underneath the print head.
- (3) With the ink jet running, check the position that the ink jet enters the gutter as shown in the following diagrams.



Ink jet $\frac{1}{3}$ from left side of the gutter wall.

Front View



Ink jet in centre of gutter.

Side View

- (4) If the alignment is incorrect, an engineer certified by Domino can re-align the ink jet.
- (5) Replace the print head holster.

End of procedure.

i-PULSE DUO PRINT HEAD MAINTENANCE

Print Head Cleaning



WARNING:

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.

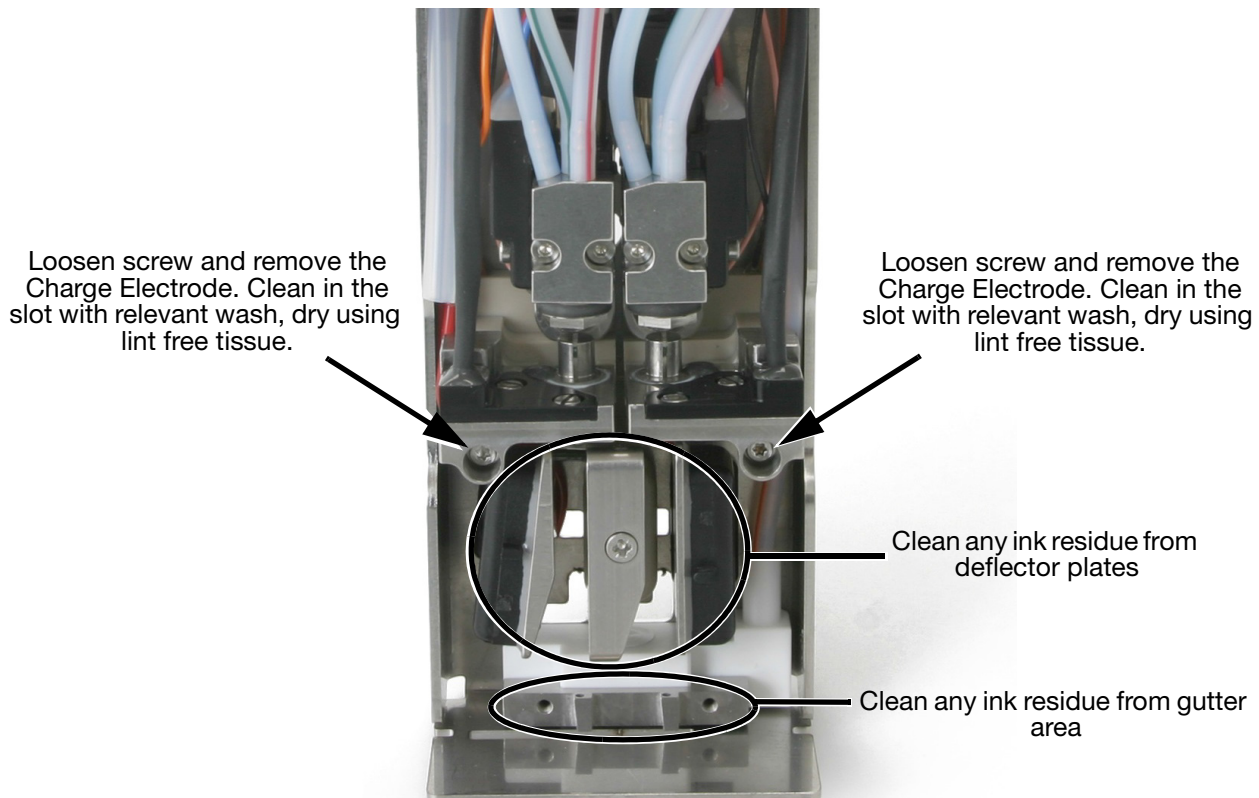
CAUTION:

The printer must be switched off before starting this procedure. If the printer is running whilst this procedure is carried out, wash will enter the gutter and may cause the viscosity of the ink to thin.

Required tools: T6 Torx Driver.

Clean the print head as follows:

- (1) If the printer is on, press and hold the standby button  for 2 seconds and wait for the printer to shut down.
- (2) Disconnect the power lead from the rear of the printer.
- (3) At the print head, remove the cover by unscrewing the thumb screw on the rear of the print head and sliding the cover off.
- (4) Follow the annotated instructions in the following illustration.



i-Pulse Duo Print Head Cleaning

- (5) Replace the print head cover and tighten the thumb screw.
- End of procedure.

Gutter Cleaning



WARNING:

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.

To clean the i-Pulse Duo gutter:

- (1) Ensure the printer is on and in the Idle state.
- (2) Unscrew the thumb screw on the rear of the print head and slide the print head cover off.
- (3) On the TouchPanel, select *Home > Setup > Advanced > Gutter*.
- (4) Tick the *Gutter pump at maximum speed* tick box.
- (5) Select the  icon on the Information Bar.
- (6) Swipe the screen from right to left twice to navigate to the *Live status* screen.
- (7) Observe the *Vacuum pressure (mbar)* value.

Ink level	Ok	Viscosity lower limit (cP)	2.33	
Ink sensor level value	60	Ink temperature (°C)	20.0	Modulation level (V)
Pressure (mBar)	3000	Vacuum pressure (mBar)	-310	Modulation mode
Actual viscosity (cP)	0.00	Feed valve	closed	Modulation mode
Viscosity upper limit (cP)	5.33	Bleed valve	closed	Head temperature (°C)
Heater state	Enabled	Wash valve	closed	Nozzle valve
Gutter pump speed (rpm)	0	Viscometer valve	closed	
Make-up level	Ok	Make-up valve	closed	
Make-up sensor level value	60			
Pump speed (rpm)	0			
Target viscosity (cP)	3.83			

Vacuum Pressure (mbar) Value

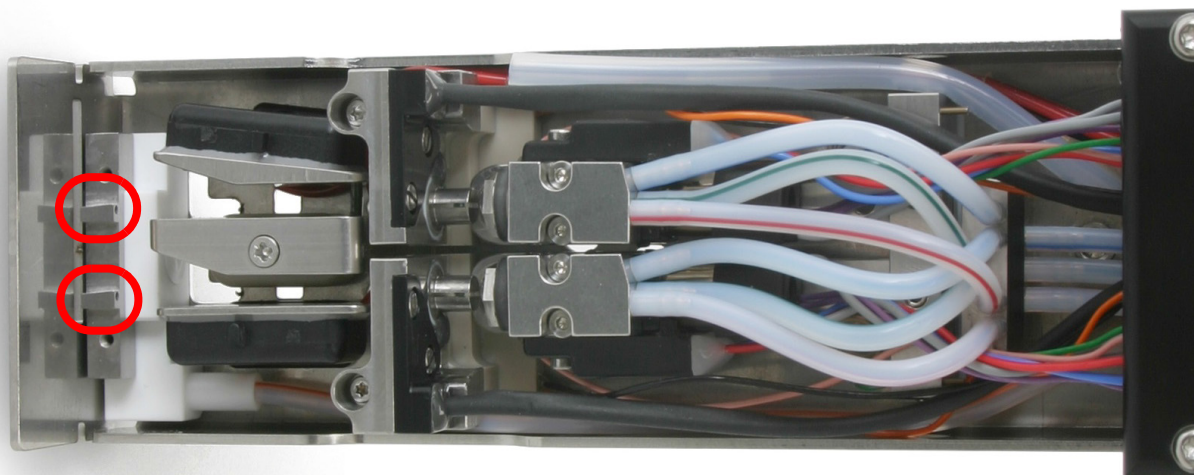
- (8) If the *Vacuum pressure (mbar)* value is greater than -180 it is likely that both gutter pipes are partially blocked with dried ink. To check each individual gutter pipe put on a latex or nitrile glove. Use a gloved finger to block each gutter pipe in turn to see if the *Vacuum pressure (mbar)* value changes.

- (9) If one or both gutter pipes are blocked with dried ink, squirt a small amount of solvent into the blocked gutter pipe and wait for the *Gutter pump speed (rpm)* value to stabilise.



CAUTION:

Risk of sending ink viscosity outside of printer operating limit. The ink viscosity will thin and go out of the printers operating range if a large amount of solvent enters the gutter.



Gutter Pipe Location

- (10) If the *Vacuum pressure (mbar)* value is now lower than -180 continue to the next step. If the *Vacuum pressure (mbar)* value is still greater than -180, squirt a small amount of solvent into the blocked gutter pipe and wait for the *Gutter pump speed (rpm)* value to stabilise.
 - (11) Select the  icon at the bottom of the screen.
 - (12) Select *Home > Setup > Advanced > Gutter*.
 - (13) Untick the *Gutter pump at maximum speed* tick box.
 - (14) Replace the print head cover and tighten the thumb screw.
- End of procedure.

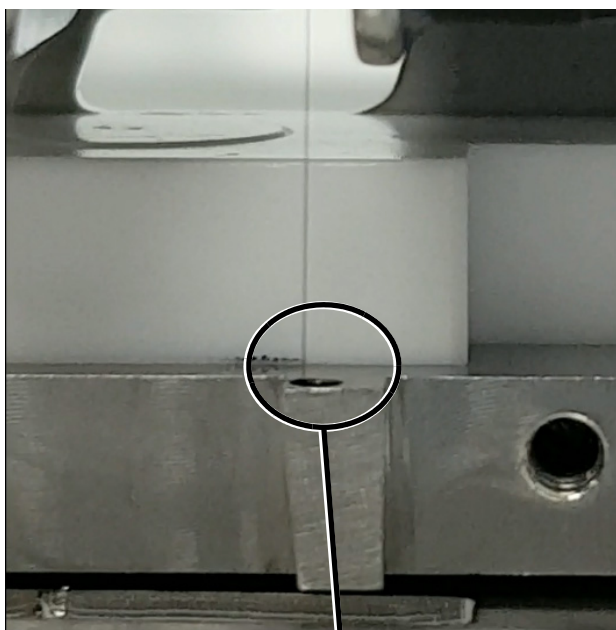
Ink Jet Alignment Check



- WARNINGS:**
- (1) Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.
 - (2) The print head must be placed in a wash station, or positioned over a beaker made of a conducting material and securely connected to earth (ground) in case the jet is misaligned out of the gutter. The electrostatic charges on the ink drops used for printing can cause a fire hazard.

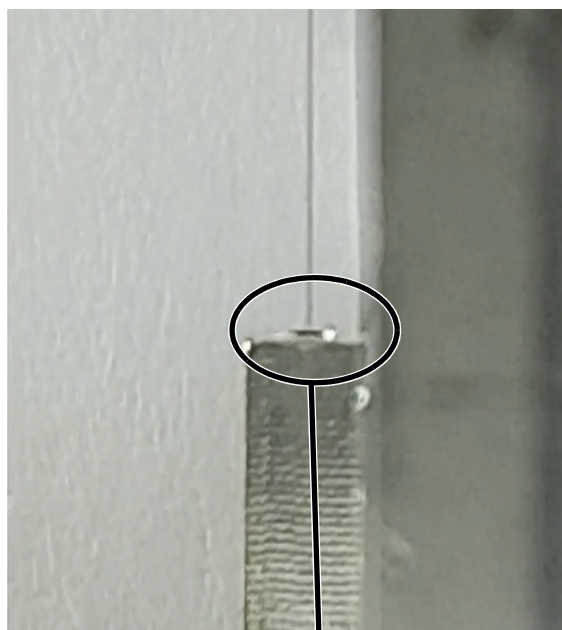
Check the ink jet alignment as follows:

- (1) Remove the print head cover by unscrewing the locking screw on the rear of the print head and sliding the cover off.
- (2) Fit the print head into a wash station, or place a beaker made of a conducting material and connected to earth (ground) underneath the print head.
- (3) With both ink jets running, check the position that each ink jet enters the gutter as shown in the following diagrams.



Ink jets $\frac{1}{4}$ from the left side of the gutter wall.

Front View



Ink jets in the centre of the gutter.

Side View

- (4) If the alignment is incorrect, an engineer certified by Domino can re-align the ink jet.
- (5) Replace the print head holster.

End of procedure.

i-PULSE RS PRINT HEAD MAINTENANCE

Print Head Cleaning

**WARNING:**

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.

CAUTION:

The printer must be switched off before starting this procedure. If the printer is running whilst this procedure is carried out, wash will enter the gutter and may cause the viscosity of the ink to thin.

The i-Pulse RS print head requires regular cleaning when using heavily pigmented ink. Check the print head before every shift and clean as necessary.

To clean the i-Pulse RS print head:

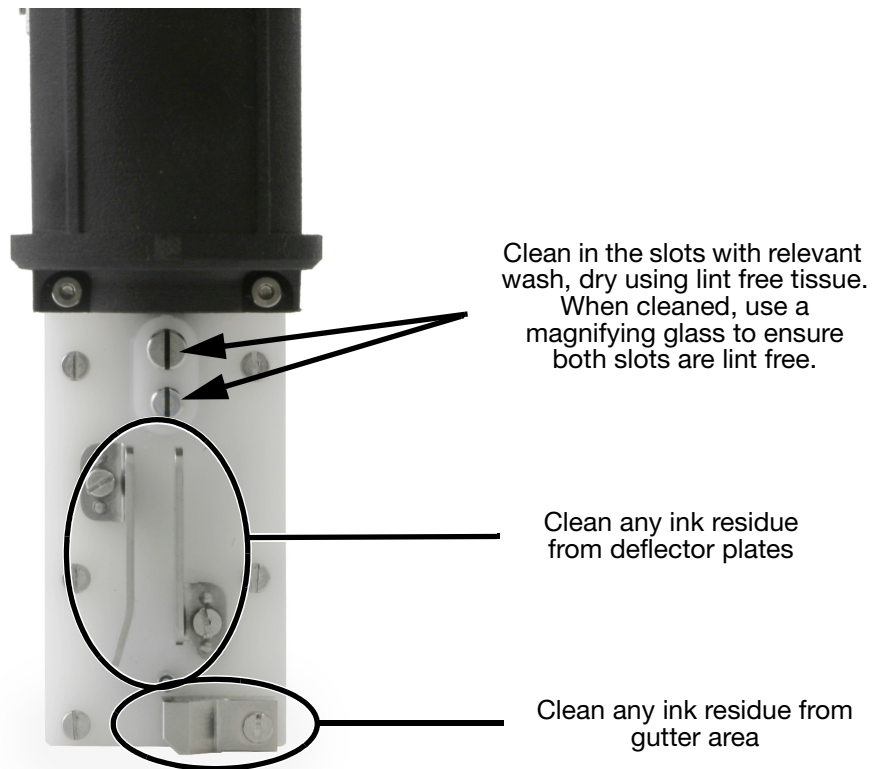
- (1) If the printer is on, press and hold the standby button  for 2 seconds and wait for the printer to shut down.
- (2) Twist and remove the print head cover.



Remove Print Head Cover

MAINTENANCE

- (3) Follow the annotated instructions on the following page. If required, place an earthed beaker underneath the print head to catch any waste fluid.



i-Pulse RS Print Head Cleaning

- (4) Replace the print head cover.
- (5) Dispose of waste solvent by following local waste disposal regulations.
- End of procedure.

Gutter Cleaning



WARNING:

Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.

To clean the i-Pulse RS gutter:

- (1) Ensure the printer is on and in the Idle state.
- (2) Twist and remove the print head cover.



Remove Print Head Cover

- (3) On the TouchScreen, select *Home > Setup > Advanced > Gutter*.
- (4) Tick the *Gutter pump at maximum speed* tick box.
- (5) Select the  icon on the Information Bar.
- (6) Swipe the screen from right to left twice to navigate to the *Live status* screen.

(7) Observe the *Vacuum pressure (mbar)* value.

Overall equipment efficiency

START

STATUS

Idle

STOP

Ink level	Ok	Viscosity lower limit (cP)	2.33		
Ink sensor level value	60	Ink temperature (°C)	20.0	Modulation level (V)	0
Pressure (mBar)	3000	Vacuum pressure (mBar)	-310	Modulation mode	Auto
Actual viscosity (cP)	0.00	Feed valve	closed	BUP time	0
Viscosity upper limit (cP)	5.33	Bleed valve	closed	Head temperature (°C)	42.0
Heater state	Enabled	Wash valve	closed	Nozzle valve	closed
Gutter pump speed (rpm)	0	Viscometer valve	closed		
Make-up level	Ok	Make-up valve	closed		
Make-up sensor level value	60				
Pump speed (rpm)	0				
Target viscosity (cP)	3.83				

⏮

Ink level

Ok

Vacuum pressure (mBar)

-310

Live status

○ ○ ●

Jet 1-Head temperature (°C)

42.0

⏭

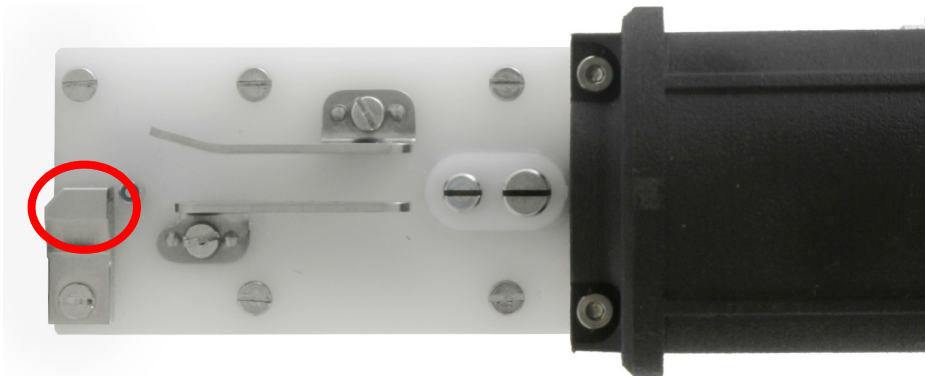
Vacuum Pressure (mbar) Value

(8) If the *Vacuum pressure (mbar)* value is greater than -180 it is likely that the gutter pipe is partially blocked with dried ink. Squirt a small amount of solvent into the gutter pipe and wait for the *Vacuum pressure (mbar)* value to stabilise.



CAUTION:

Risk of sending ink viscosity outside of printer operating limit. The ink viscosity will thin and go out of the printers operating range if a large amount of solvent enters the gutter.



Gutter Pipe Location

MAINTENANCE

(9) If the *Vacuum pressure (mbar)* value is now lower than -180 continue to the next step. If the *Vacuum pressure (mbar)* value is still greater than -180 repeat step (8).

(10) Select the  icon at the bottom of the screen.

(11) Select *Home > Setup > Advanced > Gutter*.

(12) Untick the *Gutter pump at maximum speed* tick box.

(13) Replace the print head cover.

End of procedure.

Ink Jet Alignment Check



WARNINGS:

- (1) Protective equipment such as gloves and glasses must be worn when this procedure is carried out. Physical contact with printer ink or make-up can cause skin or eye damage.
- (2) The print head must be placed in a wash station or positioned over a beaker made of a conducting material and securely connected to earth (ground) in case the jet is misaligned out of the gutter. The electrostatic charges on the ink drops used for printing can cause a fire hazard.

To check the alignment of the i-Pulse RS ink jet:

- (1) Twist and remove the print head cover.



Remove Print Head Cover

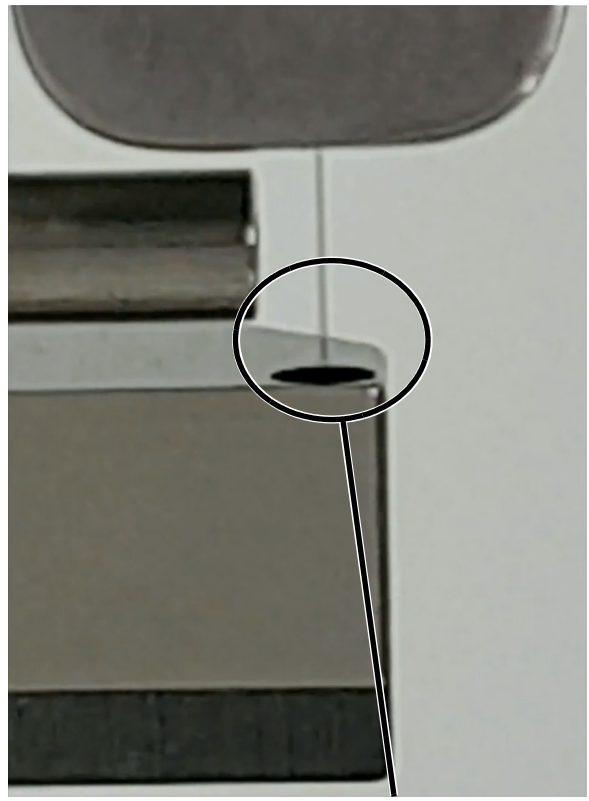
- (2) Fit the print head into a wash station, or place a beaker made of a conducting material and connected to earth (ground) underneath the print head.

- (3) With the ink jet running, check the position that the ink jet enters the gutter as shown in the diagram below.



Ink jet nearly touching left side of the gutter wall.

Front View



Ink jet in centre of gutter.

Side View

- (4) If the alignment is incorrect, an engineer certified by Domino can re-align the ink jet.
(5) Replace the print head cover.

End of procedure.

SOFTWARE

Upgrade with USB Flash Drive



CAUTION:

Risk of damage to printer. Disconnect all printer cables, connections, ancillaries, Domino Cloud Interface (DCI), beacon, alarm port, shaft encoder and sensors etc. This ensures no communication or electrical signals can be received during the upgrade process. Failure to remove all connections before the upgrade process starts may corrupt the printer's software.

- Notes: (1) *Non-standard files which have been uploaded manually onto the printer may be lost during a software upgrade. For example, trial inks or development rasters. If these files are not available in the printer after the upgrade, contact the Domino Technical Support Group (TSG) and check if the files are compatible with the new software revision. Re-install the files manually after the upgrade is complete.*
- (2) *The "Print Delay" setting will not persist during a software upgrade. Manually reset the "Print Delay" setting after the upgrade is complete.*
- (3) *Do not directly upgrade the printer to a software revision that is more than 2 releases ahead of the printers current software revision.*

To upgrade the printer's software:

- (1) Copy the upgrade file to the root folder of a USB flash drive.
- (2) Disconnect all printer cables, connections, ancillaries, Domino Cloud Interface (DCI), beacon, alarm port, shaft encoder and sensors etc. This ensures no communication or electrical signals can be received during the upgrade process. Failure to remove all connections before the upgrade process starts may corrupt the printer's software.
- (3) Power on the printer and wait for it to fully boot-up into the Idle state.



CAUTION:

Risk of damage to printer. Do not sequence the ink jet on and do not insert any USB device during the boot-up/start-up sequence.

- (4) Insert the USB flash drive into a USB port on the printer cabinet or TouchScreen.
- (5) Wait for the *Available upgrades* pop up screen to appear and select Yes.
- (6) Select the upgrade file.
- (7) The TouchScreen will now display a summary of the release notes for the upgrade file. Select the *Arrow* icon to continue the upgrade.
- (8) Select when the upgrade should take place, either:
 - Now (will restart printer automatically).
 - The next time the printer is restarted.
- (9) Select the *Arrow* icon.
- (11) Select *Upgrade now*.
- (12) The printer will now begin the upgrade process which will take between 11 and 22 minutes to complete.

MAINTENANCE

- (13) If the upgrade fails a “Rollback” button is available to rollback the software to a previous version. You can then try the upgrade process again.

End of procedure.

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PART 6 : OPTION PACKS

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OPTION PACKS

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PERFORMANCE PACK OVERVIEW

Performance packs are additional packs which can be installed when the printer is manufactured or installed at a later date to upgrade the printer's functionality.

The table below lists the performance pack options:

Pack Name and Part Number	Printer Availability	Description
Extended Raster Pack - EPT030749SP	Ax150i	The extended raster pack adds an extended range of rasters to the printer. The new rasters cover a broad range of general purpose applications.
Professional Printing Pack - EPT030750SP	Ax350i Ax550i	The professional printing pack adds advanced features for complex labels, high speed applications and high quality 2D coding.
Advanced Security Pack - EPT035038SP	Ax350i Ax550i	The advanced security pack adds security features to the printer.
Operator Assistance Pack - EPT030751SP	Ax350i Ax550i	The operator assistance pack adds security features and a barcode scanner for data entry. These features minimise the risk of operator error on the production line.
Green Pack - EPT030753SP	Ax350i Ax550i	The Green Pack adds a Peltier condenser to reduce make-up consumption by up to 20%. Power consumption is also lowered with the addition of configurable energy save modes.
Continuous Printing Pack - EPT035039SP	Ax350i Ax550i	The continuous printing pack adds features for printing long labels on high speed continuous print surfaces such as cable, wire and extrusion applications. (See Continuous Printing Pack Product Manual available at: https://mydomino.domino-printing.com)

Installation and Migration



WARNING:

High voltage electricity. Risk of death or injury from electric shock. Only trained engineers, certified by Domino can install the Green Pack.

CAUTION:

Risk of damage to printer. Only trained engineers certified by Domino can install the Green Pack.

The installation procedure for most performance packs is simple and can be completed by the user. The procedure to migrate performance packs from one printer to another is also simple easy to complete by the user.

However, to prevent damage to the printer or injury to personnel, only a trained engineer certified by Domino is permitted to install the Green Pack. The procedures bellow describe installation and migration of all performance packs except the Green Pack.

Installation

To install a performance pack:

- (1) Insert the performance pack USB into a USB port on the printer cabinet or TouchScreen.
- (2) On the TouchScreen select, *Setup > Packs management*.
- (3) Select the *Install* button.
- (4) Follow the on screen instructions.
- (5) Remove the performance pack USB.
- (6) Press and hold the standby button  for 2 seconds and restart the printer.

End of procedure.

Migration

To remove a performance pack from the current printer and install it on a different printer:

- (1) Insert the correct performance pack USB into a USB port on the printer cabinet or TouchScreen.
- (2) On the TouchScreen select, *Setup > Packs management > Migrate Packs*.
- (3) Select the *Migrate* button.
- (4) Follow the on screen instructions.
- (5) Remove the performance pack USB.
- (6) Press and hold the standby button  for 2 seconds and restart the printer.
- (7) Insert the performance pack USB into a USB port on the printer cabinet or TouchScreen that the pack is being migrated to.
- (8) On the TouchScreen select, *Setup > Packs management*.
- (9) Select the *Install* button.
- (10) Follow the on screen instructions.
- (11) Remove the performance pack USB.
- (12) Press and hold the standby button  for 2 seconds and restart the printer.

End of procedure.

PERFORMANCE PACK DETAILS

Extended Raster Pack

Note: Available for Ax150i printers only.

Software requirements: Version 01.00.20 or higher.

The extended raster pack is a performance pack that adds extra rasters to cover a broad range of general purpose applications.

The new rasters will be available when creating or editing label designs.



Extended Raster Pack

Professional Printing Pack

Note: Available for Ax350i/Ax550i printers only.

Software requirements: Version 01.00.20 or higher.

The professional printing pack is a performance pack that adds advanced features for complex labels, high speed applications and high quality 2D coding:

- High speed (ST and HQ) rasters.
- Segmented labels.
- Lua script engine.



Professional Printing Pack

Segmented Labels

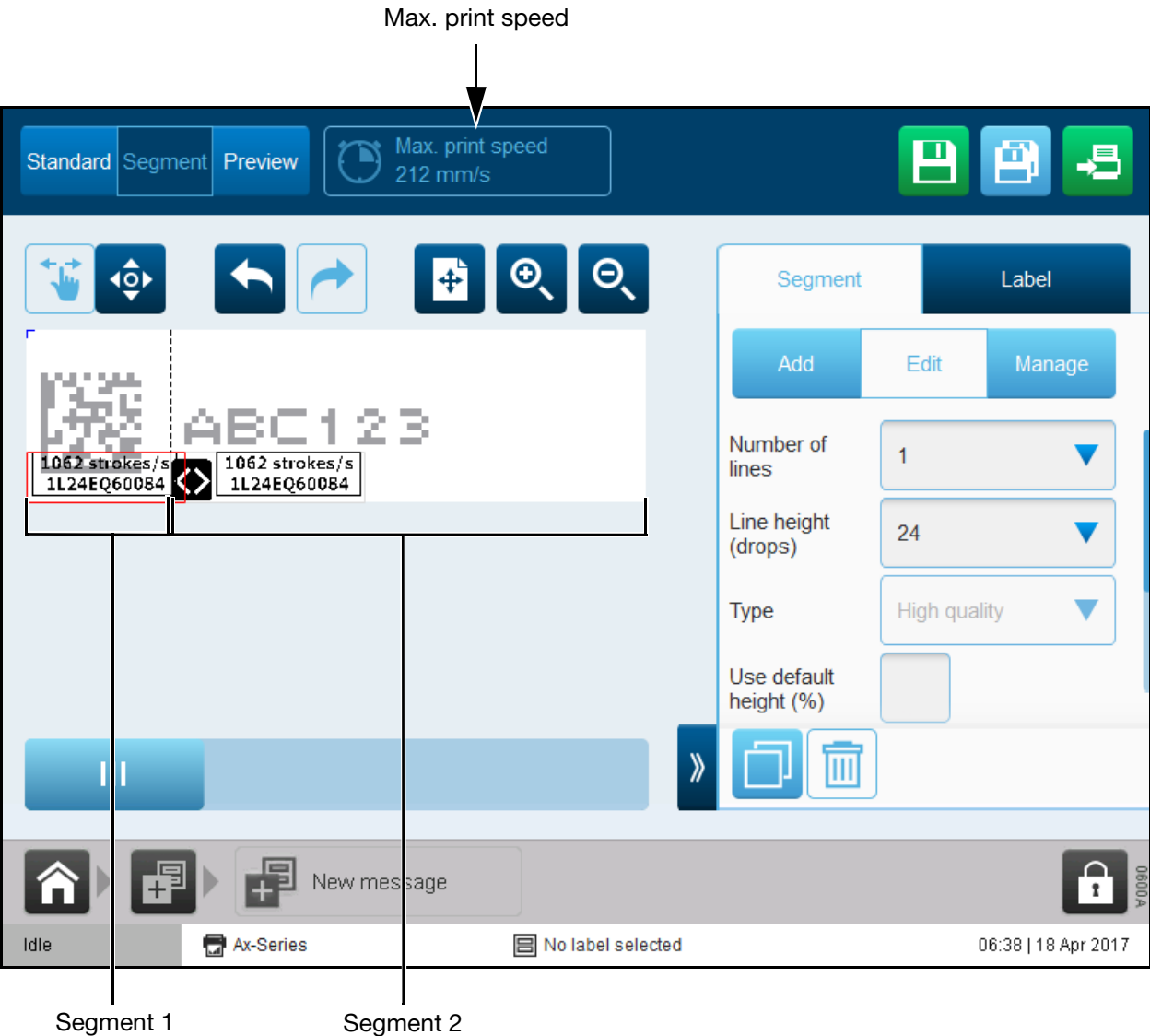
Label segmentation enables multiple rasters, print heights and stroke pitches to be used within a single label. This enables the user to optimise print quality when both text and 2D barcode elements are used in the same label.

Segmented Label Example

The illustration below shows an example of a label design with a barcode element and a text element in two different segments.

Segment 1 contains a data matrix barcode that requires a high density print to ensure it can be properly scanned. To achieve this the *Height (%)* and *Stroke pitch (mm/stroke)* settings for this segment have been adjusted to move the ink drops closer together.

Segment 2 contains a text element that does not require a high density print. The *Height (%)* and *Stroke pitch (mm/stroke)* settings for segment 2 have been left at the default values to reduce ink consumption and optimise the label print speed.



Segment Example

Add a Segment

To add a segment into a label design:

- (1) In the *Label Creator*, select *Segment* at the top of the screen.
- (2) In the side menu, select the *Segment* tab.
- (3) Select the *Add* menu.
- (4) Select the *Add Segment* icon.



- (5) The following segment settings can now be defined:

Setting Name	Description
Number of lines	Select the required number of lines.
Line height (drops)	Select the required line height in ink drops.
Type	Select the required segment quality.
Overall height (mm)	Select the height of the label segment.
Use default height (%)	Tick the tick box to use the default print height setting. Untick the tick box to display the <i>Height (%)</i> setting.
Height (%)	<i>Note:</i> This setting is only valid if the <i>Use default height (%)</i> tick box is not ticked. Set the print height percentage.
Use default stroke pitch	Tick the tick box to use the default stroke pitch setting. Untick the tick box to display the <i>stroke pitch (mm/stroke)</i> setting.
Stroke pitch (mm/stroke)	<i>Note:</i> This setting is only valid if the <i>Use default stroke pitch</i> tick box is not ticked. Set the distance between print strokes. A stroke is the line of ink drops which is used to make up each printed character.

End of procedure.

LUA Script

Import LUA Script Files

Script elements written in LUA programming language can be imported into the printer using the file manager. Once imported, the script element can then be added into the label design, see [page 4-44](#).

To import a script:

- (1) Save the script file to a USB drive.
- (2) Insert the USB drive into a USB slot on the TouchScreen or printer cabinet.
- (3) On the TouchScreen, select *Home > Setup > File manager*.
- (4) Open the USB folder.
- (5) Copy the script file and navigate to the *Scripts* folder.
- (6) Select *Paste*.

End of procedure.

Behavioural Configuration

The settings on the behavioural configuration screen define the loop time of each specific script.

To set the loop time of a specific script:

- (1) Select *Home > Setup > Global print settings > Behavioural configuration*
- (2) Select *Add new script*
- (3) Configure the following settings:

Setting Name	Description
Script path	Enter the location of the script file.
Instance name	Enter the name of the script file.
Loop time (ms)	Define the loop time.

End of procedure.

Reset Script Variables

To reset scripts:

- (1) Select *Home > Setup > Global Print Settings > Content*
- (2) Select *Reset All* to reset persistent script variables.

End of procedure.

Operator Assistance Pack

Note: Available for Ax350i/Ax550i printers only.

Hardware requirements: An available USB Type A port on the printer cabinet or TouchScreen.

The operator assistance pack is a performance pack that adds the following features to minimise the risk of operator error on the production line:

- Simple security manager.
- Barcode scanning for label selection and data entry.
- Label templates for consistent code appearance.

The simple security manager contains four user groups as described in the table below. If further security features are required, the advanced security pack is available, see [page 6-14](#).

User Group	Explanation
Admin	Access to all printer settings and functions.
Supervisor	Access to the following printer functions: • Label creator. • Edit labels. • Print optimisation. • Change printer state. • Label select. • Acknowledge alerts. • Prompted data entry via UI keyboard.
Operator	Access to the following printer functions: • Print optimisation. • Change printer state. • Label select. • Acknowledge alerts. • Prompted data entry via UI keyboard.
Barcode Scanner	Access to the following printer functions: • Print optimisation. • Change printer state. • Acknowledge alerts. • Label select via barcode scanner. • Prompted data entry via barcode scanner.

Security Manager

Enable Simple Security Mode

When the simple security mode is enabled, only users with a valid password will be able to operate the printer through the user interface.

Note: When the security mode is enabled, you will be immediately logged out of the printer. Before enabling simple security mode, ensure you have a valid password to log into the printer, see [“Log In” on page 6-10](#).

To enable simple security mode:

- (1) Select *Home > Setup > Security*.
- (2) Select the *Security mode* drop down setting.
- (3) Select *Simple*.

End of procedure.

Disable Security Mode

To disable security mode:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security*.
- (3) Select the *Security mode* drop down setting.
- (4) Select *Off*.

End of procedure.

Log In

To log in:

- (1) Select the *Padlock* icon in the bottom right corner of the user interface.
- (2) Select *Log in*.
- (3) Enter the password for the required user group. The default password for each user group is defined in the table below:

Note: Passwords are case sensitive.

User Group	Password
admin	admin
supervisor	sv
operator	op
barcodescanner	scanner

End of procedure.

Log Out

To log out:

- (1) Select the *Padlock* icon in the bottom right corner of the user interface.
- (2) Select *Log out*.

End of procedure.

Enable Auto Log-in

To enable auto log-in:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security*.
- (3) Tick the *Enable auto log-in* tick box.
- (4) Select the *Auto login user* drop down setting.
- (5) Select the user group that will automatically be logged in when the printer is turned on.

End of procedure.

Disable Auto Log-in

To disable auto log-in:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security*.
- (3) Untick the *Enable auto log-in* tick box.

End of procedure.

Change Current User Group Password

To change the current user group's password:

- (1) Select the *Padlock* icon in the bottom right corner of the user interface.
- (2) Select *Change password*.
- (3) Enter the following information:

Setting Name	Explanation
Current password	Enter the current password.
New password	Enter the new password.
Retype password	Retype the new password.

- (4) Select *Save*.

End of procedure.

Reset Forgotten Password

If a user group password is forgotten, an admin level user can reset the password.

If the admin level password is forgotten, contact your local Domino support office.

To reset a forgotten user group password:

- (1) Log in as an admin level user, [page 6-10](#).
- (2) Select *Home > Setup > Security > Users*.
- (3) Select the user profile that requires a new password.
- (4) Select *Change password*.
- (5) Enter the following information:

Setting Name	Explanation
New password	Enter the new password.
Retype password	Retype the new password.

- (6) Select *Save*.

End of procedure.

Change User Group Settings

To change a user group's settings:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security > Groups*.
- (3) Select the user group that requires changing.
- (4) The functions and settings that the user group has access to can now be changed and customised.
- (5) Select *Save*.

End of procedure.

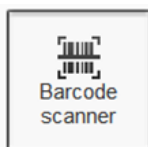
Barcode Scanner

The Operator Assistance Pack contains a barcode scanner and scanner user profile. When logged into the barcode scanner user profile, the user can select what label is sent to print by scanning a barcode containing the correct label name. If the label contains a prompted field, the user will be prompted to enter data which can also be scanned from a barcode.

Label Selection and Data Entry via Barcode Scanner

To select a label and enter prompted data using the barcode scanner:

- (1) log into the barcode scanner user group, see [page 6-10](#).
- (2) Select the *Barcode scanner* icon on the UI home screen.



- (3) Use the barcode scanner to scan a barcode that contains the label name.
- (4) Select *Send to print*.
- (5) If the label design contains a prompted element, the user will be prompted to scan another barcode that contains the required data.

End of procedure.

Advanced Security Pack

Note: Available for Ax350i/Ax550i printers only.

Hardware requirements: An available USB Type A port on the printer cabinet or TouchScreen.

The advanced security pack is a performance pack that adds the advanced and simple security mode to the printer's user interface.

Security Manager

Enable Simple Security Mode

When simple security mode is active, only users with a valid password will be able to operate the printer through the user interface.

Note: You will be immediately logged out of the printer when the security mode is enabled. Before enabling simple security mode, ensure you have a valid password to log into the printer, see ["Log In" on page 6-15](#).

To enable simple security mode:

- (1) Select *Home > Setup > Security*.
- (2) Select the *Security mode* drop down setting.
- (3) Select *Simple*.

End of procedure.

Enable Advanced Security Mode

When advanced security mode is active, only users with a valid user name and password will be able to operate the printer through the user interface. Users who are part of the admin group will also have the capability to set password policy, create new user and group profiles and edit user and group profiles.

Note: You will be immediately logged out of the printer when the security mode is enabled. Before enabling advanced security mode, ensure you have a valid password to log into the printer, see ["Log In" on page 6-15](#).

To enable the advanced security mode:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security*.
- (3) Select the *Security mode* drop down setting.
- (4) Select *Advanced*.

End of procedure.

Disable Security Mode

To disable security mode:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security*.
- (3) Select the *Security mode* drop down setting.
- (4) Select *Off*.

End of procedure.

Log In

To log in:

- (1) Select the *Padlock* icon in the bottom right corner of the user interface.
- (2) Select *Log in*.
- (3) Enter the log in details for the required user. The default usernames and passwords are defined in the table below:

Note: Usernames and passwords are case sensitive.

Username	Password	Explanation
admin	admin	Access to all printer settings and functions.
supervisor	sv	Access to the following printer functions: • Start/Stop printing • Label finder • Label creator • Edit current label • Print optimisation • Printer status • Label select • Prompted data entry via UI keyboard • Acknowledge alerts
operator	op	Access to the following printer functions: • Start/Stop printing • Print optimisation • Printer status • Label select • Prompted data entry via UI keyboard • Acknowledge alerts

End of procedure.

Log Out

To log out:

- (1) Select the *Padlock* icon in the bottom right corner of the user interface.
- (2) Select *Log out*.

End of procedure.

Enable Auto Log-in

To enable auto log-in:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security*.
- (3) Tick the *Enable auto log-in* tick box.
- (4) Select the *Auto login user* drop down setting.
- (5) Select the user that will automatically be logged in when the printer is turned on.

End of procedure.

Disable Auto Log-in

To disable auto log-in:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security*.
- (3) Untick the *Enable auto log-in* tick box.

End of procedure.

Change Current User Password

To change the current user's password:

- (1) Select the *Padlock* icon in the bottom right corner of the user interface.
- (2) Select *Change password*.
- (3) Enter the following information:

Setting Name	Explanation
Current password	Enter the current user password.
New password	Enter the new password.
Retype password	Retype the new password.

- (4) Select *Save*.

End of procedure.

Reset Forgotten Password

If a user profile password is forgotten, an admin level user can reset the password.

If the admin level user password is forgotten, contact your local Domino support office.

To reset a forgotten password:

- (1) Log in as an admin level user, [page 6-15](#).
- (2) Select *Home > Setup > Security > Users*.
- (3) Select the user profile that requires a new password.
- (4) Select *Change password*.
- (5) Enter the following information:

Setting Name	Explanation
New password	Enter the new password.
Retype password	Retype the new password.

- (6) Select *Save*.

End of procedure.

Set Password Policy

Note: This procedure is possible in Advanced Security mode only.

To set the password policy for advanced security mode:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security > Password policy*.
- (3) The settings described below can now be defined:

Setting Name	Explanation
Password expiry warning (days)	Define the number of days before the password expiry warning is displayed.
Minimum number of unrepeated passwords	Define the minimum number of different passwords a user must have before the same password can be reused.
Minimum length	Define the minimum password length.
Min upper case	Define the minimum number of upper case characters that must be used.
Min lower case	Define the minimum number of lower case characters that must be used.
Min numeric	Define the minimum number of numeric characters that must be used.
Min special	Define the minimum number of defined special characters that must be used.
Max repeated	Define the maximum number of times characters can be repeated.
Max ID characters	Define the maximum number of consecutive characters that can be used for a user name.
Special characters	Define which special characters can be used for passwords.

End of procedure.

Add New User

Note: This procedure is possible in Advanced Security mode only.

To add a new user:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security > Users*.
- (3) Select the *Add new user* icon.
- (4) The settings described in the table below can now be changed:

Setting Name	Explanation
User name	Define the name of the user.
Password	Enter the user's password.
Retype password	Retype the user's password.
Groups	Select which user groups the user belongs to.
Status	Select the user's status: • Active • Dormant • Locked
Must change password	When a new user is added, this tick box will be selected. It is not possible to de-select this tick box until the new user has logged on and changed their password.
Forename	Enter the user's first name (optional).
Surname	Enter the user's last name (optional).
Department	Enter the user's department (optional).
Inactivity timeout (minutes)	Define the number of minutes the printer will be inactive for before the user is logged out.
Account expiry enabled	Select the tick box to enable the account expiry date.
Account expiry date	Enter the date the user account will expire on if the <i>Account expiry date</i> is enabled.
Password expiry (days)	Enter the number of days the current password can be used before it expires.

- (5) Select *Save*.

End of procedure.

Change User Settings

Note: This procedure is possible in Advanced Security mode only.

To change a user's settings:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security > Users*.
- (3) Select the user that requires changing.
- (4) To change the user's password, select *Change password* at the bottom of the screen.
- (5) The settings described in the table below can also be changed:

Setting Name	Explanation
User name	Displays the user name, this name cannot be changed.
Groups	Select which user groups the user belongs to.
Status	Select the user's status: • Active • Dormant • Locked
Must change password	Select the tick box to force a new user to change their password the first time they log in.
Forename	Enter the user's first name (optional).
Surname	Enter the user's last name (optional).
Department	Enter the user's department (optional).
Inactivity timeout (minutes)	Define the number of minutes the printer will be inactive for before the user is logged out.
Account expiry enabled	Select the tick box to enable the account expiry date.
Account expiry date	Enter the date on which the user account will expire on if <i>Account expiry date</i> is enabled.
Password expiry (days)	Enter the number of days the current password can be used before it expires.

- (6) Select *Save*.

End of procedure

Delete User

Note: This procedure is possible in Advanced Security mode only.

To delete or change a user:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security > Users*.
- (3) Select the red *Bin* icon next to the user to be deleted.

End of procedure.

Add New User Group

Note: This procedure is possible in Advanced Security mode only.

To add a new user group:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security > Groups*.
- (3) Select the *Add new group* icon.
- (4) Select the *Group name* entry box and enter a name for the user group.
- (5) Select *Save*.
- (6) Select which printer functions and settings members of the user group will be allowed access to.
- (7) Select *Save*.

End of procedure.

Delete User Group

Note: This procedure is possible in Advanced Security mode only.

To delete a user group:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security > Groups*.
- (3) Select the user group to be deleted.
- (4) Select *Delete Group*.

End of procedure.

Change User Group Settings

Each user group can be customised to allow or block the user group's access to specific printer settings.

To change the printer settings a user group has access to:

- (1) Log into the printer as an admin level user.
- (2) Select *Home > Setup > Security > Groups*.
- (3) Select the user group that requires changing.
- (4) A list of printer settings and UI screens will now be displayed. Change the value of each drop down box to define the settings members of the user group are allowed or not allowed to access.
- (5) Select *Save*.

End of procedure

Green Pack



WARNING: High voltage electricity. Risk of death or injury from electric shock. Only trained engineers, certified by Domino, can install the Green Pack.

CAUTION: Risk of damage to printer. Only trained engineers certified by Domino can install the Green Pack.

Note: Available for Ax350i/Ax550i printers only.

Hardware requirements: Revision 5 main PCB or higher.

Software requirements: Version 01.00.20 or higher.

The Green Pack is a performance pack that reduces make-up usage by up to 20% with the addition of a Peltier condenser. Power consumption is also lowered with the addition of configurable energy save modes:

- Peltier condenser to capture fluid evaporation in the printer cabinet.
- Print head and ink system energy save mode (configurable).
- Graphs to display make-up and ink usage on the user interface.

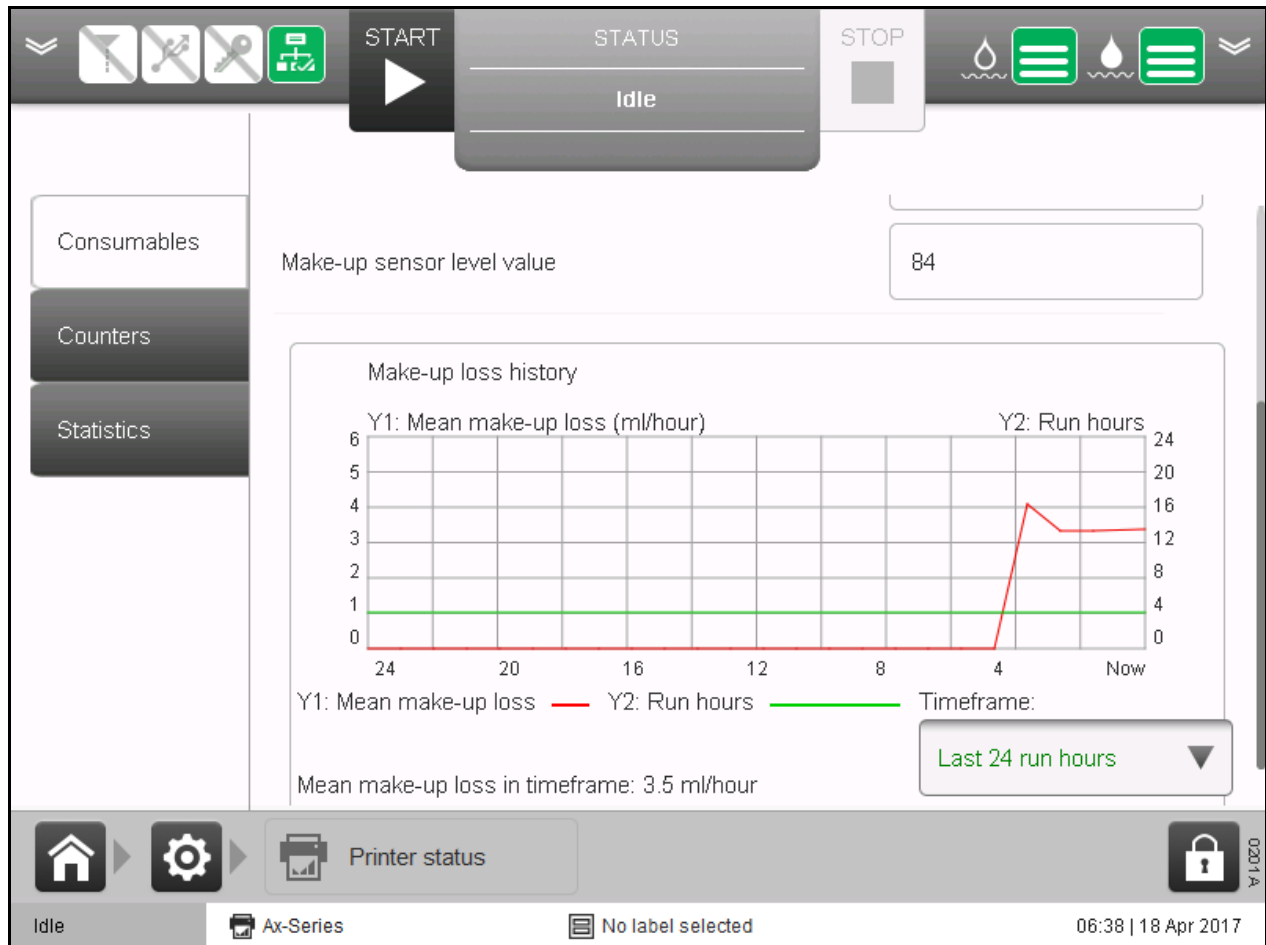


Green Pack

Make-up Loss Graph

The make-up loss graph displays how many millilitres per hour of make-up has been consumed by the printer.

To view the make-up loss graph, select *Home > Setup > Printer status > Consumables*.



Make-up Loss Graph

Energy Save Options

The printer has several energy saving features which can be enabled, disabled and customised on the *Power options* screen.

To enable or disable the printer's energy saving features:

- (1) Select *Home > Setup > Production line setup > Power options*.
- (2) Tick the *Energy save enabled* tick box.
- (3) The energy save features described below can now be customised.

Auto Jet Off

If enabled, the ink jet will sequence off when a specified amount of time has passed after the last print trigger signal is received.

The settings described in table below are used to setup the auto jet off feature.

Setting Name	Explanation
Auto jet off enabled	Enable or disable auto jet off.
Auto jet off time (mins)	Define the amount of time from the last print trigger signal received to when the ink jet will sequence off.

Auto Sleep

If enabled, the printer will enter sleep mode after a specified amount of time has passed since the last print trigger signal is received or since the ink jet is sequenced off. This depends on which *Action for auto sleep* option is selected.

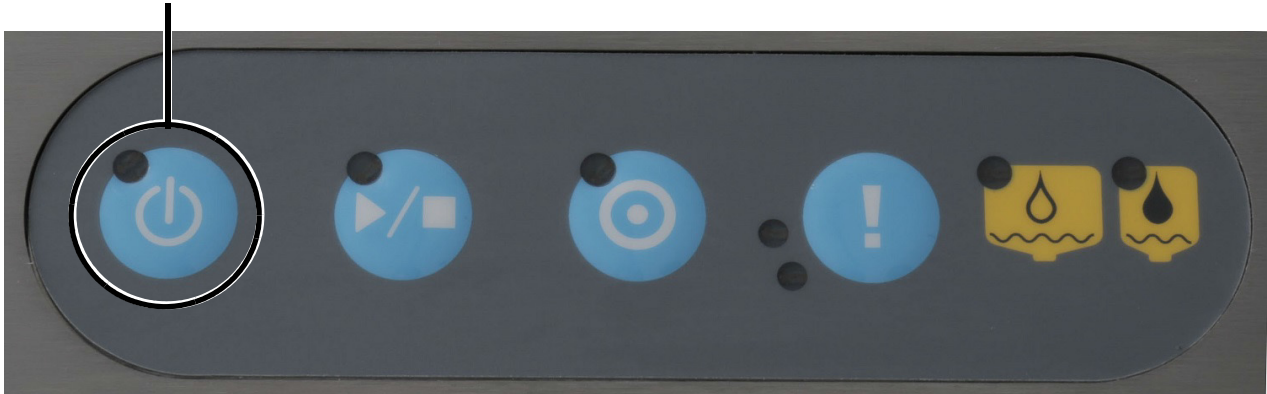
The settings described in table below are used to setup the auto sleep feature.

Setting Name	Explanation
Auto sleep enabled	Enable or disable auto sleep.
Action for auto sleep	Define the action that triggers auto sleep: No print - The printer will enter sleep mode after a specified amount of time has passed since the last print trigger signal is received. Jet off - The printer will enter sleep mode after a specified amount of time has passed since the ink jet was last sequenced off.
Auto sleep time (mins)	Define the amount of time from the last print trigger signal received, or from the time the ink jet was last sequenced off, to when the printer will enter sleep mode.

Action for Power Button (Standby Button)

The *Action for power button* setting defines what action will occur when the standby button on the printer cabinet is pressed.

Standby Button



Standby Button Location

To change the standby button action:

- (1) Select *Home > Setup > Production line setup > Power options*.
- (2) Select the *Action for power button* drop down setting.
- (3) Select one of the options described below:

Setting Name	Explanation
Sleep	The printer will enter sleep mode when the standby button is pressed.
Shut down	The printer will shut down when the standby button is pressed.
Prompt	The printer will ask the user if it should enter sleep mode or shut down when the standby button is pressed.

End of procedure.

CONNECTIVITY PACK OVERVIEW

As standard, Ax-Series printers are equipped with a power socket and two product detect / shaft encoder sockets. Ax350i/Ax550i printers are also equipped with a TouchScreen socket.

Connectivity packs can be fitted to Ax-Series printers to equip them with additional external sockets. Connectivity packs can be fitted when the printer is manufactured, or fitted at a later date by an engineer certified by Domino as an upgrade.

The table below lists the connectivity pack options:

Pack Name and Part Number	Printer Availability	Description
Status Pack - EPT022072SP	Ax150i Ax350i Ax550i	The Status Pack adds the following external connections: • Beacon connector: 5-Way Socket • Alarm connector: 7-Way Socket, 1A, 30V maximum
RS232 Pack - EPT022073SP	Ax150i Ax350i Ax550i	The RS232 Pack adds the following external connections: • Serial RS232 Port: 8-Way Socket
GPIO Pack - EPT022074SP	Ax150i Ax350i Ax550i	The GPIO pack adds the following external connections: • GPIO Port: 14-way Socket
Comms Pack - EPT022075SP	Ax150i Ax350i Ax550i	The Comms Pack adds the following external connections: • Ethernet Port • USB Port
Extended Comms Pack - EPT022070SP	Ax350i Ax550i	The extended comms pack adds the following external connections: • Powered HMI Port: 7-way Socket • Control Port: 8-way Socket • USB port
Extended GPIO Pack - EPT022071SP	Ax350i Ax550i	The extended GPIO pack adds the following external connections: • User Port: 25-Way D-Sub Socket • User Port: 37-Way D-Sub Socket

CONNECTIVITY PACK DETAILS

Extended Comms Pack



WARNING: High voltage electricity. Risk of death or injury from electric shock. Only trained engineers, certified by Domino can install this pack.

CAUTION: Risk of damage to printer. Only trained engineers certified by Domino can install this pack.

Note: Available for Ax350i/Ax550i printers only.

The extended comms pack is a connectivity pack that adds the following external connections to the printer cabinet:

- Powered HMI Port
- External Control Port
- USB Port

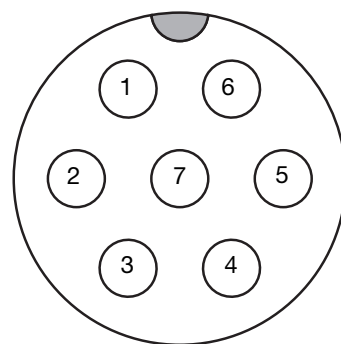
For powered HMI, external control and USB port locations, see [page 3-13](#).

Powered HMI Port

The powered HMI port provides a 24V power supply and communication for a Domino TouchPanel Type 1 or Type 2.

Socket type: 7 way DIN Socket IEC60130-9.

Pin	Signal	Description
1	+24V	24V supply at 500mA nominal
2	0V	Power GND
3	TX1+	Transmit +
4	TX1-	Transmit -
5	RX1+	Receive +
6	RX1-	Receive -
7	NC	Not Connected
Shell	NC	Ground via panel mounting



*Powered HMI Port
(external cabinet view)*

External Control Port

The external control port enables external equipment to monitor the printer's on/off status and power the printer on/off.



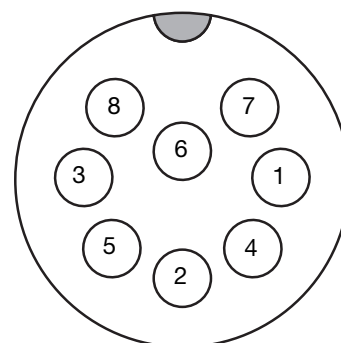
CAUTION: *Input voltages above 5V will damage the printer's main PCB.*

If voltage free contacts are used to activate the inputs, pin 7 (5V+ OUT) can be used as the high voltage level.

The signals described below are 3v3 TTL compatible. Inputs are 5V TTL compatible.

Socket type: 8 way DIN Socket IEC60130-9.

Pin	Signal	Description
1	PWR_ON_IN	Remote Power On Input. A high pulse will turn the printer on.
2	PWR_OFF_IN	Remote Power Off Input. A high pulse will request that the printer is turned off.
3	PWR_STAT_OUT	Power Status Output. 0V output when the printer is off. 3v3 output when the printer is on.
4	PRINT_EN_IN	This input is for future use.
5	PRINT_STATUS_OUT	This output is for future use.
6	CUSTOM_OUT	This output is for future use.
7	5V+ OUT	+5V STANDBY (<100mA).
8	0V	Signal Ground.
Shell	NC	Ground via panel mounting.



*External Control Port
(external cabinet view)*

USB Port

An external USB type A port which can be used to transfer files to and from the printer.

Status Pack



WARNING: **High voltage electricity. Risk of death or injury from electric shock. Only trained engineers, certified by Domino, can install this pack.**

CAUTION: *Risk of damage to printer. Only trained engineers certified by Domino can install this pack.*

The Status Pack is a connectivity pack that adds the following external connections:

- 5 way socket for a beacon.
- 7 way socket for an alarm (Requires: L014981 - Cable X29 Interlock 9m BCP7).

For beacon and alarm socket locations, see [page 3-13](#).

Beacon Socket

The beacon socket provides an external 24V DC FET switched output to drive each of the lamps in a standard Domino beacon.

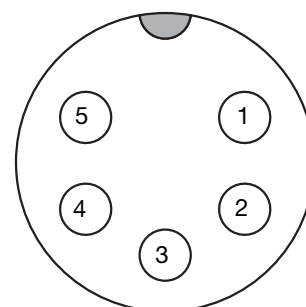
The beacon lamps illuminate in a solid state as described below:

Lamp Colour	Description
Green	Printer is printing or ready to print.
Red	There is a fault, the printer has stopped, an alert is displayed on the user interface.
Amber	The printer requires attention, the printer is still printing, an alert is displayed on the user interface.
Blue	The printer is on but in an idle state. The printer is not ready for printing.

Socket type: 5 way DIN Socket IEC60130-9.

The beacon socket pin assignments are shown below:

Pin	Signal	Description
1	Blue	Lamp driver 24V FET - switched
2	Amber	Lamp driver 24V FET - switched
3	Green	Lamp driver 24V FET - switched
4	Red	Lamp driver 24V FET - switched
5	0V	Ground
Shell	NC	Ground via panel mounting



*Beacon Socket
(external cabinet view)*

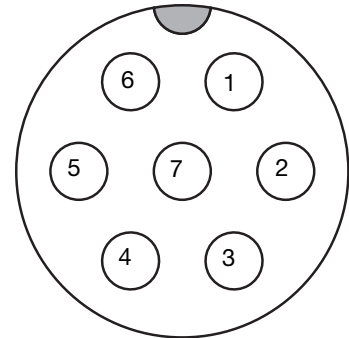
Alarm Socket

The alarm socket provides an output to trigger a customer supplied alarm.

Socket type: 7 way DIN Socket IEC60160-9.

The alarm socket pin assignments are shown below:

Pin	Signal	Description
1	ALARM1_NC	Green (normally closed)
2	ALARM1_NO	Green (normally open)
3	ALARM2_NC	Red (normally closed)
4	ALARM2_NO	Red (normally open)
5	ALARM3_NC	Amber (normally closed)
6	ALARM3_NO	Amber (normally open)
7	COM	Externally supplied common
Shell	NC	Ground via panel mounting



*Alarm Socket
(external cabinet view)*

RS232 Pack



WARNING: **High voltage electricity. Risk of death or injury from electric shock. Only trained engineers, certified by Domino, can install this pack.**

CAUTION: *Risk of damage to printer. Only trained engineers certified by Domino can install this pack.*

The RS232 Pack is a connectivity pack that adds an 8-Way RS232 plug to the printer cabinet. The RS232 plug can be used to send information to the printer, such as labels and variable information from an external control system. Information can also be sent to the external control system from the printer. Basic printer control can also be performed using the RS232 plug.

For the RS232 plug location, see [page 3-13](#).

EDC Serial Protocol Setup

To setup and enable EDC serial protocol for the printer's RS232 port:

- (1) Select *Home > Setup > Printer network > Protocol settings*.
- (2) Select the *Protocol Setting* drop down setting and select *EDC Serial*.
- (3) Select the *Serial Com* drop down setting and select *COM1*.

Note: *COM2 is for the 37-way user port (if fitted).*

- (4) Select *Advanced*.
- (5) The advanced serial settings can be set using the drop down boxes.
- (6) Select *Save*.
- (7) Define the EDC serial protocol settings.
- (8) Select the *Protocol enabled* tick box.

End of procedure.

Codenet Protocol Setup

To setup and enable codenet protocol for the printer's RS232 port:

- (1) Select *Home > Setup > Printer network > Protocol settings*.
- (2) Select the *Protocol Setting* drop down setting and select *Codenet*.
- (3) Select the *Serial Com* drop down setting and select *COM1*.

Note: *COM2 is for the 37-way user port (if fitted).*

- (4) Select *Advanced*.
- (5) The advanced serial settings can be set using the drop down boxes.
- (6) Select *Save*.
- (7) Define the codenet protocol settings.
- (8) Select the *Protocol enabled* tick box.

End of procedure.

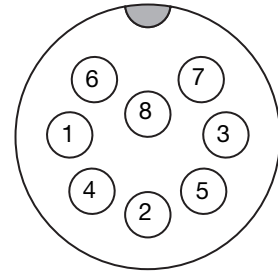
RS232 Plug

The RS232 plug provides external RS232 level serial connection to user equipment.

Plug type: 8 way DIN plug IEC60160-9.

The RS232 plug pin assignments are shown below.

Pin	Signal	Description
1	RS232_DCD	Data Carrier Detect
2	RS232_TXD	Transmit Data
3	RS232_RXD	Receive Data
4	RS232_DSR	Data Set Ready
5	0V	Signal GND
6	RS232_DTR	Data Terminal Ready
7	RS232_CTS	Clear To Send
8	RS232_RTS	Request To Send
Shell	NC	Ground via panel mounting



*RS232 Plug
(external cabinet view)*

Comms Pack



WARNING: High voltage electricity. Risk of death or injury from electric shock. Only trained engineers, certified by Domino can, install this pack.

CAUTION: Risk of damage to printer. Only trained engineers certified by Domino can install this pack.

The comms pack is a connectivity pack that adds an external USB and Ethernet connection to the printer cabinet:

For USB and Ethernet port locations, see [page 3-13](#).

Codenet Protocol Setup

To setup and enable codenet protocol for the printer's Ethernet port:

- (1) Select *Home > Setup > Printer network > Protocol settings*.
- (2) Select the *Protocol Setting* drop down setting and select *Codenet*.
- (3) Select the *Protocol Mode* drop down setting and select *TCP*.
- (4) Define the codenet protocol settings.
- (5) Select the *Protocol enabled* tick box.

End of procedure.

IO Packs

GPIO Pack


WARNING:

High voltage electricity. Risk of death or injury from electric shock. Only trained engineers, certified by Domino can install this pack.

CAUTION:

Risk of damage to printer. Only trained engineers certified by Domino can install this pack.

The GPIO pack is a connectivity pack that adds a 14 way GPIO port to the printer cabinet. The GPIO port can be connected to customer supplied equipment to enable control of basic printer functions and output of printer acknowledgements. Additionally, up to 15 labels can be selected from the printer's internal memory and sent to print.

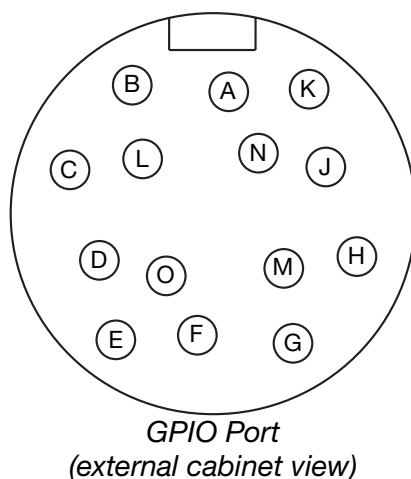
To setup the GPIO port functions, see [page 6-40](#).

For the GPIO port location, see [page 3-13](#).

GPIO Port

Socket type: Panel mount 14 way Lumberg.

Note: See [page 6-42](#) to assign configurable input and output pin functions.



Pin	Signal	Description
A	0V	GROUND
B	INPUT 0+	CONFIGURABLE INPUT
C	INPUT 0-	CONFIGURABLE INPUT
D	INPUT 1	CONFIGURABLE INPUT
E	INPUT 2	CONFIGURABLE INPUT
F	INPUT 3	CONFIGURABLE INPUT
G	+24V DC	24V DC EXTERNAL SUPPLY FOR INPUTS
H	0V	COMMON EXTERNAL REFERENCE FOR INPUTS
J	OUTPUT 0	CONFIGURABLE OUTPUT

OPTION PACKS

Pin	Signal	Description
K	OUTPUT 1	CONFIGURABLE OUTPUT
L	OUTPUT 2	CONFIGURABLE OUTPUT
M	OUTPUT 3	CONFIGURABLE OUTPUT
N	+24V DC	24V DC EXTERNAL SUPPLY FOR OUTPUTS
O	0V	COMMON EXTERNAL REFERENCE FOR OUTPUTS
Shell	GROUND	EMC GROUND VIA PANEL MOUNTING

Extended IO Pack


WARNING:

High voltage electricity. Risk of death or injury from electric shock. Only trained engineers, certified by Domino can install this pack.

CAUTION:

Risk of damage to printer. Only trained engineers certified by Domino can install this pack.

Note: Available for Ax350i/Ax550i printers only.

The extended IO pack is a connectivity pack that adds two user port sockets to the printer cabinet. The user ports can be connected to customer supplied equipment to enable control of basic printer functions and output of printer acknowledgements. Additionally, up to 255 labels can be selected from the printer's internal memory and sent to print.

To setup the user port functions, see [page 6-40](#).

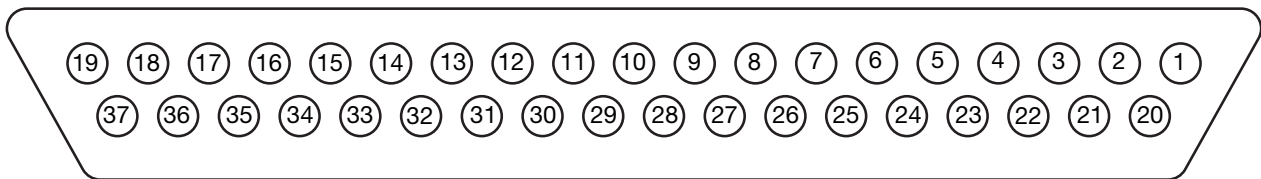
For user port locations, see [page 3-13](#).

37 Way User Port

The 37 way user port provides the following external connections to user equipment:

- Configurable opto-isolated inputs.
- Configurable opto-isolated outputs.
- Isolated and non-isolated power and ground reference.

Socket type: Vertical PCB mount 37W D-Sub waterproof (IP67) socket.



*37 Way User Port Socket Connections
(external cabinet view)*

Note: See [page 6-42](#) to assign configurable input and output pin functions.

Pin	Signal Name
1	CONFIGURABLE INPUT
2	CONFIGURABLE INPUT
3	CONFIGURABLE INPUT
4	CONFIGURABLE INPUT
5	CONFIGURABLE INPUT
6	CONFIGURABLE INPUT
7	CONFIGURABLE INPUT
8	CONFIGURABLE INPUT
9	CONFIGURABLE INPUT

OPTION PACKS

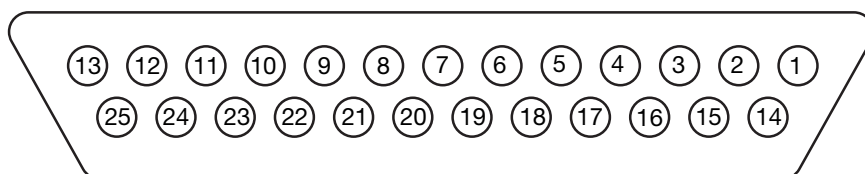
Pin	Signal Name
10	CONFIGURABLE INPUT
11	CONFIGURABLE INPUT
12	CONFIGURABLE INPUT
13	ISOLATED GROUND
14	CONFIGURABLE OUTPUT
15	CONFIGURABLE OUTPUT
16	CONFIGURABLE OUTPUT
17	CONFIGURABLE OUTPUT
18	OUTPUT COMMON
19	ISOLATED +24V OUTPUT
20	ISOLATED +24V OUTPUT
21	NOT USED
22	NOT USED
23	ISOLATED GROUND
24	DIGITAL GROUND
25	SCREEN EMC GROUND
26	NOT USED
27	RS232 CTS INPUT
28	RS232 RTS OUTPUT
29	NOT USED
30	NOT USED
31	NOT USED
32	RS232 TX OUTPUT
33	RS232 RX INPUT
34	RS232 DTR OUTPUT
35	DIGITAL GROUND
36	RS232 DSR INPUT
37	NOT USED

25 Way User Port

The 25 way user port provides the following external connections to user equipment:

- Configurable opto-isolated inputs.
- Configurable opto-isolated or open collector outputs.
- Isolated and non-isolated power and ground reference.

Socket type: Vertical PCB mount 25W D-Sub waterproof (IP67/8) socket.



*25 Way User Port Socket Connections
(external cabinet view)*

Note: See [page 6-42](#) to assign configurable input and output pin functions.

Pin	Signal Name
1	DIGITAL GROUND
2	CONFIGURABLE INPUT
3	CONFIGURABLE INPUT
4	CONFIGURABLE INPUT
5	CONFIGURABLE INPUT
6	CONFIGURABLE OUTPUT
7	CONFIGURABLE OUTPUT
8	CONFIGURABLE OUTPUT
9	NOT USED
10	NOT USED
11	NOT USED
12	NOT USED
13	ISOLATED GROUND
14	NOT USED
15	NOT USED
16	NOT USED
17	NOT USED
18	OUTPUT COMMON (OPTO)
19	ISOLATED +24V OUTPUT
20	ISOLATED +24V OUTPUT

OPTION PACKS

Pin	Signal Name
21	NOT USED
22	NOT USED
23	ISOLATED GROUND
24	DIGITAL GROUND
25	NOT USED

Using IO Packs

The following pages in this chapter explain the printer settings that are available to setup and use the GPIO Pack and Extended IO Pack.

Set Up

To view and configure the settings available on the set up screen:

- (1) Select *Home > Setup > IO Port > Set up.*
- (2) The settings available on the Set up screen are described in the three tables below.

Table 1: Input Settings

Setting	Explanation
GPI debounce time (us)	Define the input debounce time.
GPI queue depth	Define the maximum number of data packets allowed in the input queue. Range 4 - 1023
GPI queue empty alert	Tick the tick box to raise a printer alert when the external data queue is empty.
GPI queue first element received alert	Tick the box to raise a printer alert when the first data packet enters the external data queue.
GPI queue overflow alert	Tick the box to raise a printer alert when there are too many entries in the external data queue.

Table 2: Other Settings

Setting	Explanation
EDC history queue depth	External data packets sent to the printer can be saved in a history queue. The history queue allows the printer to check if duplicate data is received and to raise an alert if required. This setting defines the maximum number of data packets that can be saved in the history queue. Range: 0 - 2048
EDC history queue maximum duplicates	Define the number of duplicate data packets allowed before a printer alert is triggered. For example, if this setting is set to ten, then ten duplicate data packets will be allowed before the alert is triggered. Range: 0 - 100
EDC history queue full action	Define the printers action when the EDC history queue becomes full: Reset - Clear the history queue and start again. Overwrite - Overwrite the history queue.

Table 2: Other Settings

Setting	Explanation
EDC history queue preserve over power off	Tick the box to preserve the EDC history queue when the printer is powered off. If the box is not ticked, the EDC history queue will be deleted when the printer is powered off.
Label select: alert on empty slot selection	Tick the box to show an alert when a empty label slot is selected. See “External Label Selection” on page 6-47.
Custom shift register: alert on invalid bit count	<i>Note: The custom shift register is a data transfer feature used in special applications such as Tetrapak.</i> Tick the box to show an amber alert when the number of bits read by the custom shift register since the last product detect is not a multiple of 16.

End of procedure.

Assignment

Inputs

Assign a function to each input pin to ensure the correct function is triggered.

Notes: (1) *Inputs and Outputs that are labelled “B-Pin” are valid for the GPIO port J19 (if fitted), see [page 6-34](#).*

(2) *The inputs and outputs that are labelled “Ext25” are valid for the 25 way user port J24 (if fitted), see [page 6-38](#).*

(3) *The inputs and outputs that are labelled “Ext37” are valid for the 37 way user port J23 (if fitted), see [page 6-36](#).*

To assign pins to specific input functions:

- (1) Select *Home > Setup > IO Port > Assignment*.
- (2) Select the *Active level* drop down setting for each input pin and select if a High or Low signal will activate the input function.
- (3) Select the *Function* drop down setting for each pin and select a function as described in the table below.

Table 3: Input Functions

Function	Explanation
Unassigned	The pin will not be assigned to any function.
Label selection	The pin will be assigned to external label selection. When selecting <i>Label selection</i> for the first time, the alert “no message is selected for printing” will be displayed. Select only the minimum number of pins required for external label selection. See page 6-47 for more information about external label selection.
Reset print counter 1	Reset <i>Resettable Counter 1</i> on the printer’s <i>Statistics</i> menu. Resettable Counter 1 can be viewed on the printer’s user interface by selecting <i>Home > Setup > Printer status > Statistics</i>
Reset print counter 2	Reset <i>Resettable Counter 2</i> on the printer’s <i>Statistics</i> menu. Resettable Counter 2 can be viewed on the printer’s user interface by selecting <i>Home > Setup > Printer status > Statistics</i>
Print inverted	Invert the current printing label.
Print reversed	Reverse the current printed label.
Global print inverted	Invert all labels that are sent to print until this input is triggered again.
Global print reversed	Reverse all labels that are sent to print until this input is triggered again.
Jet control	Sequence the ink jet on or off.
Reset all data	Reset all counter elements in the currently printing label. All counters will reset to the value that was specified when the counters were created.
Reset data 1	Reset counter element 1 in the currently printing label. The counter will reset to the value that was specified when the counter was created.

Table 3: Input Functions

Function	Explanation
Reset data 2	Reset counter element 2 in the currently printing label. The counter will reset to the value that was specified when the counter was created.
Reset data 3	Reset counter element 3 in the currently printing label. The counter will reset to the value that was specified when the counter was created.
Reset data 4	Reset counter element 4 in the currently printing label. The counter will reset to the value that was specified when the counter was created.
Update print counter 1	Update counter element 1 in the currently printing label. The counter will be updated by the step size that was specified when the counter was created.
Update print counter 2	Update counter element 2 in the currently printing label. The counter will be updated by the step size that was specified when the counter was created.
Update print counter 3	Update counter element 3 in the currently printing label. The counter will be updated by the step size that was specified when the counter was created.
Update print counter 4	Update counter element 4 in the currently printing label. The counter will be updated by the step size that was specified when the counter was created.
Print abort	The printer will stop printing the label at the end of the current print stroke.
Print enable	Enable or disable printing, this input will not turn the ink jet on or off.
CSR clock	This interrupt is for the Custom Shift Register (CSR) feature used in conjunction with the CSR data input. This is used in special applications such as Tetrapak.
CSR data	This input is for the custom shift register feature mainly used in special applications like Tetrapak.
Reset backlash	Reset the encoder backlash count to zero. See “Encoder Mode Setup” on page 3-44.

(4) The number for each pin will turn green to indicate when an input is received.



Input Received



Input NOT Received

End of procedure.

Outputs

Assign a function to each output pin to ensure the correct function is triggered.

Notes: (1) Inputs and Outputs that are labelled “B-Pin” are valid for the GPIO port J19 (if fitted), see [page 6-34](#).

(2) The inputs and outputs that are labelled “Ext25” are valid for the 25 way user port J24 (if fitted), see [page 6-38](#).

(3) The inputs and outputs that are labelled “Ext37” are valid for the 37 way user port J23 (if fitted), see [page 6-36](#).

To assign pins to specific output functions:

- (1) Select *Home > Setup > IO Port > Assignment*.
- (2) Scroll down the screen to the list of *Output* functions.
- (3) Select the *Active level* drop down setting for each output pin and select if a High or Low signal will be sent when the output function is active.
- (4) Adjust the *Pulse time (ms)* value for each pin to set the length of the output pulse.
- (5) Select the *Function* drop down setting for each output pin and select a function as described in the table below.

Table 4: Output Functions

Function	Explanation
Unassigned	The pin will not be assigned to any function.
Print ready	The output will be switched when the printer is ready to print.
Good print	The output will be switched when printing is good.
Bad print	The output will be switched when an alert that could affect printing the right message on the right product occurs. Examples of alerts that will trigger the output include, Print stretch detected, Long print, Check print quality, Detected too Many Products, Some Ignored, Print Trigger Occurred Too Early, and Print GO occurred While Printing.
EDC buffer low	The output will be switched when the amount of data in the EDC (external data capture) starts to run low.
EDC buffer high	The output will be switched when the amount of data in the EDC (external data capture) is high.
External ack	The output will be switched after an input has been received that expects an acknowledgement.
End of Label	The output will be switched after a label has finished printing.
Counter 1 Rollover	The output will be switched when the rollover of counter element 1 in the currently printing label occurs.
Counter 2 Rollover	The output will be switched when the rollover of counter element 2 in the currently printing label occurs.
Counter 3 Rollover	The output will be switched when the rollover of counter element 3 in the currently printing label occurs.

Table 4: Output Functions

Function	Explanation
Counter 4 Rollover	The output will be switched when the rollover of counter element 4 in the currently printing label occurs.
Reset all data outputs	The output will be switched when the command to reset all external data queues is sent.
Reset data output 1	The output will be switched when the command to reset external data queue 1 is sent.
Reset data output 2	The output will be switched when the command to reset external data queue 2 is sent.
Reset data output 3	The output will be switched when the command to reset external data queue 3 is sent.
Reset data output 4	The output will be switched when the command to reset external data queue 4 is sent.

(6) To send a test output signal press the *Test* button for each output.

*Test Button*

(7) The number next to each output will turn green to indicate when the output is active.

*Active**Not Active*

End of procedure.

Monitor

The Monitor screen is used to monitor input and output statistics. The External Data Capture (EDC) queues and the product detect queue can also be cleared on this screen. The settings and information available on the Monitor screen are explained in the table below.

To view the IO Port monitor screen:

- (1) Select *Home > Setup > IO Port > Monitor*.
- (2) The settings described in the table below can now be changed:

Function	Explanation
'EdcSerial' queue state	Displays the number of data packets transmitted over serial connection that are stored in the external data queue. Select <i>Clear</i> to clear the external data queue.
'Edc TcpSink' Queue state	Displays the number of data packets transmitted over Ethernet TCP/IP connection that are stored in the external data queue. Select <i>Clear</i> to clear the external data queue.
Current PD frequency	Displays the product detect sensor frequency as also displayed on the line movement graph. To view the line movement graph select <i>Home > Setup > Production line setup > Line movement</i> .
EDC history reset	Data packets received by the printer can be stored in the historic log. This allows new data to be checked against historic data to avoid printing duplicate information. This setting displays the number of historic logs stored in the printer. Select <i>Clear</i> to clear the historic log.
Encoder count	Displays the number of shaft encoder pulse counts.
Encoder frequency	Displays the shaft encoder pulse count frequency in Khz as also displayed on the line movement graph. To view the line movement graph select <i>Home > Setup > Production line setup > Line movement</i> .
PD queue state	Displays the number of print trigger signals that are queued. Select <i>Clear</i> to clear the queue.

End of procedure.

External Label Selection

External devices can be connected to the user ports (if fitted) or GPIO port (if fitted) to select labels for printing. External label selection uses assigned input pins as a binary reference to select labels from the printers internal memory and send them to print.

- Notes:*
- (1) *Up to four input pins on the GPIO port can be assigned to label selection. If all four input pins are assigned to label selection, up to 15 labels can be selected.*
 - (2) *Up to eight input pins on the 25-way user port and 37 way user port can be assigned to label selection. If all eight input pins are assigned to label selection, up to 255 labels can be selected.*
 - (3) *The number of pins available for label selection will be reduced if pins are assigned to other functions.*

Assign Input Pins to Label Selection

To assign input pins to label selection:

- (1) Select *Home > Setup > IO Port > Assignment*.
- (2) Select each required input pin and select *Label selection*.

End of procedure.

Assign Labels to Slot Numbers

Assign each label a slot number to make the label available for external label selection. When a binary signal is received via the assigned input pins, the number received will select the label that is assigned to that slot number and sent the label to print.

To assign a label to a slot number:

- (1) Select *Home > Setup > IO Port > Assignment*.
- (2) Assign the required number of input pins to *Label selection*.
- (3) Select *Label select*.
- (4) Either select the  icon to automatically populate the label slots with labels. Or, select each label slot and select specific labels to assign to each slot.
- (5) Select *ON*.

End of procedure.

Disable External Label Selection

To disable external label selection:

- (1) Select *Home > Setup > IO Port > Label select*
- (2) Select the icon at the top of the screen to clear the label slots.
- (3) Select *Assignment*.
- (4) Un-assign the input pins, or assign the input pins to different functions.

End of procedure.

Binary Inputs for Label Selection

The table below lists the valid binary inputs for label selection using the GPIO port.

Label Slot Number	Pin B+C	Pin D	Pin E	Pin F
Input Binary Value	1	2	3	4
Label Slot Number	Inputs Selected			
No Label Selected	0	0	0	0
Slot 1	1	0	0	0
Slot 2	0	1	0	0
Slot 3	1	1	0	0
Slot 4	0	0	1	0
Slot 5	1	0	1	0
Slot 6	0	1	1	0
Slot 7	1	1	1	0
Slot 8	0	0	0	1
Slot 9	1	0	0	1
Slot 10	0	1	0	1
Slot 11	1	1	0	1
Slot 12	0	0	1	1
Slot 13	1	0	1	1
Slot 14	0	1	1	1
Slot 15	1	1	1	1

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